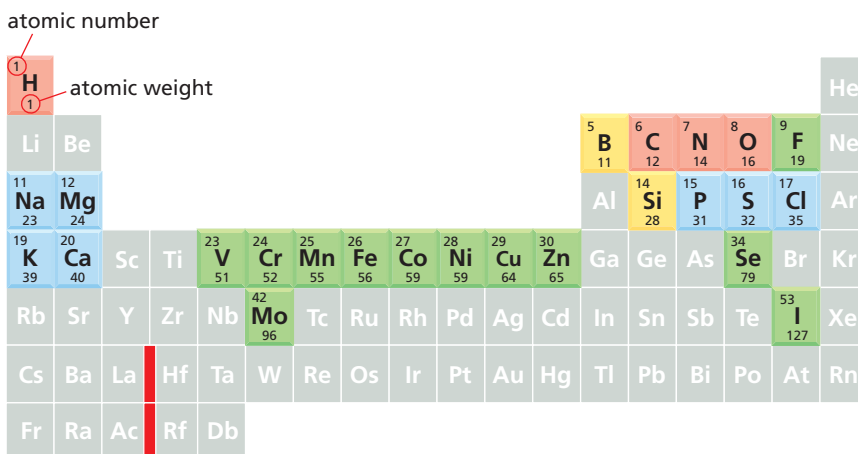


CHAPTER 2

Much of the information in this chapter is summarized—and in some cases further elaborated—in the nine two-page Panels with which the chapter ends (Panels 2-1 to 2-9). Although the Panels will be cited at appropriate places in the text, they should also be useful for refreshing background knowledge when reading later chapters.

THE CHEMICAL COMPONENTS OF A CELL

Living organisms are made of only a small selection of the 92 naturally occurring elements, four of which—carbon (C), hydrogen (H), nitrogen (N), and oxygen (O)—make up 96.5% of an organism's weight ([Figure 2-1](#)). The *atoms* of these



The Chemical Components of a Cell

Catalysis and the Use of Energy by Cells

How Cells Obtain Energy from Food

Figure 2-1 The main elements in cells, highlighted in the periodic table. When ordered by their atomic number and arranged in this manner, elements fall into vertical columns that show similar properties.

The four elements highlighted in *red* constitute 99% of the total number of atoms present in the human body (and 96.5% of its weight). An additional seven elements, highlighted in *blue*, together represent about 0.9% of the total atoms in our bodies. The elements shown in *green* are required in trace amounts by humans. It remains unclear whether those elements shown in *yellow* are essential in humans.

The chemistry of life is therefore predominantly the chemistry of lighter elements. The atomic weights shown here are those of the most common isotope of each element. The vertical *red* line marks a break in the periodic table where a group of large atoms with similar chemical properties is omitted.

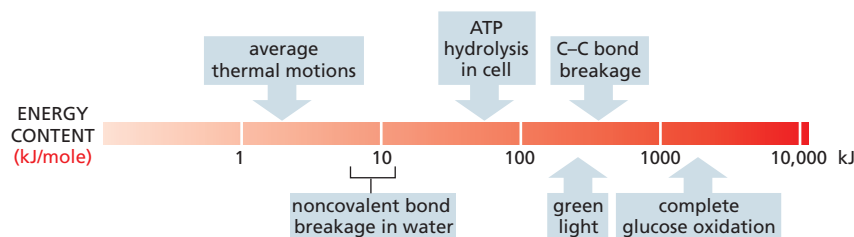


Figure 2-2 Some energies important for cells. A crucial property of any bond—covalent or noncovalent—is its strength. *Bond strength* is measured by the amount of energy that must be supplied to break it, expressed in units of either kilojoules per mole (kJ/mole) or kilocalories per mole (kcal/mole). Thus if 100 kJ of energy must be supplied to break 6×10^{23} bonds of a specific type (that is, 1 mole of these bonds), then the strength of that bond is 100 kJ/mole. Note that, in this diagram, energies are compared on a logarithmic scale. (Typical strengths and lengths of the main classes of chemical bonds are given in Table 2-1, later in text.)

One joule (J) is the amount of energy required to move an object a distance of 1 meter (m) against a force of 1 newton (N). This measure of energy is derived from the SI units (Système International d'Unités) universally employed by physical scientists. A second unit of energy, often used by cell biologists, is the kilocalorie (kcal): 1 calorie (cal) is the amount of energy needed to raise the temperature of 1 gram (g) of water by 1°C. One kilojoule (kJ) is equal to 0.239 kcal.

elements are linked together by **covalent bonds** to form *molecules* (see [Panel 2-1](#), pp. 94–95). Because covalent bonds are typically 100 times stronger than the thermal energies within a cell, they resist being pulled apart by thermal motions, and they are normally broken only during biologically catalyzed chemical reactions that are of use to the cell. *Noncovalent bonds* are much weaker ([Figure 2-2](#)), but sets of them allow molecules to recognize each other and reversibly associate, which is critical for the vast majority of biological functions.

Water Is Held Together by Hydrogen Bonds

Because 70% of the weight of a cell is water, the reactions that make life possible occur in an aqueous environment. Life on Earth is thought to have begun in shallow bodies of water that had concentrated essential molecules, and the conditions in that primeval environment have left a permanent stamp on the chemistry of all living things.

The chemical properties of water are reviewed in [Panel 2-2](#) (pp. 96–97). In a water molecule (H_2O), the two H atoms are linked to the O atom by covalent bonds that are highly polar, inasmuch as the O atom attracts electrons more strongly than does the H atom. Consequently, there is a preponderance of positive charge on the two H atoms and of negative charge on the O atom. When a positively charged region of one water molecule (that is, one of its H atoms) approaches a negatively charged region (that is, the O atom) of a second water molecule, the electrical attraction between them can result in a **hydrogen bond** ([Figure 2-3A](#)). These bonds are much weaker than covalent bonds and are easily broken by the random thermal motions that reflect the heat energy of the molecules. Thus, each bond lasts only a very short time. But the combined effect of many weak bonds can be profound. For example, each water molecule can form hydrogen bonds through its two H atoms to two other water molecules, producing a network in which hydrogen bonds are being continually broken and formed. It is only because of these hydrogen bonds that link water molecules together that water is a liquid at room temperature—with a high boiling point and high surface tension—rather than a gas.

Hydrogen bonds are not limited to water, and they are central to much of biology. This bond represents a special form of polar interaction in which an electropositive hydrogen atom is shared by two electronegative atoms. The hydrogen in this bond can be viewed as a proton that has partially dissociated from a donor atom, allowing it to be shared by a second, acceptor atom. Unlike a typical electrostatic interaction, this bond is highly directional—being strongest when a straight line can be drawn between all three of the involved atoms ([Figure 2-3B](#)).

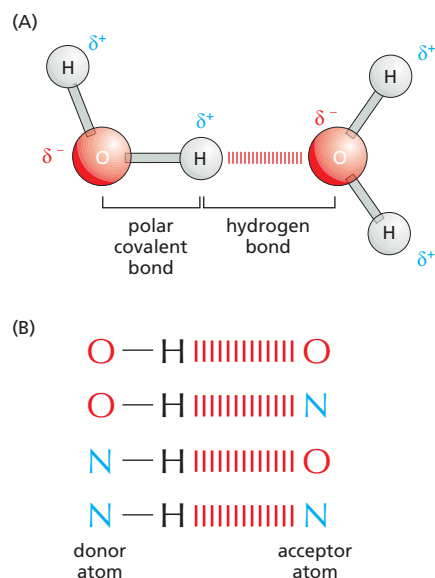


Figure 2-3 The noncovalent hydrogen bond. (A) A hydrogen bond forms between two water molecules. The slight positive charge associated with the hydrogen atom is electrically attracted to the slight negative charge of the oxygen atom. This causes water to exist as a large hydrogen-bonded network (see [Panel 2-2](#), pp. 96–97). (B) In cells, hydrogen bonds commonly form between molecules that contain an oxygen or nitrogen. The atom bearing the hydrogen is considered the H-bond donor, and the atom that interacts with the hydrogen is the H-bond acceptor. This type of dipole–dipole interaction is of critical importance in biology. For this reason, and because it is highly directional, the hydrogen bond receives special attention among the set of noncovalent attractions that we discuss next.

Molecules, such as alcohols, that contain polar bonds and that can form hydrogen bonds with water dissolve readily in water. Molecules carrying charges (ions) likewise interact favorably with water. Such molecules are termed **hydrophilic**, meaning that they are water-loving. Many of the molecules in the aqueous environment of a cell necessarily fall into this category, including sugars, DNA, RNA, and most proteins. **Hydrophobic** (water-fearing) molecules, by contrast, are uncharged and form few or no hydrogen bonds, and so do not dissolve in water. Hydrocarbons are an important example. In these molecules, all of the H atoms are covalently linked to C atoms by a largely nonpolar bond; thus, they cannot form effective hydrogen bonds to other molecules (see Panel 2-1, pp. 94–95). This makes the hydrocarbon as a whole hydrophobic—a property that is exploited in cells, whose membranes are constructed from molecules that have long hydrocarbon tails, as we shall see in Chapter 10.

Four Types of Noncovalent Attractions Help Bring Molecules Together in Cells

Much of biology depends on the specific binding between different molecules caused by three types of **noncovalent bonds**—hydrogen bonds, electrostatic attractions (ionic bonds), and van der Waals attractions—combined with a fourth factor that can push molecules together: the hydrophobic force.

Electrostatic attractions are strongest when the atoms involved are fully charged, or ionized. But a weaker electrostatic attraction occurs between molecules that contain polar covalent bonds. Like hydrogen bonds, electrostatic attractions are extremely important in biology. For example, any large molecule with many polar groups will have a pattern of partial positive and negative charges on its surface. When such a molecule encounters a second molecule with a complementary set of charges, the two will be drawn to each other by electrostatic attraction.

In addition to hydrogen bonds and electrostatic attractions, a third type of noncovalent bond, called a **van der Waals attraction**, comes into play when any two atoms approach each other closely. These weak, nonspecific interactions are due to fluctuations in the distribution of electrons in every atom, which can generate a transient attraction when the atoms are in very close proximity. These attractions occur in all types of molecules, even those that are nonpolar.

The relative lengths and strengths of these three types of noncovalent bonds are compared to the length and strength of covalent bonds in [Table 2-1](#), both in the presence and in the absence of water. Note that, because water forms competing interactions with the involved molecules, the strength of both electrostatic attractions and hydrogen bonds is greatly weakened inside of the cell.

The fourth effect that often brings molecules together in water is not, strictly speaking, a bond at all. However, a very important **hydrophobic force** is caused by a pushing of nonpolar surfaces out of the hydrogen-bonded water network, where they would otherwise physically interfere with the highly favorable

TABLE 2-1 Covalent and Noncovalent Chemical Bonds

Bond type		Length (nm)	Strength (kJ/mole ^{**})	
			In vacuum	In water
Covalent		0.10	377 (90)	377 (90)
Noncovalent	Ionic*	0.25	335 (80)	12.6 (3)
	Hydrogen	0.17	16.7 (4)	4.2 (1)
	van der Waals attraction (per atom)	0.35	0.4 (0.1)	0.4 (0.1)

*An ionic bond is an electrostatic attraction between two fully charged atoms.
^{**}Values in parentheses are kcal/mole. 1 kJ = 0.239 kcal and 1 kcal = 4.18 kJ.

interactions between water molecules. Bringing any two nonpolar surfaces together reduces their contact with water, and in this sense, the force is nonspecific. Nevertheless, we shall see in Chapter 3 that hydrophobic forces are central to the proper folding of protein molecules.

The properties of the four types of noncovalent attractions are presented in **Panel 2-3** (pp. 98–99). Although each individual noncovalent attraction would be much too weak to be effective in the face of thermal motions, the energies of these noncovalent attractions can sum to create a strong force between two separate molecules. Thus, it is an entire set of noncovalent attractions that enables the complementary surfaces of two macromolecules to hold the two macromolecules together (**Figure 2-4**).

Some Polar Molecules Form Acids and Bases in Water

One of the simplest kinds of chemical reaction, and one that has a considerable significance for cells, takes place when a molecule containing a highly polar covalent bond between a hydrogen and another atom dissolves in water. The hydrogen atom in such a molecule has given up its electron almost entirely to the companion atom, and so exists as an almost naked positively charged hydrogen nucleus; in other words, a **proton** (H^+). When this polar molecule becomes surrounded by water molecules, the proton will be attracted to the partial negative charge on the O atom of an adjacent water molecule. The proton can easily dissociate from its original partner and associate instead with the oxygen atom of the water molecule, generating a **hydronium ion** (H_3O^+) (**Figure 2-5A**). The reverse reaction also takes place very readily, so in an aqueous solution protons are constantly flitting to and fro between one molecule and another.

Substances that release protons when they dissolve in water, thus forming H_3O^+ , are termed **acids**. The higher the concentration of H_3O^+ , the more acidic the solution. H_3O^+ is present even in pure water, at a concentration of 10^{-7} M, as a result of the movement of protons from one water molecule to another (**Figure 2-5B**). By convention, the H_3O^+ concentration is usually referred to as the H^+ concentration, even though most protons in an aqueous solution are present as H_3O^+ . As explained in Panel 2-2, to avoid the use of unwieldy numbers the concentration of H_3O^+ is expressed using a logarithmic scale called the **pH scale**. Pure water has a pH of 7.0 and is said to be neutral; that is, neither acidic ($\text{pH} < 7$) nor basic ($\text{pH} > 7$).

Acids are characterized as being strong or weak, depending on how readily they give up their protons to water. Strong acids, such as hydrochloric acid (HCl), easily lose their protons. Acetic acid, on the other hand, is a weak acid because it holds on to its proton more tightly when dissolved in water. Many of the acids important in the cell—such as molecules containing a carboxyl (COOH) group—are weak acids.

Because the proton of a hydronium ion can be passed readily to many types of molecules in cells, altering their character, the concentration of H_3O^+ inside a cell (the acidity) must be closely regulated. Acids—especially weak acids—will give

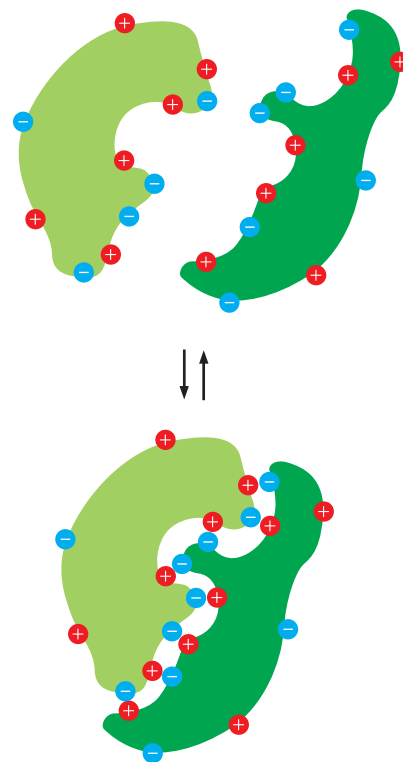


Figure 2-4 Schematic indicating how two macromolecules with complementary surfaces can bind tightly to one another through noncovalent interactions. Noncovalent chemical bonds have less than 1/20 the strength of a covalent bond. They are able to produce tight binding only when many of them are formed simultaneously. Although only electrostatic attractions are illustrated here, in reality all four noncovalent forces often contribute to holding two macromolecules together (**Movie 2.1**).

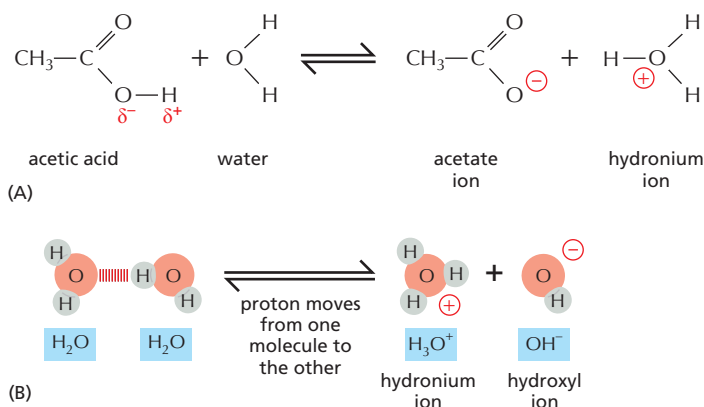


Figure 2-5 How protons readily move in aqueous solutions. (A) The reaction that takes place when a molecule of acetic acid dissolves in water. At pH 7, nearly all of the acetic acid is present as acetate ion. (B) Water molecules continually exchange protons with each other to form hydronium and hydroxyl ions. These ions in turn rapidly recombine to form water molecules.

up their protons more readily if the concentration of H_3O^+ in solution is low and will tend to receive them back if the concentration in solution is high.

The opposite of an acid is a **base**. Any molecule capable of accepting a proton from a water molecule is called a base. Sodium hydroxide (NaOH) is basic (the term *alkaline* is also used) because it dissociates readily in aqueous solution to form Na^+ ions and OH^- ions. Because of this property, NaOH is called a strong base. More important in living cells, however, are the weak bases—those that have a weak tendency to reversibly accept a proton from water. Many biologically important molecules contain an amino (NH_2) group. This group is a weak base that can generate OH^- by taking a proton from water: $-\text{NH}_2 + \text{H}_2\text{O} \rightarrow -\text{NH}_3^+ + \text{OH}^-$ (see Panel 2-2, pp. 96–97).

Because an OH^- ion combines with an H_3O^+ ion to form two water molecules, any increase in the OH^- concentration forces a decrease in the concentration of H_3O^+ , and vice versa. Thus the product of the two values, $[\text{OH}^-] \times [\text{H}_3\text{O}^+]$, is always 10^{-14} (moles/liter)². A pure solution of water contains an equal concentration (10^{-7} M) of both ions, rendering it neutral. The interior of a cell is also kept close to neutrality by the presence of **buffers**: weak acids and bases that can release or take up protons near pH 7, keeping the environment of the cell relatively constant under a variety of conditions.

A Cell Is Formed from Carbon Compounds

Having briefly reviewed the ways that atoms combine into molecules and how these molecules behave in an aqueous environment, we now examine the main classes of small molecules found in cells. We shall see that a few categories of molecules, formed from a handful of different elements, give rise to all the extraordinary richness of form and behavior shown by living things.

If we disregard water and inorganic ions such as potassium, nearly all the molecules in a cell are based on carbon. Carbon is outstanding among all the elements in its ability to form large molecules; silicon is a poor second. Because carbon is small and has four electrons and four vacancies in its outermost shell, a carbon atom can form four covalent bonds with other atoms. Most important, one carbon atom can join to other carbon atoms through highly stable covalent C–C bonds to form chains and rings and hence generate large and complex molecules with no obvious upper limit to their size. The carbon compounds made by cells are called *organic molecules*. In contrast, all other molecules, including water, are said to be *inorganic*.

Certain combinations of atoms, such as the methyl ($-\text{CH}_3$), hydroxyl ($-\text{OH}$), carboxyl ($-\text{COOH}$), carbonyl ($-\text{C}=\text{O}$), phosphate ($-\text{PO}_3^{2-}$), sulfhydryl ($-\text{SH}$), and amino ($-\text{NH}_2$) groups, occur repeatedly in the molecules made by cells. Each such **chemical group** has distinct chemical and physical properties that influence the behavior of the molecule in which the group occurs. The most common chemical groups and some of their properties are summarized in Panel 2-1 (pp. 94–95).

Cells Contain Four Major Families of Small Organic Molecules

The small organic molecules of the cell are carbon-based compounds that have masses in the range of 100–1000 daltons and contain up to 30 or so carbon atoms. They are usually found free in solution and have many different fates. Some are used as *monomer* subunits to construct the giant polymeric *macromolecules* that make up most of the mass of the cell—proteins, nucleic acids, and large polysaccharides. Others act as energy sources and are broken down and transformed into other small molecules in a maze of intracellular metabolic pathways. Many small molecules have more than one role in the cell; for example, acting both as a potential subunit for a macromolecule and as an energy source. Small organic molecules account for only about one-tenth of the total mass of organic matter in a cell, but they are very diverse. Nearly 4000 different kinds of small organic molecules have been detected in the well-studied bacterium, *Escherichia coli*.

All organic molecules are synthesized from and are broken down into the same set of simple compounds. As a consequence, the compounds in a cell are

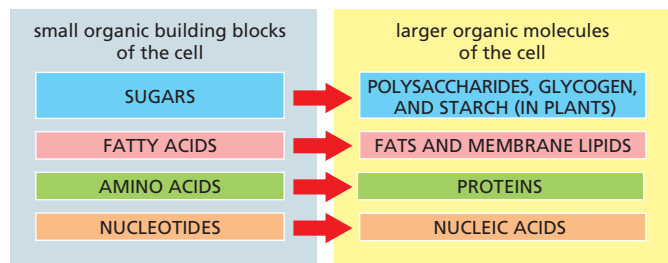
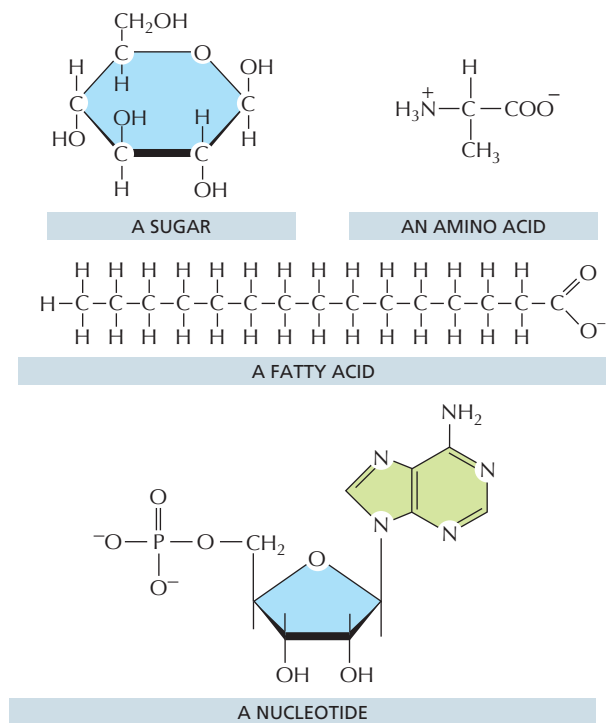


Figure 2-6 The four main families of small organic molecules in cells. These small molecules form the monomeric building blocks, or subunits, for most of the macromolecules and other assemblies of the cell. Some, such as the sugars and the fatty acids, are also energy sources. Their structures are outlined here and shown in more detail in the Panels at the end of this chapter and in Chapter 3.

chemically related and most can be classified into a few distinct families. Broadly speaking, cells contain four major families of small organic molecules: the *sugars*, the *fatty acids*, the *nucleotides*, and the *amino acids* (Figure 2-6). Although many compounds present in cells do not fit into these categories, these four families of small organic molecules, together with the macromolecules made by linking them into long chains, account for a large fraction of the cell mass.

Amino acids and the proteins that they form will be the subject of Chapter 3. A summary of the structures and properties of the remaining three families—sugars, fatty acids, and nucleotides—is presented in Panels 2-4, 2-5, and 2-6, respectively (see pp. 100–105).

The Chemistry of Cells Is Dominated by Macromolecules with Remarkable Properties

By weight, **macromolecules** are the most abundant carbon-containing molecules in a living cell (Figure 2-7). They are the principal components from which a cell is constructed, and they also determine the most distinctive properties of living organisms. The macromolecules in cells are polymers that are constructed

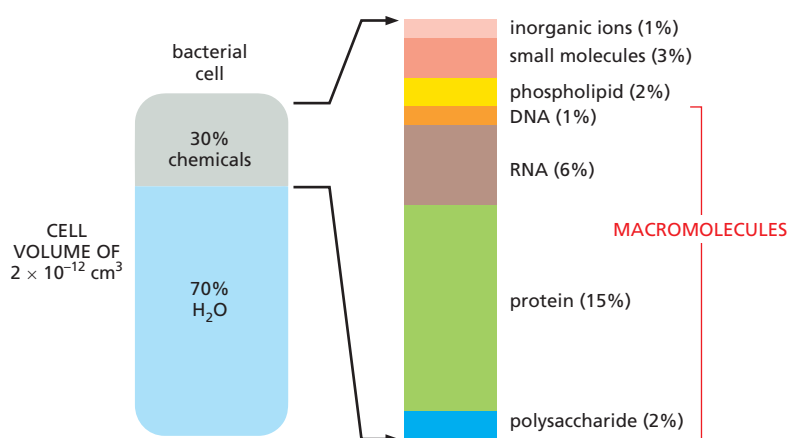


Figure 2-7 The distribution of molecules in cells. The approximate composition of a bacterial cell is shown by weight. The composition of an animal cell is similar, even though its volume is roughly 1000 times greater. Note that macromolecules dominate. The major inorganic ions include Na^+ , K^+ , Mg^{2+} , Ca^{2+} , and Cl^- .

by covalently linking small organic molecules (called *monomers*) into long chains (Figure 2-8). They have remarkable properties that could not have been predicted from their simple constituents.

Proteins are abundant and spectacularly versatile, performing thousands of distinct functions in cells. Many proteins serve as *enzymes*, the catalysts that facilitate the many covalent bond-making and bond-breaking reactions that the cell needs. Enzymes catalyze all of the reactions in which cells extract energy from food molecules, for example. Other proteins are used to build structural components, such as tubulin, a protein that self-assembles to make the cell's long microtubules, or histones, proteins that compact the DNA in chromosomes. Many proteins serve as signaling devices, producing networks that control cell functions. Yet other proteins act as molecular motors to produce force and movement, as for myosin in muscle. We shall describe the remarkable chemistry that underlies these diverse roles throughout this book.

Although the chemical reactions that add subunits to each polymer are different in detail for proteins, nucleic acids, and polysaccharides, they share important features. Each polymer grows by the addition of a monomer onto the end of a growing chain in a *condensation reaction*, in which one molecule of water is lost with each subunit added (Figure 2-9). The stepwise polymerization of monomers into a long chain is a simple way to manufacture a large, complex molecule, because the subunits are added by the same reaction performed over and over again by the same set of enzymes. Apart from some of the polysaccharides, most macromolecules are made from a limited set of monomers that are slightly different from one another; for example, the 20 different amino acids from which proteins are made. It is critical to life that the polymer chain is not assembled at random from these subunits; instead, the subunits are added in a precise order, or *sequence*. The elaborate mechanisms that allow enzymes to accomplish this task are described in detail in Chapters 5 and 6.

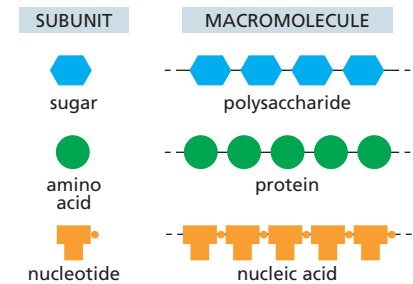


Figure 2-8 Three families of macromolecules. Each is a polymer formed from small molecules (called monomers) linked together by covalent bonds. There are two types of nucleic acid: RNA and DNA.

Noncovalent Bonds Specify Both the Precise Shape of a Macromolecule and Its Binding to Other Molecules

Most of the covalent bonds in a macromolecule allow rotation of the atoms they join, giving the polymer chain great flexibility. In principle, this allows a macromolecule to adopt an almost unlimited number of shapes, or *conformations*, as random thermal energy causes the polymer chain to writhe and rotate. However, the shapes of most biological macromolecules are highly constrained because of the many weak *noncovalent bonds* that form between different parts of the same molecule. If these noncovalent bonds are formed in sufficient numbers, the polymer chain can strongly prefer one particular conformation, determined by the linear sequence of monomers in its chain. Most protein molecules and many of the small RNA molecules found in cells fold tightly into a highly preferred conformation in this way (Figure 2-10).

The four types of noncovalent interactions important in biological molecules were presented earlier (see also Panel 2-3, pp. 98-99). In addition to folding biological macromolecules into unique shapes, they can also add up to create a strong attraction between two different molecules (see Figure 2-4). This form of

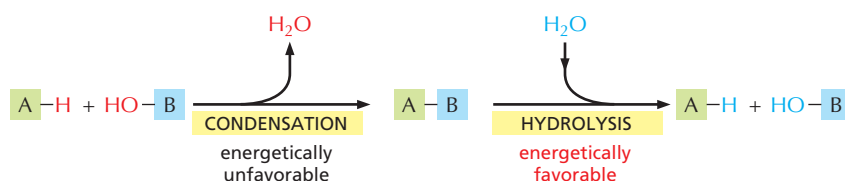


Figure 2-9 Condensation and hydrolysis as opposite reactions. The macromolecules of the cell are polymers that are formed from subunits (or monomers) by a condensation reaction, and they are broken down by hydrolysis. The condensation reactions are all energetically unfavorable; thus, polymer formation requires an energy input, as will be described in the text.

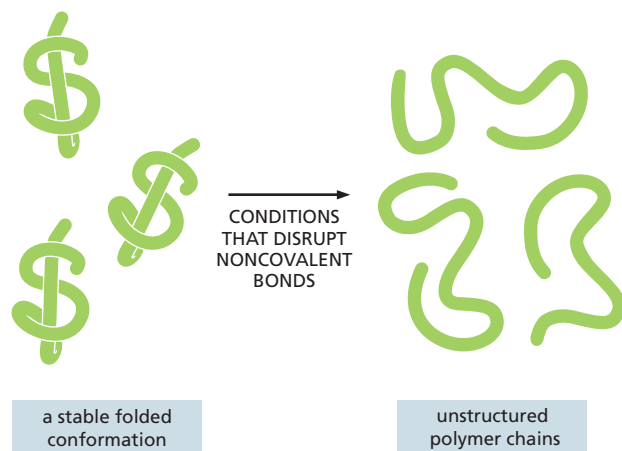


Figure 2–10 Proteins and RNA molecules are folded into a particularly stable three-dimensional shape, or **conformation**. If the noncovalent bonds maintaining the stable conformation are disrupted, the molecule becomes a flexible chain that loses its biological activity.

molecular interaction provides for great specificity, inasmuch as the close multi-point contacts required for strong binding make it possible for a macromolecule to select out—through binding—just one of the many thousands of types of molecules present inside a cell. Moreover, because the strength of the binding depends on the number of noncovalent bonds that are formed, interactions of almost any affinity are possible—allowing rapid dissociation where appropriate.

As we discuss next, binding of this type underlies all biological catalysis, making it possible for proteins to function as enzymes. In addition, noncovalent interactions allow macromolecules to be used to build larger structures, thereby forming intricate machines with multiple moving parts that perform such complex tasks as DNA replication and protein synthesis (**Figure 2–11**).

Summary

Living organisms are autonomous, self-propagating chemical systems. They are formed from a distinctive and restricted set of small carbon-based molecules that are essentially the same for every living species. Each of these small molecules is composed of a set of atoms linked to each other in a precise configuration through covalent bonds. The main categories are sugars, fatty acids, amino acids, and nucleotides.

Most of the dry mass of a cell consists of macromolecules that have been produced as linear polymers of amino acids (proteins) or nucleotides (DNA and RNA), covalently linked to each other in an exact order. Most of the protein molecules and many of the RNAs fold into a particular conformation that is determined

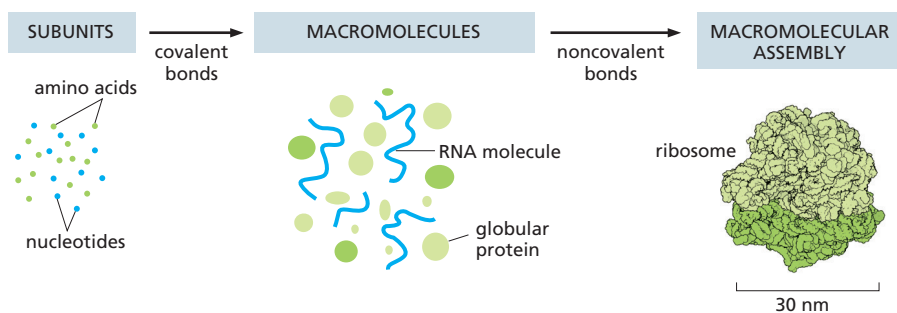
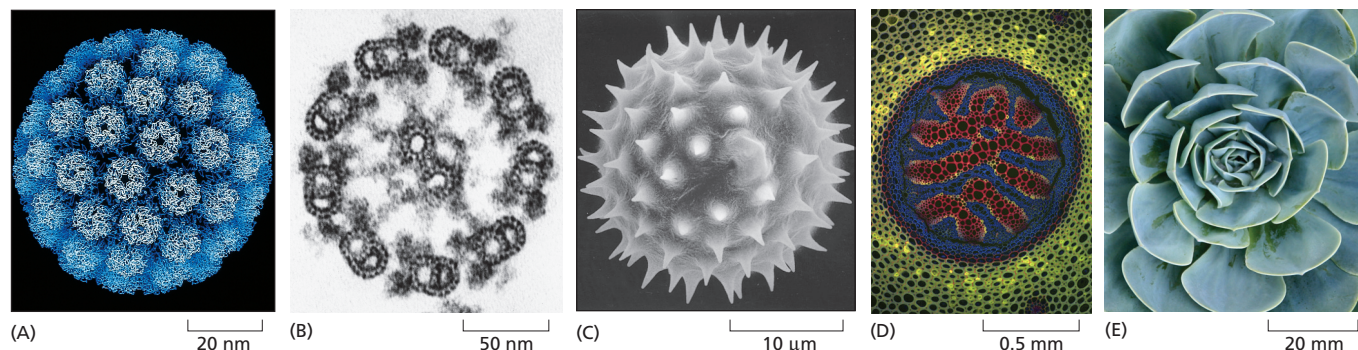


Figure 2–11 Small molecules are covalently linked to form macromolecules, which in turn can assemble through noncovalent interactions to form large complexes. Small molecules, proteins, and a ribosome are drawn approximately to scale. Ribosomes are a central part of the machinery that the cell uses to make proteins: each ribosome is formed as a complex of about 90 macromolecules (protein and RNA molecules).



by their sequence of subunits. This folding process creates unique surfaces, and it depends on a large set of weak attractions produced by noncovalent forces between atoms. These forces are of four types: electrostatic attractions, hydrogen bonds, van der Waals attractions, and an attraction between nonpolar groups caused by their hydrophobic expulsion from water. The same set of weak forces governs the specific binding of a macromolecule to both small molecules and other macromolecules, producing the myriad associations between biological molecules that generate the structure and the chemistry of a cell.

CATALYSIS AND THE USE OF ENERGY BY CELLS

One property of living things above all makes them seem almost miraculously different from nonliving matter: they create and maintain order in a universe that is tending always to greater disorder (Figure 2-12). To create this order, the cells in a living organism must perform a never-ending stream of chemical reactions. In some of these reactions, small organic molecules—amino acids, sugars, nucleotides, and lipids—are being taken apart or modified to supply the many other small molecules that the cell requires. In other reactions, small molecules are being used to construct an enormously diverse range of proteins, nucleic acids, and other macromolecules that endow living systems with all of their most distinctive properties. Each cell can be viewed as a tiny chemical factory, performing many millions of reactions every second.

Figure 2-12 Biological structures are highly ordered. Well-defined, ornate, and beautiful spatial patterns can be found at every level of organization in living organisms. In order of increasing size: (A) protein molecules in the coat of a virus (a parasite that, although not technically alive, contains the same types of molecules as those found in living cells); (B) the regular array of microtubules seen in a cross section of a sperm tail; (C) surface contours of a pollen grain (a single cell); (D) cross section of a fern stem, showing the patterned arrangement of cells; and (E) a spiral arrangement of leaves in a succulent plant. (A, courtesy of Robert Grant, Stéphane Crainic, and James M. Hogle; B, courtesy of Lewis Tilney; C, courtesy of Colin MacFarlane and Chris Jeffree; D, courtesy of Jim Haseloff; E, courtesy of Aron van de Selenib.)

Cell Metabolism Is Organized by Enzymes

The chemical reactions that a cell carries out would normally proceed at an appreciable rate only at much higher temperatures than those existing inside cells. For this reason, each reaction requires a specific boost in chemical reactivity. This requirement is crucial, because it allows the cell to control its chemistry. The control is exerted through specialized biological *catalysts*. These are almost always proteins called *enzymes*, although RNA catalysts also exist, called *ribozymes*. Each enzyme accelerates, or *catalyzes*, just one of the many possible kinds of reactions that a particular molecule might undergo. Enzyme-catalyzed reactions are connected in series, so that the product of one reaction becomes the starting material, or *substrate*, for the next (Figure 2-13). Long linear reaction pathways are in turn linked to one another, forming a maze of interconnected reactions that enable the cell to survive, grow, and reproduce.

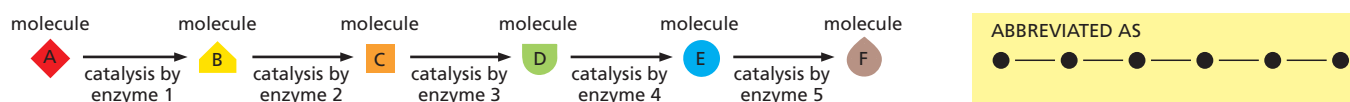


Figure 2-13 How a set of enzyme-catalyzed reactions generates a metabolic pathway. Each enzyme catalyzes a particular chemical reaction, leaving the enzyme unchanged. In this example, a set of enzymes acting in series converts molecule A to molecule F, forming a metabolic pathway. (For a diagram of many of the reactions in a human cell, abbreviated as shown, see Figure 2-62.)

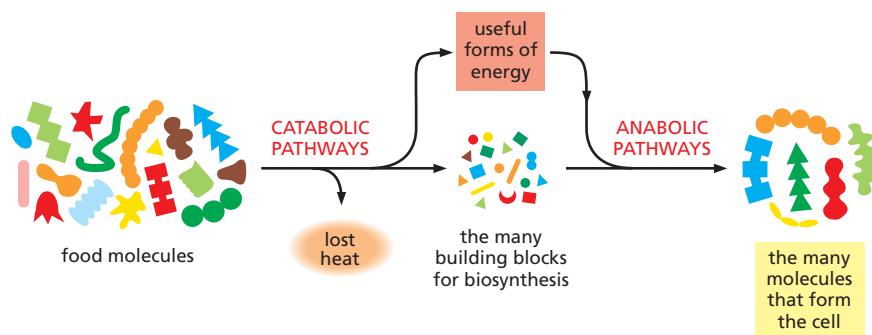


Figure 2-14 Schematic representation of the relationship between catabolic and anabolic pathways in metabolism. Catabolism produces both the building blocks and the energy required for biosynthesis. As indicated, a major portion of the energy stored in the chemical bonds of food molecules is dissipated as heat. As also suggested in this diagram, the mass of food required by any organism that derives all of its energy from catabolism is much greater than the mass of the molecules that it can produce by anabolism.

Two opposing streams of chemical reactions occur in cells: (1) the *catabolic* pathways break down foodstuffs into smaller molecules, thereby generating both a useful form of energy for the cell and some of the small molecules that the cell needs as building blocks; (2) the *anabolic*, or *biosynthetic*, pathways use the small molecules plus the energy harnessed by catabolism to drive the synthesis of the many other molecules that form the cell. Together these two sets of reactions constitute the **metabolism** of the cell (Figure 2-14).

The many details of cell metabolism form the traditional subject of *biochemistry*. Most of these details need not concern us here. But the general principles by which cells obtain energy from their environment and use it to create order are central to cell biology. We begin with a discussion of why a constant input of energy is needed to sustain all living things.

Biological Order Is Made Possible by the Release of Heat Energy from Cells

The universal tendency of things to become disordered is a fundamental law of physics—the *second law of thermodynamics*—which states that in the universe or in any isolated system (a collection of matter that is completely isolated from the rest of the universe), the degree of disorder always increases. This law has such profound implications for life that we will restate it in several ways.

For example, we can present the second law in terms of probability by stating that systems will change spontaneously toward those arrangements that have the greatest probability. If we consider a box of 100 coins all lying heads-up, a series of accidents that disturbs the box will tend to move the arrangement toward a mixture of 50 heads and 50 tails. The reason is simple: there is a huge number of possible arrangements of the individual coins in the mixture that can achieve the 50-50 result, but only one possible arrangement that keeps all of the coins oriented heads-up. Because the 50-50 mixture is therefore the most probable, we say that it is more “disordered.” For the same reason, it is a common experience that one’s living space will become increasingly disordered without intentional effort: the movement toward disorder is a *spontaneous process*, requiring a periodic effort to reverse it (Figure 2-15).

The amount of disorder in a system can be quantified and expressed as the **entropy** of the system: the greater the disorder, the greater the entropy. Thus, another way to express the second law of thermodynamics is to say that systems will change spontaneously toward arrangements with greater entropy.

Living cells—by surviving, growing, and forming complex organisms—are generating order and thus might appear to defy the second law of thermodynamics. How is this possible? The answer is that a cell is not an isolated system: it takes in energy from its environment in the form of the chemical bonds in food

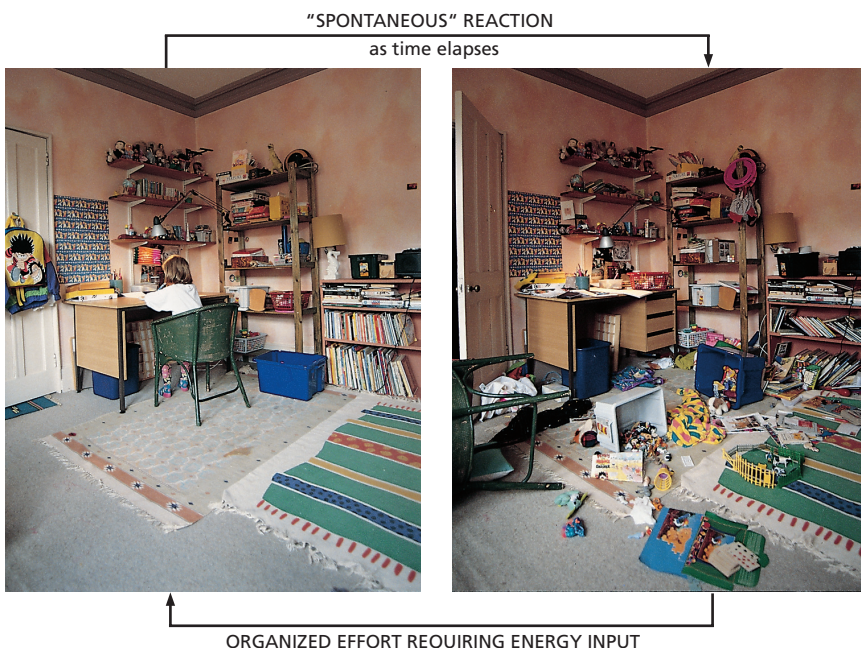


Figure 2-15 An everyday illustration of the spontaneous drive toward disorder.

Reversing this tendency toward disorder requires an intentional effort and an input of energy: it is not spontaneous. In fact, from the second law of thermodynamics, we can be certain that the human intervention required will release enough heat to the environment to more than compensate for the reordering of the items in this room.

or as photons from the Sun (or even, as in some chemosynthetic bacteria, from inorganic molecules alone). It then uses this energy to generate order within itself. Critically, during the chemical reactions that generate order, the cell converts part of the energy it uses into heat. The heat is discharged into the cell's environment and disorders the surroundings. As a result, the total entropy—that of the cell plus its surroundings—increases, as demanded by the second law of thermodynamics.

To understand the principles governing these energy conversions, think of a cell surrounded by a sea of matter representing the rest of the universe. As the cell lives and grows, it creates internal order. But it constantly releases heat energy as it synthesizes molecules and assembles them into cell structures. Heat is energy in its most disordered form—the random jostling of molecules. When the cell releases heat to the sea, it increases the intensity of molecular motions there (thermal motion)—thereby increasing the randomness, or disorder, of the sea. The second law of thermodynamics is satisfied because the increase in the amount of order inside the cell is always more than compensated for by an even greater decrease in order (increase in entropy) in the surrounding sea of matter (**Figure 2-16**).

Where does the heat that the cell releases come from? Here we encounter another important law of thermodynamics. The *first law of thermodynamics*

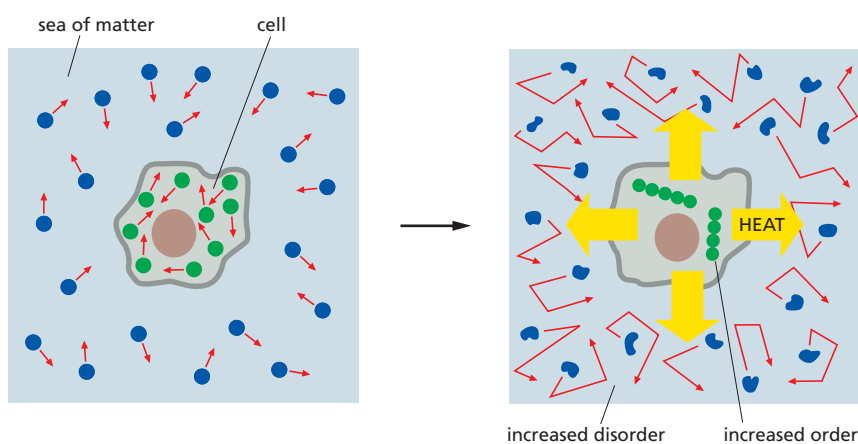


Figure 2-16 A simple thermodynamic analysis of a living cell.

In the diagram on the left, the molecules of both the cell and the rest of the universe (the sea of matter) are depicted in a relatively disordered state. In the diagram on the right, the cell has taken in energy from food molecules and released heat through reactions that order the molecules the cell contains. The heat released increases the disorder in the environment around the cell (depicted by jagged arrows and distorted molecules, indicating increased molecular motions caused by heat). As a result, if enough heat is released, the second law of thermodynamics—which states that the amount of disorder in the universe must always increase—is satisfied as the cell grows and divides. For a detailed discussion, see Panel 2-7 (pp. 106–107).

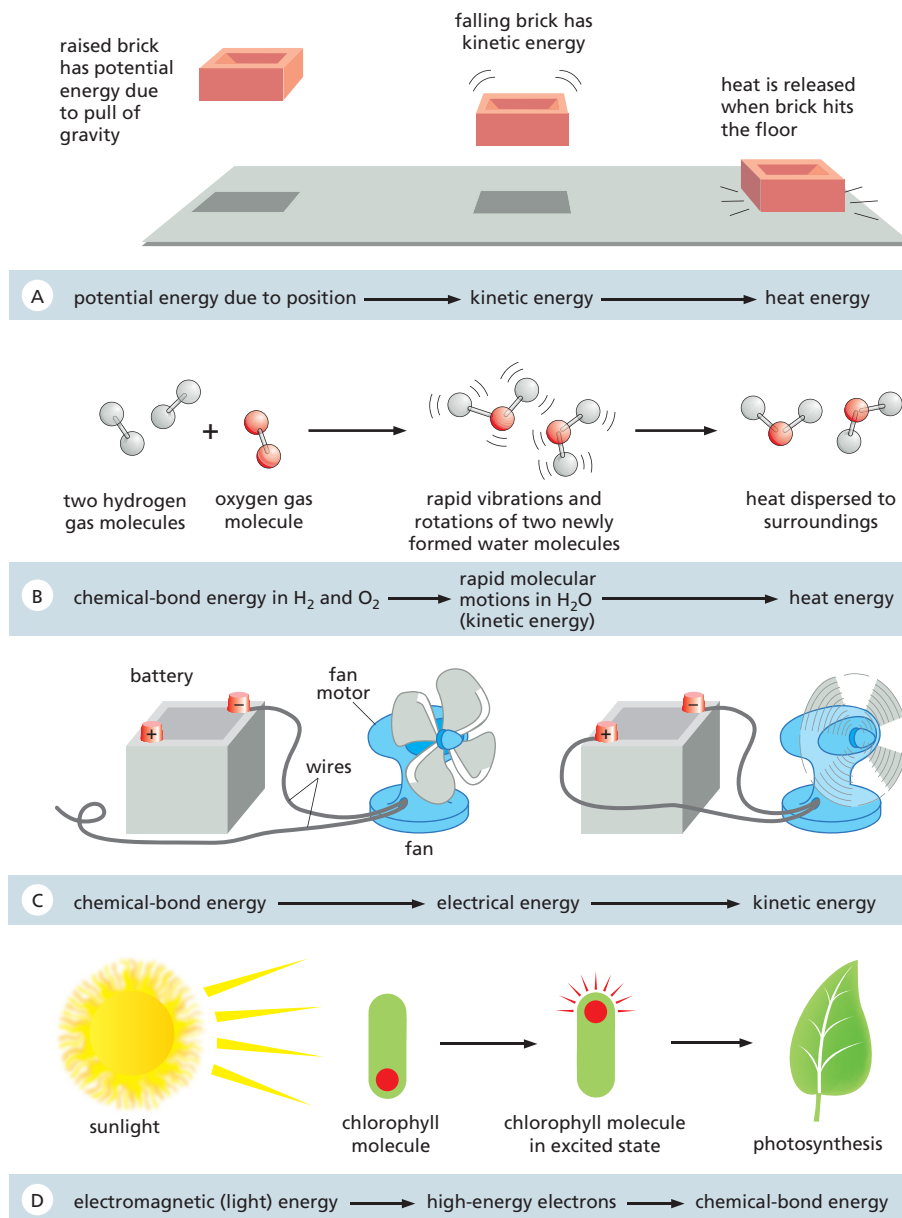
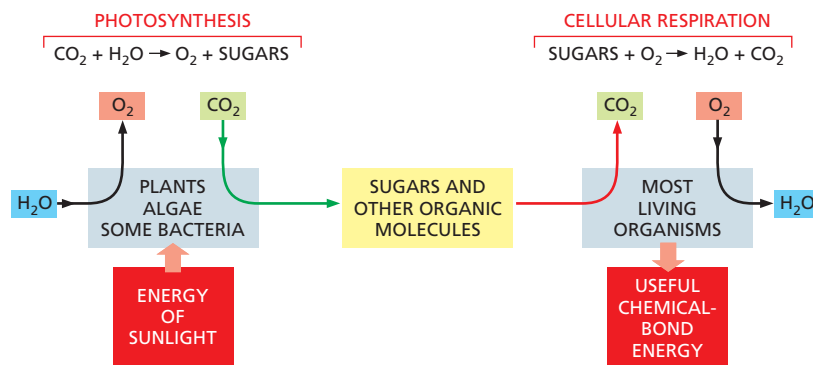


Figure 2-17 Some interconversions between different forms of energy.

(A) We can use the height and weight of the brick to predict exactly how much heat will be released when it hits the floor. (B) The large amount of chemical-bond energy released when water (H_2O) is initially converted to very rapid thermal motions in the two new H_2O molecules; however, collisions with other H_2O molecules almost instantaneously spread this kinetic energy evenly throughout the surroundings (heat transfer), making the new H_2O molecules indistinguishable from all the rest. (C) Through coupled processes to be described later, cells can convert chemical-bond energy into kinetic energy to drive, for example, molecular motor proteins; however, this occurs without the intermediate conversion of chemical energy to electrical energy that a man-made appliance such as this fan requires. (D) Some cells can also harvest the energy from sunlight to form chemical bonds via photosynthesis.

states that energy can be converted from one form to another, but that it cannot be created or destroyed. **Figure 2-17** illustrates some interconversions between different forms of energy. The amount of energy in different forms will change as a result of the chemical reactions inside the cell, but the first law tells us that the total amount of energy must always be the same. For example, an animal cell takes in foodstuffs and converts some of the energy present in the chemical bonds between the atoms of these food molecules (chemical-bond energy) into the random thermal motion of molecules (heat energy)—it is this heat that keeps our bodies warm.

The cell cannot derive any benefit from the heat energy it releases unless the heat-generating reactions inside the cell are directly linked to the processes that generate molecular order. It is the tight *coupling* of heat production to an increase in order that distinguishes the metabolism of a cell from the wasteful burning of fuel in a fire. Later, we illustrate how this coupling occurs. For now, it is sufficient to recognize this critical fact: a direct linkage of the “controlled burning” of food molecules to the generation of biological order



is required for cells to create and maintain an island of order in a universe tending toward chaos.

Cells Obtain Energy by the Oxidation of Organic Molecules

All animal and plant cells are powered by energy stored in the chemical bonds of organic molecules, whether they are sugars that a plant has photosynthesized as food for itself or the mixture of large and small molecules that an animal has eaten. Organisms must extract this energy in usable form to live, grow, and reproduce. In both plants and animals, energy is extracted from food molecules by a process of gradual oxidation, or controlled burning.

Earth's atmosphere contains a great deal of oxygen, and in the presence of oxygen the most energetically stable form of carbon is CO₂ and that of hydrogen is H₂O. A cell is therefore able to obtain energy from sugars or other organic molecules by allowing their carbon and hydrogen atoms to combine with oxygen to produce CO₂ and H₂O, respectively—a process called **aerobic respiration**.

Photosynthesis (discussed in detail in Chapter 14) and respiration are complementary processes (Figure 2–18). This means that the transactions between plants and animals are not all one way. Plants, animals, and microorganisms have existed together on this planet for so long that many of them have become an essential part of the others' environments. The oxygen released by photosynthesis is consumed in the combustion of organic molecules during aerobic respiration. And some of the CO₂ molecules that are fixed today into organic molecules by photosynthesis in a green leaf were yesterday released into the atmosphere by the respiration of an animal—or by the respiration of a fungus or bacterium decomposing dead organic matter. We therefore see that carbon utilization forms a huge cycle that involves the *biosphere* (all of the living organisms on Earth) as a whole (Figure 2–19). Similarly, atoms of nitrogen, phosphorus, and sulfur move between the living and nonliving worlds in cycles that involve plants, algae, animals, fungi, and bacteria.

Figure 2–18 Photosynthesis and respiration as complementary processes in the living world.

Photosynthesis converts the electromagnetic energy in sunlight into chemical-bond energy in sugars and other organic molecules. Plants, algae, and cyanobacteria obtain the carbon atoms that they need for this purpose from atmospheric CO₂ and the hydrogen from water, producing sugars and releasing O₂ gas as a by-product. The organic molecules produced by photosynthesis in turn serve as food for other organisms. Many of these organisms carry out aerobic respiration, a process that uses O₂ to form CO₂ from the same carbon atoms that had been taken up as CO₂ and converted into sugars by photosynthesis. In the process, the organisms that respire obtain the chemical-bond energy that they need to survive.

The first cells on Earth are thought to have been capable of neither photosynthesis nor respiration (discussed in Chapter 14). However, photosynthesis must have preceded respiration on Earth, because there is strong evidence that billions of years of photosynthesis were required before O₂ had been released in sufficient quantity to create an atmosphere rich in this gas. (Earth's atmosphere currently contains 21% O₂.)

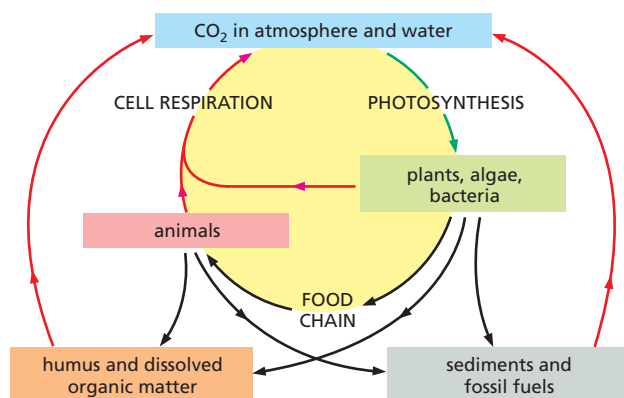


Figure 2–19 How carbon atoms cycle through the biosphere.

Individual carbon atoms are incorporated into organic molecules of the living world by the photosynthetic activity of bacteria, algae, and plants. They pass to animals, microorganisms, and organic material in soil and oceans in cyclic paths. CO₂ is restored to the atmosphere when organic molecules are oxidized by cells during respiration or burned by humans as fossil fuels. In this diagram, the *green arrow* denotes an uptake of CO₂, whereas a *red arrow* indicates CO₂ release.

As indicated in Chapter 1, the total biomass on Earth is estimated to contain ~550 gigatons (10¹⁵ grams) of carbon (Gt C), of which 450 Gt C are plants, 70 are bacteria, 7 are archaea, and 2 are animals (see Figure 1–14).

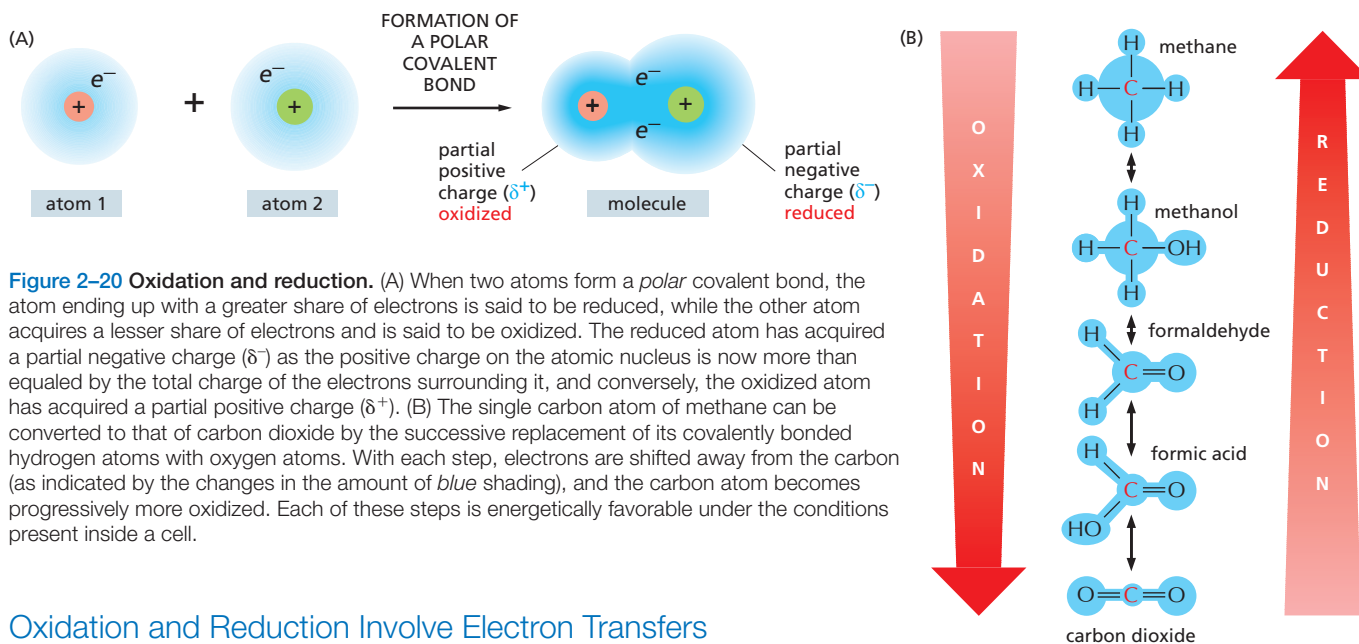


Figure 2-20 Oxidation and reduction. (A) When two atoms form a *polar* covalent bond, the atom ending up with a greater share of electrons is said to be reduced, while the other atom acquires a lesser share of electrons and is said to be oxidized. The reduced atom has acquired a partial negative charge (δ^-) as the positive charge on the atomic nucleus is now more than equaled by the total charge of the electrons surrounding it, and conversely, the oxidized atom has acquired a partial positive charge (δ^+). (B) The single carbon atom of methane can be converted to that of carbon dioxide by the successive replacement of its covalently bonded hydrogen atoms with oxygen atoms. With each step, electrons are shifted away from the carbon (as indicated by the changes in the amount of *blue* shading), and the carbon atom becomes progressively more oxidized. Each of these steps is energetically favorable under the conditions present inside a cell.

Oxidation and Reduction Involve Electron Transfers

The cell does not oxidize organic molecules in one step, as occurs when organic material is burned in a fire. Through the use of enzyme catalysts, metabolism takes these molecules through a large number of reactions that only rarely involve the direct addition of oxygen. Before we consider some of these reactions and their purpose, we discuss what is meant by the process of oxidation.

Oxidation refers to more than the addition of oxygen atoms; the term applies more generally to any reaction in which electrons are transferred from one atom to another. Oxidation in this sense refers to the removal of electrons, and **reduction**—the converse of oxidation—means the addition of electrons. Thus, Fe^{2+} is oxidized if it loses an electron to become Fe^{3+} , and a chlorine atom is reduced if it gains an electron to become Cl^- . Because the number of electrons is conserved (no loss or gain) in a chemical reaction, oxidation and reduction always occur simultaneously; that is, if one molecule gains an electron in a reaction (reduction), a second molecule loses the electron (oxidation). When a sugar molecule is oxidized to CO_2 and H_2O , for example, the O_2 molecules involved in forming H_2O gain electrons and thus are said to have been reduced.

Why is a “gain” of electrons referred to as a “reduction”? The term arose before anything was known about the movement of electrons. Originally, reduction reactions involved a liberation of oxygen—for example, when metals are extracted from ores by heating—which caused the samples to become lighter; in other words, “reduced” in mass.

It is important to recognize that the terms “oxidation” and “reduction” apply even when there is only a partial shift of electrons between atoms linked by a covalent bond (Figure 2-20). When a carbon atom becomes covalently bonded to an atom with a strong affinity for electrons, such as oxygen, chlorine, or sulfur, for example, it gives up more than its equal share of electrons and forms a *polar* covalent bond. Because the positive charge of the carbon nucleus is now somewhat greater than the negative charge of its electrons, the atom acquires a partial positive charge and is said to be oxidized. Conversely, a carbon atom in a C-H linkage has slightly more than its share of electrons, and so it is said to be reduced.

When a molecule in a cell picks up an electron (e^-), it often picks up a proton (H^+) at the same time (protons being freely available in water). The net effect in this case is to add a hydrogen atom to the molecule.



Even though a proton plus an electron is involved (instead of just an electron), such *hydrogenation* reactions are reductions, and the reverse *dehydrogenation*

reactions are oxidations. It is especially easy to tell whether an organic molecule is being oxidized or reduced: reduction is occurring if its number of C–H bonds increases, whereas oxidation is occurring if its number of C–H bonds decreases (see Figure 2–20B).

Cells use enzymes to catalyze the oxidation of organic molecules in small steps, through a sequence of reactions that allows useful energy to be harvested. We now need to explain how enzymes work and some of the constraints under which they operate.

Enzymes Lower the Activation-Energy Barriers That Block Chemical Reactions

Consider the reaction



Once ignited, the paper burns readily, releasing to the atmosphere both energy as heat and water and carbon dioxide as gases. The reaction is irreversible, as the smoke and ashes never spontaneously retrieve these entities from the heated atmosphere and reconstitute themselves into paper. When the paper burns, its chemical energy is dissipated as heat—not lost from the universe, as energy can never be created or destroyed, but irretrievably dispersed in the chaotic random thermal motions of molecules. At the same time, the atoms and molecules of the paper become dispersed and disordered. In the language of thermodynamics, there has been a loss of *free energy*; that is, of energy that can be harnessed to do work or drive chemical reactions. This loss reflects a reduction of orderliness in the way the energy and molecules were stored in the paper.

We shall discuss free energy in more detail shortly, but the general principle is clear enough intuitively: chemical reactions proceed spontaneously only in the direction that leads to a loss of free energy. In other words, the spontaneous direction for any reaction is the direction that goes “downhill,” where a “downhill” reaction is one that is *energetically favorable*.

Although the most energetically favorable form of carbon under ordinary conditions is CO_2 , and that of hydrogen is H_2O , a living organism does not disappear in a puff of smoke, and the paper book in your hands does not burst into flames. This is because the molecules both in the living organism and in the book are in a relatively stable state, and they cannot be changed to a state of lower energy without an input of energy; in other words, a molecule requires **activation energy**—a kick over an energy barrier—before it can undergo a chemical reaction that leaves it in a more stable state (Figure 2–21). In the case of a burning book, the activation energy can be provided by the heat of a lighted match. For the molecules in the watery solution inside a cell, the kick is delivered by an unusually energetic random collision with surrounding molecules.

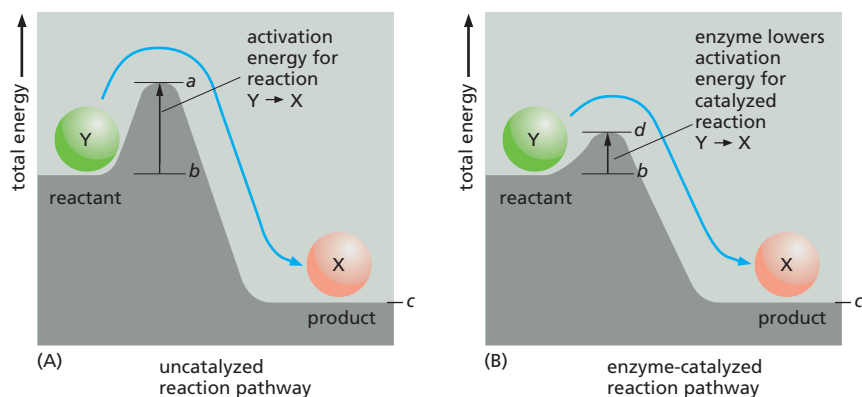


Figure 2–21 The important principle of activation energy. (A) Compound Y (a reactant) is in a relatively stable state, and energy is required to convert it to compound X (a product), even though X is at a lower overall energy level than Y. This conversion will not take place, therefore, unless compound Y can acquire enough activation energy (*energy a minus energy b*) from its surroundings to undergo the reaction that converts it into compound X. This energy may be provided by means of an unusually energetic collision with other molecules. For the reverse reaction, $X \rightarrow Y$, the activation energy will be much larger (*energy a minus energy c*); this reaction will therefore occur much more rarely. Activation energies are always positive; note, however, that the total energy change for the energetically favorable reaction $Y \rightarrow X$ is *energy c minus energy b*, a negative number. (B) Energy barriers for specific reactions can be lowered by catalysts, as indicated by the line marked *d*. Enzymes are particularly effective catalysts because they greatly reduce the activation energy for the reactions they perform.

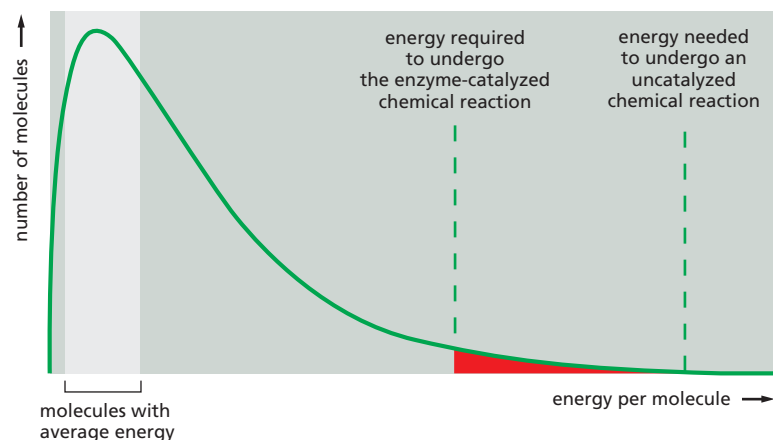


Figure 2-22 Lowering the activation energy greatly increases the probability of a reaction. At any given instant, a population of identical substrate molecules will have a range of energies, distributed as shown on the graph. The varying energies come from collisions with surrounding molecules, which make the substrate molecules jiggle, vibrate, and spin. For a molecule to undergo a chemical reaction, the energy of the molecule must exceed the activation-energy barrier for that reaction (*dashed lines*). For most biological reactions, this almost never happens without enzyme catalysis. Even with enzyme catalysis, the substrate molecules must experience a particularly energetic collision to react (*red shaded area*). Raising the temperature will also increase the number of molecules with sufficient energy to overcome the activation energy needed for a reaction; but in marked contrast to enzyme catalysis, this effect is nonselective, speeding up all reactions (**Movie 2.2**).

The chemistry in a living cell is tightly controlled, because the kick over energy barriers is greatly aided by a specialized class of proteins—the **enzymes**. Each enzyme binds tightly to one or more molecules, called **substrates**, and holds them in a way that greatly reduces the activation energy of a particular chemical reaction that the bound substrates can undergo. A substance that can lower the activation energy of a reaction is termed a **catalyst**; catalysts increase the rate of chemical reactions because they allow a much larger proportion of the random collisions with surrounding molecules to kick the substrates over the energy barrier, as illustrated in **Figure 2-22**. Enzymes are among the most effective catalysts known: some are capable of speeding up reactions by factors of 10^{14} or more. Enzymes thereby allow reactions that would not otherwise occur to proceed rapidly at normal temperatures.

Enzymes Can Drive Substrate Molecules Along Specific Reaction Pathways

An enzyme cannot change the equilibrium point for a reaction. The reason is simple: when an enzyme (or any catalyst) lowers the activation energy for the reaction $Y \rightarrow X$, of necessity it also lowers the activation energy for the reverse reaction $X \rightarrow Y$ by exactly the same amount (see **Figure 2-21**). The forward and backward reactions will therefore be accelerated by the same factor by an enzyme, and the equilibrium point for the reaction will be unchanged (**Figure 2-23**).

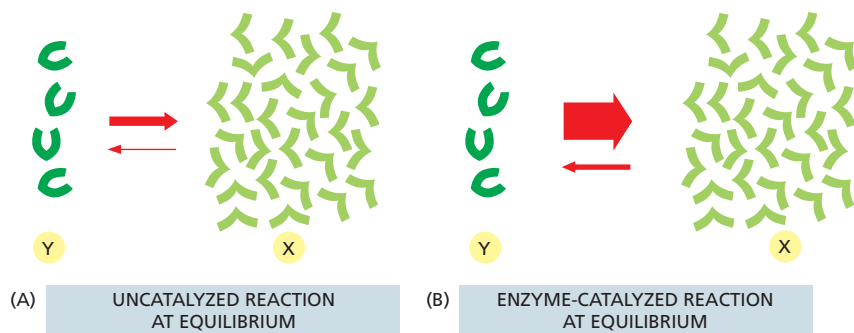


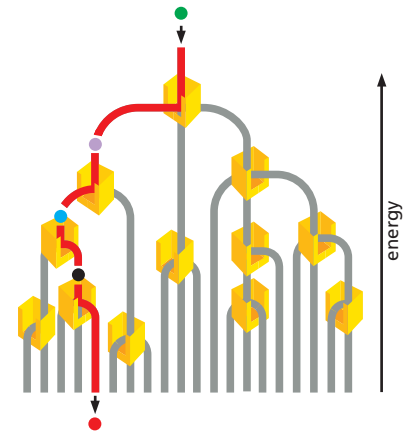
Figure 2-23 Enzymes cannot change the equilibrium point for reactions. Enzymes, like all catalysts, speed up the forward and backward rates of a reaction by the same factor. Therefore, for both the catalyzed and the uncatalyzed reactions shown here, the number of molecules undergoing the transition $X \rightarrow Y$ is equal to the number of molecules undergoing the transition $Y \rightarrow X$ when the ratio of X molecules to Y molecules is 7 to 1. In other words, the two reactions will eventually reach exactly the same equilibrium point, although the catalyzed reaction will reach equilibrium much faster.

Figure 2-24 Directing substrate molecules through a specific reaction pathway by enzyme catalysis. A substrate molecule in a cell (green ball) is converted into a different molecule (red ball) by means of a series of enzyme-catalyzed reactions. As indicated (yellow box), several reactions are energetically favorable at each step, but only one is catalyzed by each enzyme. Sets of enzymes thereby determine the exact reaction pathway that is followed by each molecule inside the cell.

Thus no matter how much an enzyme speeds up a reaction, it cannot change its direction.

Despite the above limitation, enzymes steer all of the reactions in cells through specific reaction paths. This is because enzymes are both highly selective and very precise, usually catalyzing only one particular reaction. In other words, each enzyme selectively lowers the activation energy of only one of the several possible chemical reactions that its bound substrate molecules could undergo. In this way, sets of enzymes can direct each of the many different molecules in a cell along a particular reaction pathway (Figure 2-24).

The success of living organisms is attributable to a cell's ability to make enzymes of many types, each with precisely specified properties. Each enzyme has a unique shape containing an *active site*, a pocket or groove in the enzyme into which only particular substrates will fit (Figure 2-25). Like all other catalysts, enzyme molecules themselves remain unchanged after participating in a reaction and therefore can function over and over again. In Chapter 3, we discuss further how enzymes work.



How Enzymes Find Their Substrates: The Enormous Rapidity of Molecular Motions

An enzyme will often catalyze the reaction of thousands of substrate molecules every second. This means the enzyme must be able to bind a new substrate molecule in a fraction of a millisecond. But both enzymes and their substrates are present in relatively small numbers in a cell. How do they find each other so fast? Rapid binding is possible because the motions caused by heat energy are enormously fast at the molecular level. These molecular motions can be classified broadly into three kinds: (1) the movement of a molecule from one place to another (*translational motion*), (2) the rapid back-and-forth movement of covalently linked atoms with respect to one another (*vibrations*), and (3) *rotations*. All three of these motions help to bring the surfaces of interacting molecules together.

The rates of molecular motions can be measured by a variety of spectroscopic techniques. A large globular protein is constantly tumbling, rotating about its axis approximately a million times per second. Molecules are also in constant translational motion, which causes them to explore the space inside the cell very efficiently by wandering through it—a process called **diffusion**. In this way, every molecule in a cell collides with a huge number of other molecules each second. As the molecules in a liquid collide and bounce off one another, an individual molecule moves first one way and then another, its path constituting a *random*

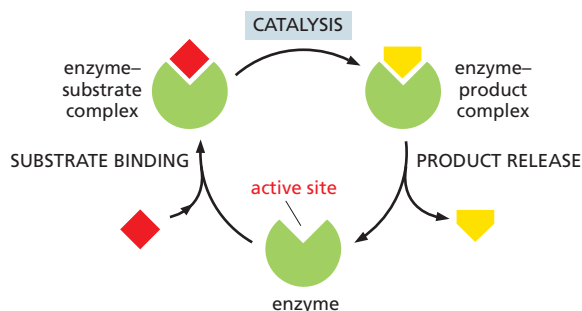


Figure 2-25 How enzymes work. Each enzyme has an active site to which one or more *substrate* molecules bind, forming an enzyme–substrate complex. A reaction occurs at the active site, producing an enzyme–product complex. The *product* is then released, allowing the enzyme to bind further substrate molecules.

walk (Figure 2-26). In such a walk, the average net distance that each molecule travels (as the “crow flies”) from its starting point is proportional to the square root of the time involved; that is, if it takes a molecule 1 second on average to travel 1 μm , it takes 4 seconds to travel 2 μm , 100 seconds to travel 10 μm , and so on.

The inside of a cell is very crowded (Figure 2-27). Nevertheless, experiments in which fluorescent dyes and other labeled molecules are injected into cells show that small organic molecules diffuse through the watery gel of the cytosol nearly as rapidly as they do through water. A small organic molecule, for example, takes only about one-fifth of a second on average to diffuse a distance of 10 μm . Diffusion is therefore an efficient way for small molecules to move the limited distances in the cell (a typical animal cell is 15 μm in diameter).

Proteins also move rapidly in cells. But because enzymes move more slowly than substrates, we can think of them as sitting still. The rate of encounter of each enzyme molecule with its substrate will depend on the concentration of the substrate molecule. For example, some abundant substrates are present at a concentration of 0.5 mM. As pure water is 55.5 M, there is only about one such substrate molecule in the cell for every 10^5 water molecules. Nevertheless, the active site on an enzyme molecule that binds this substrate will be bombarded by about 500,000 random collisions with the substrate molecule per second. (For a substrate concentration tenfold lower, the number of collisions drops to 50,000 per second, and so on.) A random collision between the active site of an enzyme and the matching surface of its substrate molecule often leads immediately to the formation of an enzyme–substrate complex. A reaction in which a covalent bond is broken or formed can then occur extremely rapidly. When one appreciates how quickly molecules move and react, the observed rates of enzymatic catalysis do not seem so amazing.

Two molecules that are held together by noncovalent bonds can also dissociate. The multiple weak noncovalent bonds that they form with each other will persist until random thermal motion causes the two molecules to separate. In general, the stronger the binding of the enzyme and substrate, the slower their rate of dissociation. In contrast, whenever two colliding molecules have poorly matching surfaces, they form few noncovalent bonds and the total energy of association will be negligible compared with that of thermal motion. In this case, the two molecules dissociate as rapidly as they come together, preventing incorrect and unwanted associations between mismatched molecules, such as between an enzyme and the wrong substrate.

The Free-Energy Change for a Reaction, ΔG , Determines Whether It Can Occur Spontaneously

Although enzymes speed up reactions, they cannot by themselves force energetically unfavorable reactions to occur. In terms of a water analogy, enzymes by themselves cannot make water run uphill. Cells, however, must do just that in order to grow and divide: they must build highly ordered and energy-rich molecules from small and simple ones. We shall see that this is done through enzymes that directly *couple* energetically favorable reactions, which release energy and produce heat, to energetically unfavorable reactions, which produce biological order.

What do cell biologists mean by the term “energetically favorable,” and how can this be quantified? According to the second law of thermodynamics, the universe tends toward maximum disorder (largest *entropy* or greatest probability). Thus, a chemical reaction can proceed spontaneously only if it results in a net increase in the disorder of the universe (see Figure 2-16). This disorder of the universe can be expressed most conveniently in terms of the *free energy* of a system, a concept we touched on earlier.

Free energy, G , is an expression of the *energy available to do work*; for example, the work of driving chemical reactions. The value of G is of interest only when a system undergoes a *change*. The **free-energy change**, denoted ΔG (delta G), is critical because, as explained in **Panel 2-7** (pp. 106–107), it is a direct measure of the

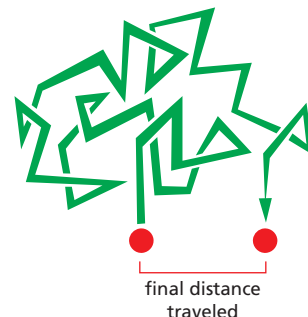


Figure 2-26 A random walk. Molecules in solution move in a random fashion as a result of the continual buffeting they receive in collisions with other molecules. This movement allows small molecules to diffuse rapidly from one part of the cell to another, as described in the text (**Movie 2.3**).

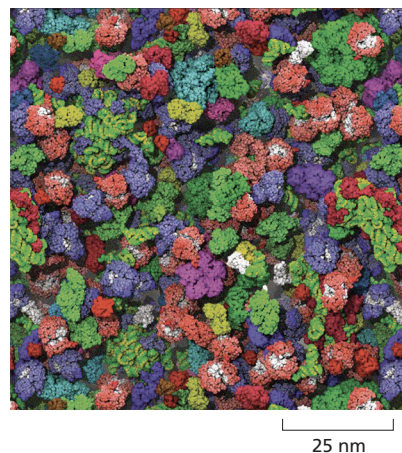


Figure 2-27 The crowded structure of the cell interior. Only the macromolecules, which are drawn to scale and displayed in different colors, are shown. Enzymes and other macromolecules diffuse relatively slowly inside the cell, in part because they interact with many other macromolecules; small molecules, by contrast, diffuse nearly as rapidly as they do in water (**Movie 2.4**). (From S.R. McGuffee and A.H. Elcock, *PLoS Comput. Biol.* 6(3):e1000694, 2010.)

amount of disorder created in the universe when a reaction takes place. *Energetically favorable reactions*, by definition, are those that decrease free energy; in other words, they have a *negative* ΔG and disorder the universe (Figure 2-28).

An example of an energetically favorable reaction on a macroscopic scale is the “reaction” by which a compressed spring relaxes to an expanded state, releasing its stored elastic energy as heat to its surroundings; an example on a microscopic scale is salt dissolving in water. Conversely, *energetically unfavorable reactions* with a *positive* ΔG —such as the joining of two amino acids to form a peptide bond—by themselves create order in the universe. Therefore, these reactions can take place only if they are coupled to a second reaction with a *negative* ΔG so large that the ΔG of the overall process is negative (Figure 2-29).

The Concentration of Reactants Influences the Free-Energy Change and a Reaction's Direction

As we have just described, a reaction $Y \leftrightarrow X$ will go in the direction $Y \rightarrow X$ when the associated free-energy change, ΔG , is negative, just as a tensed spring left to itself will relax and lose its stored energy to its surroundings as heat. For a chemical reaction, however, ΔG depends not only on the energy stored in each individual molecule but also on the concentrations of the molecules in the reaction mixture. Remember that ΔG reflects the degree to which a reaction creates a more disordered—in other words, a more probable—state of the universe. Recalling our coin analogy, it is very likely that a coin will flip from a head to a tail orientation if a jiggling box contains 90 heads and 10 tails, but this is a less probable event if the box has 10 heads and 90 tails.

The same is true for a chemical reaction. For a reversible reaction $Y \leftrightarrow X$, a large excess of Y over X will tend to drive the reaction in the direction $Y \rightarrow X$. Therefore, as the ratio of Y to X increases, the ΔG becomes more negative for the transition $Y \rightarrow X$ (and more positive for the transition $X \rightarrow Y$).

The amount of concentration difference that is needed to compensate for a given decrease in chemical-bond energy (and accompanying heat release) is not intuitively obvious. In the late nineteenth century, the relationship was determined through a thermodynamic analysis that makes it possible to separate the concentration-dependent and the concentration-independent parts of the free-energy change, as we describe next.

The Standard Free-Energy Change, ΔG° , Makes It Possible to Compare the Energetics of Different Reactions

Because ΔG depends on the concentrations of the molecules in the reaction mixture at any given time, it is not a particularly useful value for comparing the relative energies of different types of reactions. To consider reactions on a comparable basis, we need to turn to the **standard free-energy change** of a reaction, ΔG° . The ΔG° is the change in free energy under a standard condition, defined as that where the concentrations of all the reactants are set to the same fixed value of 1 mole/liter. Defined in this way, ΔG° depends only on the intrinsic characters of the reacting molecules.

For the simple reaction $Y \rightarrow X$ at 37°C, ΔG° is related to ΔG as follows:

$$\Delta G = \Delta G^\circ + RT \ln \frac{[X]}{[Y]}$$

where ΔG is in kilojoules per mole, $[Y]$ and $[X]$ denote the concentrations of Y and X in moles/liter, $\ln$ is the *natural logarithm*, and RT is the product of the gas constant, R , and the absolute temperature, T . At 37°C, $RT = 2.58 \text{ kJ mole}^{-1}$. (A mole is 6×10^{23} molecules of a substance.)

A large body of thermodynamic data has been collected that has made it possible to determine the standard free-energy change, ΔG° , for the important metabolic reactions of a cell. Given these ΔG° values, combined with additional information about metabolite concentrations and reaction pathways, it is possible to quantitatively predict the course of most biological reactions.

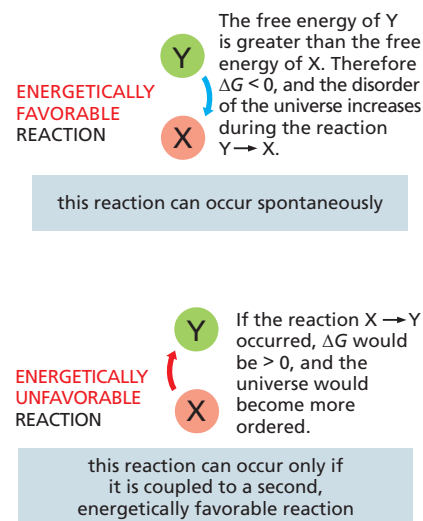


Figure 2-28 The distinction between energetically favorable and energetically unfavorable reactions, and how they relate to ΔG .

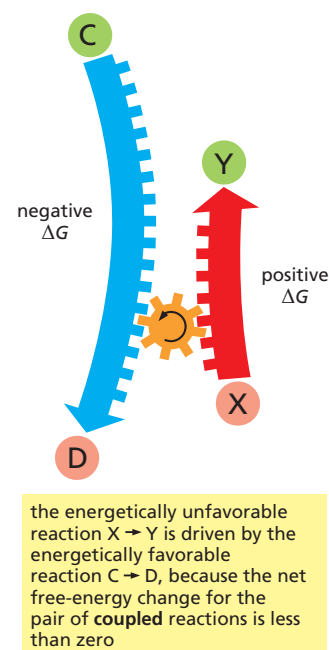
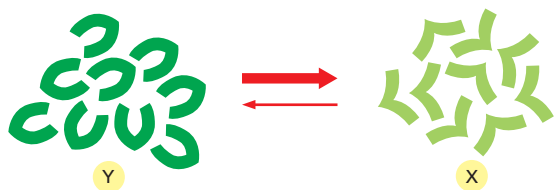


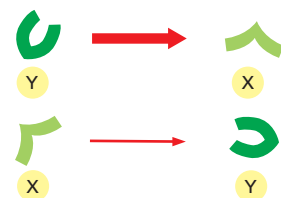
Figure 2-29 How reaction coupling is used to drive energetically unfavorable reactions.

FOR THE ENERGETICALLY FAVORABLE REACTION $Y \rightarrow X$,



if X and Y are at equal concentrations, $[Y] = [X]$, the formation of X is energetically favored. In other words, the ΔG of $Y \rightarrow X$ is negative and the ΔG of $X \rightarrow Y$ is positive. Nevertheless because of thermal bombardments, there will always be some X converting to Y.

FOR EACH INDIVIDUAL MOLECULE,

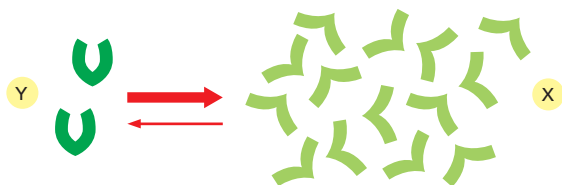


conversion of Y to X will occur often.

Conversion of X to Y will occur less often than the transition $Y \rightarrow X$, because it requires a more energetic collision.

Therefore, if one starts with an equal mixture, the ratio of X to Y molecules will increase

EVENTUALLY, there will be a large enough excess of X over Y to just compensate for the slow rate of $X \rightarrow Y$, such that the number of X molecules being converted to Y molecules each second is exactly equal to the number of Y molecules being converted to X molecules each second. At this point, the reaction will be at equilibrium.



AT EQUILIBRIUM, there is no net change in the ratio of Y to X, and the ΔG for both forward and backward reactions is zero.

Figure 2–30 Chemical equilibrium. When a reaction reaches equilibrium, the forward and backward fluxes of reacting molecules are equal and opposite. In these diagrams, the widths of the *red* arrows indicate the relative rates at which an *individual* molecule reacts.

The Equilibrium Constant and ΔG° Are Readily Derived from Each Other

Inspection of the above equation reveals that the ΔG equals the value of ΔG° when the concentrations of Y and X are equal. But as any favorable reaction proceeds, the concentrations of the products will increase as the concentration of the substrates decreases. This change in relative concentrations will cause $[X]/[Y]$ to become increasingly large, making the initially favorable ΔG less and less negative (the logarithm of a number x is positive for $x > 1$, negative for $x < 1$, and zero for $x = 1$). Eventually, when $\Delta G = 0$, a chemical **equilibrium** will be attained; here there is no net change in free energy to drive the reaction in either direction, inasmuch as the concentration effect just balances the push given to the reaction by ΔG° . As a result, the ratio of product to substrate reaches a constant value at chemical equilibrium (**Figure 2–30**).

We can define the **equilibrium constant**, K , for the reaction $Y \rightarrow X$ as

$$K = \frac{[X]}{[Y]}$$

where $[X]$ is the concentration of the product and $[Y]$ is the concentration of the reactant at equilibrium. Remembering that $\Delta G = \Delta G^\circ + RT \ln [X]/[Y]$, and that $\Delta G = 0$ at equilibrium, we see that

$$\Delta G^\circ = -RT \ln \frac{[X]}{[Y]} = -RT \ln K$$

At 37°C, where $RT = 2.58$, the equilibrium equation is therefore:

$$\Delta G^\circ = -2.58 \ln K$$

Converting this equation from the natural logarithm (ln) to the more commonly used base 10 logarithm (log), we get

$$\Delta G^\circ = -5.94 \log K$$

The above equation reveals how the equilibrium ratio of X to Y (expressed as the equilibrium constant, K) depends on the intrinsic character of the molecules (as expressed in the value of ΔG° in kilojoules per mole). The more energetically favorable a reaction, the more product will accumulate when the reaction proceeds to equilibrium. More precisely, for every 5.94 kJ/mole difference in free energy at 37°C, the equilibrium constant changes by a factor of 10 (Table 2-2). Note that this amount of free-energy difference is roughly equivalent to the free energy available from a single hydrogen bond.

More generally, for a reaction that has multiple reactants and products, such as $A + B \rightarrow C + D$,

$$K = \frac{[C][D]}{[A][B]}$$

The concentrations of the two reactants and the two products are multiplied because the rate of the forward reaction depends on the collision of A and B and the rate of the backward reaction depends on the collision of C and D. Thus, at 37°C,

$$\Delta G^\circ = -5.94 \log \frac{[C][D]}{[A][B]}$$

where ΔG° is in kilojoules per mole, and [A], [B], [C], and [D] denote the concentrations of the reactants and products in moles/liter.

The Free-Energy Changes of Coupled Reactions Are Additive

We have pointed out that unfavorable reactions can be coupled to favorable ones to drive the unfavorable ones forward (see Figure 2-29). In thermodynamic terms, this is possible because the overall free-energy change for a set of coupled reactions is the sum of the free-energy changes in each of its component steps. Consider, as a simple example, two sequential reactions



whose ΔG° values are +5 and -13 kJ/mole, respectively. If these two reactions occur sequentially, the ΔG° for the coupled reaction will be -8 kJ/mole. This means that, with appropriate conditions, the unfavorable reaction $X \rightarrow Y$ can be driven by the favorable reaction $Y \rightarrow Z$, provided that this second reaction follows the first. For example, several of the reactions in the long pathway that converts sugars into CO_2 and H_2O have positive ΔG° values. But the pathway nevertheless proceeds because the total ΔG° for the series of sequential reactions has a large negative value.

Forming a sequential pathway is not adequate for many purposes. Often the desired pathway is simply $X \rightarrow Y$, without further conversion of Y to some other product. Fortunately, there are other more general ways of using enzymes to couple reactions together. In order to explain how this is possible, we need to introduce the activated carrier molecules that we discuss next.

Activated Carrier Molecules Are Essential for Biosynthesis

To make life possible, the energy released by the oxidation of food molecules must be stored temporarily before it can be channeled into the construction of the many other molecules needed by the cell. In most cases, the energy is stored as chemical-bond energy in a small set of activated "carrier molecules," which contain one or more energy-rich covalent bonds. These molecules diffuse rapidly throughout the cell and thereby carry their bond energy from sites of energy

TABLE 2-2 Relationship Between the Standard Free-Energy Change, ΔG° , and the Equilibrium Constant

Equilibrium constant $\frac{[X]}{[Y]} = K$	Free energy of X minus free energy of Y [kJ/mole (kcal/mole)]
10^5	-29.7 (-7.1)
10^4	-23.8 (-5.7)
10^3	-17.8 (-4.3)
10^2	-11.9 (-2.8)
10^1	-5.9 (-1.4)
1	0 (0)
10^{-1}	5.9 (1.4)
10^{-2}	11.9 (2.8)
10^{-3}	17.8 (4.3)
10^{-4}	23.8 (5.7)
10^{-5}	29.7 (7.1)

Values of the equilibrium constant were calculated for the simple chemical reaction $Y \leftrightarrow X$ using the equation given in the text. The ΔG° given here is in kilojoules per mole at 37°C, with kilocalories per mole in parentheses. One kilojoule (kJ) is equal to 0.239 kilocalories (kcal) (1 kcal = 4.18 kJ). As explained in the text, ΔG° represents the free-energy difference under standard conditions (where all components are present at a concentration of 1.0 mole/liter).

From this table, we see that if there is a favorable standard free-energy change (ΔG°) of -17.8 kJ/mole (-4.3 kcal/mole) for the transition $Y \rightarrow X$, there will be 1000 times more molecules in state X than in state Y at equilibrium ($K = 1000$).

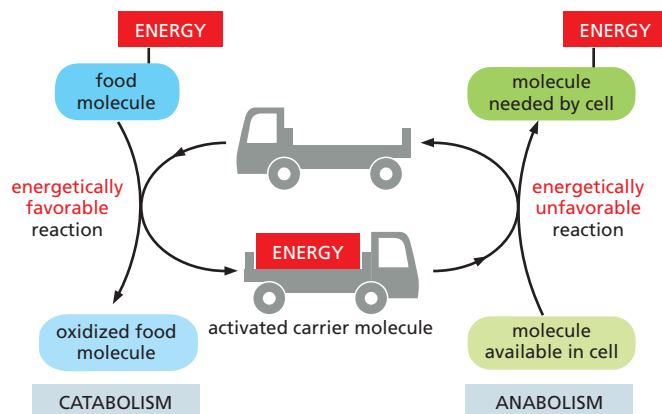


Figure 2–31 Energy transfer and the role of activated carriers in metabolism. By serving as energy shuttles, activated carrier molecules perform their function as go-betweens that link the breakdown of food molecules and the release of energy (*catabolism*) to the energy-requiring biosynthesis of small and large organic molecules (*anabolism*).

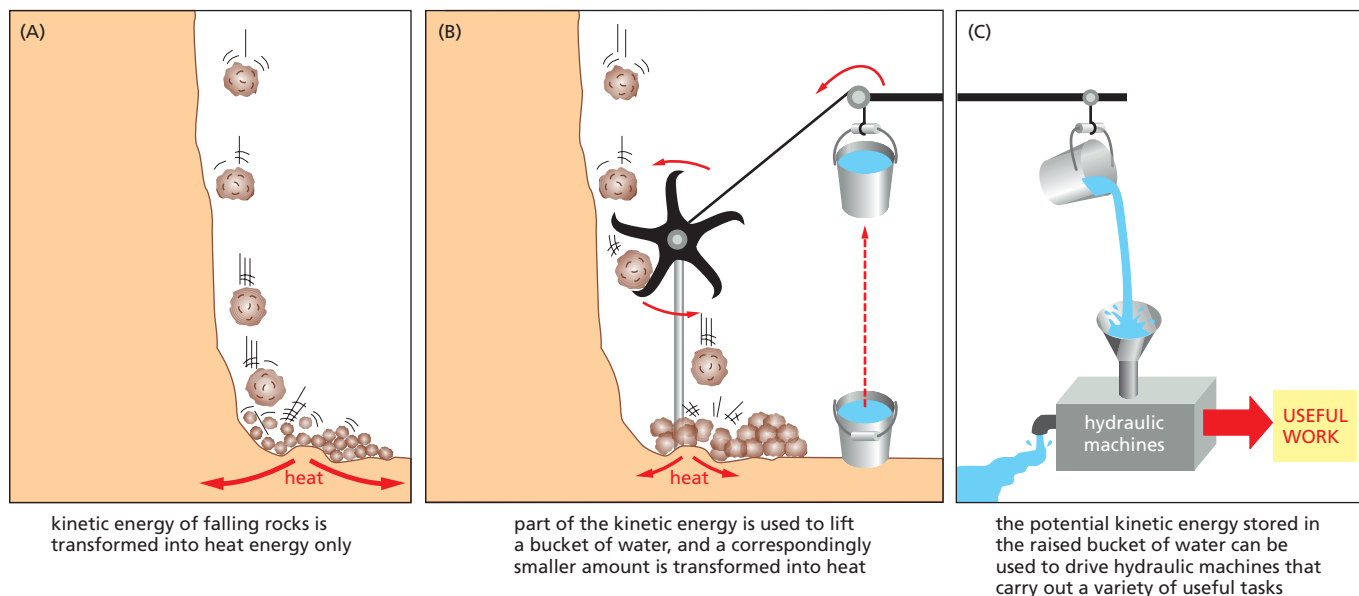
generation to the sites where the energy will be used for biosynthesis and other cell activities (**Figure 2–31**).

The **activated carriers** store energy in an easily exchangeable form, either as a readily transferable chemical group or as electrons held at a high energy level, and they can serve a dual role as a source of both energy and chemical groups in biosynthetic reactions. For historical reasons, these molecules are also sometimes referred to as *coenzymes*. The most important of the activated carrier molecules are ATP and two molecules that are closely related to each other, NADH and NADPH. As we discuss next, coupled reactions allow cells both to generate such activated carrier molecules and to use them like money to pay for reactions that otherwise could not take place.

The Formation of an Activated Carrier Is Coupled to an Energetically Favorable Reaction

Coupling mechanisms require enzymes and are fundamental to all the energy transactions of the cell. The nature of a **coupled reaction** is illustrated by a mechanical analogy in **Figure 2–32**, in which an energetically favorable chemical reaction is represented by rocks falling from a cliff. The energy of falling rocks would normally be entirely wasted in the form of heat generated by friction when the rocks hit the ground (see the falling-brick diagram in Figure 2–17). By careful design, however, part of this energy could be used instead to drive a paddle wheel

Figure 2–32 A mechanical model illustrating the principle of coupled chemical reactions. The spontaneous reaction shown in (A) could serve as an analogy for the direct oxidation of glucose to CO_2 and H_2O , which produces heat only. In (B), the same reaction is coupled to a second reaction; this second reaction is analogous to the synthesis of activated carrier molecules. The energy produced in (B) is in a more useful form than in (A) and can be used to drive a variety of otherwise energetically unfavorable reactions (C).



that lifts a bucket of water (Figure 2-32B). Because the rocks can now reach the ground only after moving the paddle wheel, we say that the energetically favorable reaction of rock falling has been directly *coupled* to the energetically unfavorable reaction of lifting the bucket of water. Note that because part of the energy is used to do work in Figure 2-32B, the rocks hit the ground with less velocity than in Figure 2-32A, and correspondingly less energy is dissipated as heat.

Similar processes occur in cells, where enzymes play the role of the paddle wheel. By mechanisms that we discuss later in this chapter, enzymes couple an energetically favorable reaction, such as the oxidation of foodstuffs, to an energetically unfavorable reaction, such as the generation of an activated carrier molecule. As in this example, the amount of heat released by the oxidation reaction is reduced by exactly the amount of energy stored in the energy-rich covalent bonds of the activated carrier molecule. And the activated carrier molecule picks up a packet of energy of a size sufficient to power a chemical reaction elsewhere in the cell.

ATP Is the Most Widely Used Activated Carrier Molecule

The most important and versatile of the activated carriers in cells is **ATP** (adenosine triphosphate). Just as the energy stored in the raised bucket of water in Figure 2-32B can drive a wide variety of hydraulic machines, ATP is a convenient and versatile store, or currency, of energy used to drive a huge variety of chemical reactions in cells. ATP is synthesized by coupling a reaction that is highly energetically favorable to an energetically unfavorable phosphorylation reaction in which a phosphate group is added to **ADP** (adenosine diphosphate). When required, ATP gives up its energy packet through its energetically favorable hydrolysis to ADP and inorganic phosphate (Figure 2-33). The regenerated ADP is then available to be used for another round of the phosphorylation reaction that forms ATP.

The two outermost phosphate groups in ATP are said to be held to the rest of the molecule by “high-energy” covalent bonds, because each of these phosphoanhydride linkages releases a great deal of free energy when hydrolyzed. The unusually large negative free-energy change for these hydrolysis reactions arises from a number of factors. The release of the terminal phosphate group when ATP forms ADP removes an unfavorable repulsion between adjacent negative charges; in addition, the inorganic phosphate ion released is stabilized both by resonance and by favorable hydrogen-bond formation with water.

The energetically favorable reaction of ATP hydrolysis is coupled to many otherwise unfavorable reactions through which needed molecules are synthesized.

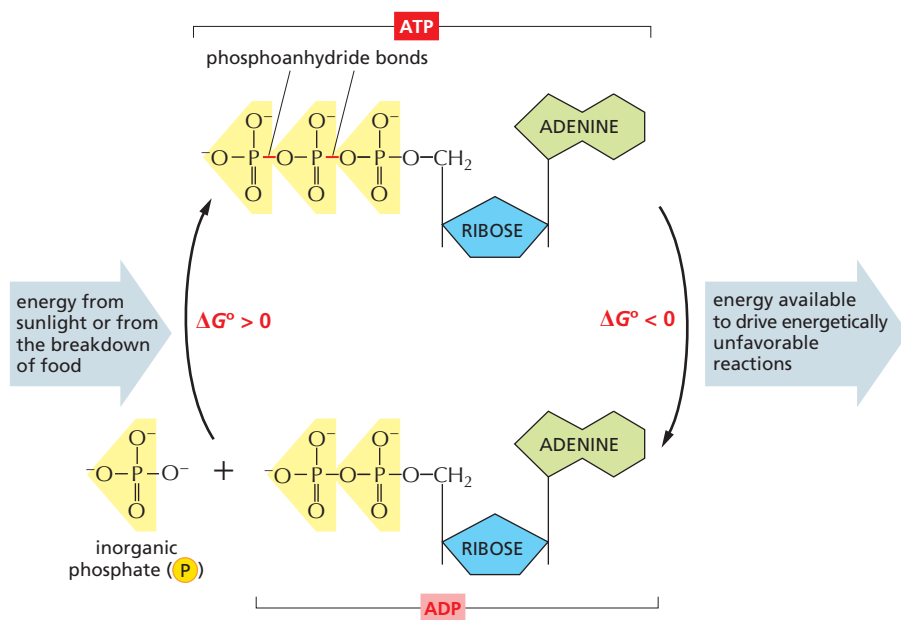


Figure 2-33 The interconversion of ATP and ADP occurs in a cycle. The two outermost phosphates in ATP are held to the rest of the molecule by “high-energy” phosphoanhydride bonds and are readily hydrolyzed or transferred to other molecules. Water can be added to ATP to form ADP and inorganic phosphate. This hydrolysis of the terminal phosphate of ATP yields between 46 and 54 kJ/mole of usable energy, depending on the intracellular conditions. (Although the ΔG° of this reaction is -30.5 kJ/mole, its ΔG inside cells is much more negative, because the ratio of ATP to the products ADP and phosphate is kept so high.)

The formation of ATP from ADP and phosphate reverses the hydrolysis reaction; because this condensation reaction is energetically unfavorable, it must be coupled to a highly energetically favorable reaction to occur.

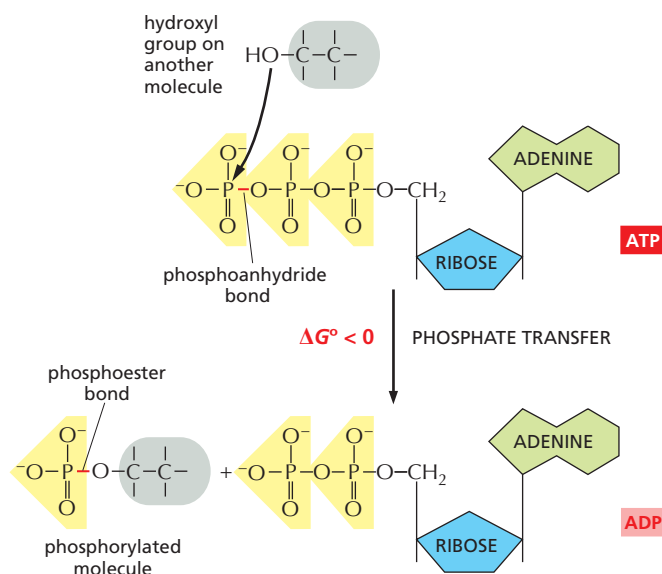


Figure 2–34 An example of a phosphate transfer reaction. Because an energy-rich phosphoanhydride bond in ATP is converted to a phosphoester bond, this reaction is energetically favorable, having a large negative ΔG . Reactions of this type are involved in the synthesis of phospholipids and in the initial steps of reactions that catabolize sugars, as well as in many other metabolic pathways.

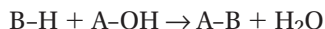
Many of these coupled reactions involve the transfer of the terminal phosphate in ATP to another molecule, as illustrated by the phosphorylation reaction in [Figure 2–34](#).

As the most abundant activated carrier in cells, ATP is the principal energy currency. To give just two examples, it supplies energy for many of the pumps that transport substances into and out of the cell (discussed in Chapter 11), and it powers the molecular motors that enable muscle cells to contract and nerve cells to transport materials from one end of their long axons to another (discussed in Chapter 16).

Energy Stored in ATP Is Often Harnessed to Join Two Molecules Together

We have previously discussed one way in which an energetically favorable reaction can be coupled to an energetically unfavorable reaction, $X \rightarrow Y$, so as to enable it to occur. In that scheme, a second enzyme catalyzes the energetically favorable reaction $Y \rightarrow Z$, pulling all of the $X \rightarrow Y$ in the process. But when the required product is Y and not Z , this mechanism is not useful.

A typical biosynthetic reaction is one in which two molecules, A and B , are joined together to produce $A-B$ in the energetically unfavorable *condensation* reaction



There is an indirect pathway that allows $B-H$ and $A-OH$ to form $A-B$, in which a coupling to ATP hydrolysis makes the reaction go. Here, energy from ATP hydrolysis is first used to convert $A-OH$ to a higher-energy intermediate compound, which then reacts directly with $B-H$ to give $A-B$. The simplest possible mechanism involves the transfer of a phosphate from ATP to $A-OH$ to make $A-O-PO_3$, in which case the reaction pathway contains only two steps:

1. $A-OH + ATP \rightarrow A-O-PO_3 + ADP$
2. $B-H + A-O-PO_3 \rightarrow A-B + \text{phosphate}$

Net result: $A-OH + ATP + B-H \rightarrow A-B + ADP + \text{phosphate}$

The condensation reaction, which by itself is energetically unfavorable, is forced to occur by being directly coupled to ATP hydrolysis in an enzyme-catalyzed reaction pathway ([Figure 2–35A](#)).

A biosynthetic reaction of exactly this type synthesizes the amino acid glutamine ([Figure 2–35B](#)). We will see shortly that similar (but more complex) mechanisms are also used to produce nearly all of the large molecules of the cell.

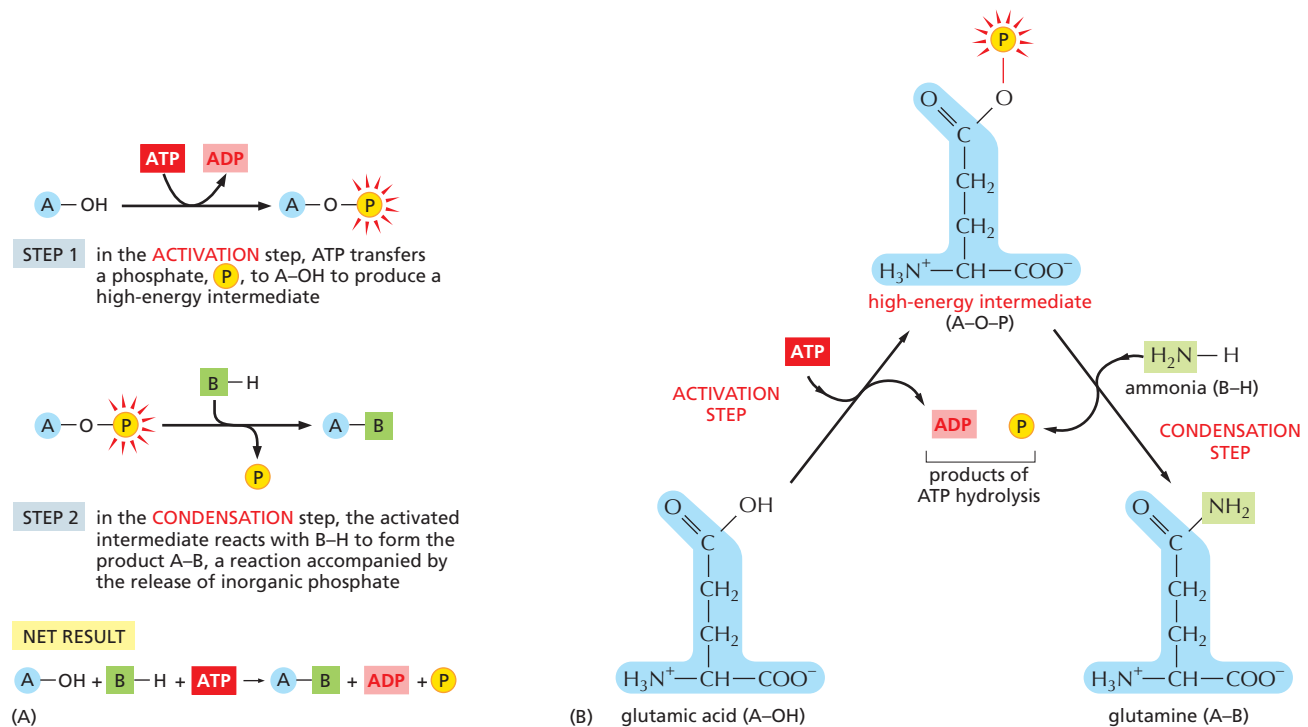


Figure 2-35 An example of an energetically unfavorable biosynthetic reaction driven by ATP hydrolysis. (A) Schematic illustration of the condensation reaction described in the text. In this set of reactions, a phosphate group is first donated by ATP to form a high-energy intermediate, A-O-PO₃, which then reacts with the other substrate, B-H, to form the product A-B. (B) Reaction showing the biosynthesis of the amino acid glutamine from glutamic acid. Glutamic acid, which corresponds to the A-OH shown in (A), is first converted to a high-energy phosphorylated intermediate, which corresponds to A-O-PO₃. This intermediate then reacts with ammonia (which corresponds to B-H) to form glutamine. In this example, both steps occur on the surface of the same enzyme, glutamine synthetase (not shown). ATP hydrolysis can drive this energetically unfavorable reaction because it produces a favorable free-energy change (ΔG° of -30.5 kJ/mole) that is larger in magnitude than the energy required for the synthesis of glutamine from glutamic acid plus NH₃ (ΔG° of $+14.2$ kJ/mole). For clarity, the glutamic acid side chain is shown in its uncharged form.

NADH and NADPH Are Important Electron Carriers

Other important activated carrier molecules participate in oxidation-reduction reactions and are commonly part of coupled reactions in cells. These activated carriers are specialized to carry electrons held at a high energy level (sometimes called “high-energy” electrons) and hydrogen atoms. The most important of these electron carriers are **NAD⁺** (nicotinamide adenine dinucleotide) and the closely related molecule **NADP⁺** (nicotinamide adenine dinucleotide phosphate). As part of an enzyme-catalyzed reaction in which a substrate molecule is oxidized, each picks up a “packet of energy” corresponding to two electrons plus a proton (H^+), and they are thereby converted to **NADH** (*reduced* nicotinamide adenine dinucleotide) and **NADPH** (*reduced* nicotinamide adenine dinucleotide phosphate), respectively (Figure 2-36). These molecules can therefore be regarded as carriers of hydride ions (the H^+ plus two electrons, or H^-).

Like ATP, NADPH is an activated carrier that participates in many important biosynthetic reactions that would otherwise be energetically unfavorable. The NADPH is produced according to the general scheme shown in Figure 2-36A. During a special set of energy-yielding catabolic reactions, two hydrogen atoms are removed from a substrate molecule. Both electrons but just one hydrogen atom (that is, a hydride ion, H^-) are added to the nicotinamide ring of NADP⁺ to form NADPH; the second hydrogen atom is released as a proton (H^+) into solution. This is a typical oxidation-reduction reaction, in which the substrate is oxidized and NADP⁺ is reduced.

NADPH readily gives up the hydride ion it carries in a subsequent oxidation-reduction reaction, because the nicotinamide ring can achieve a more stable

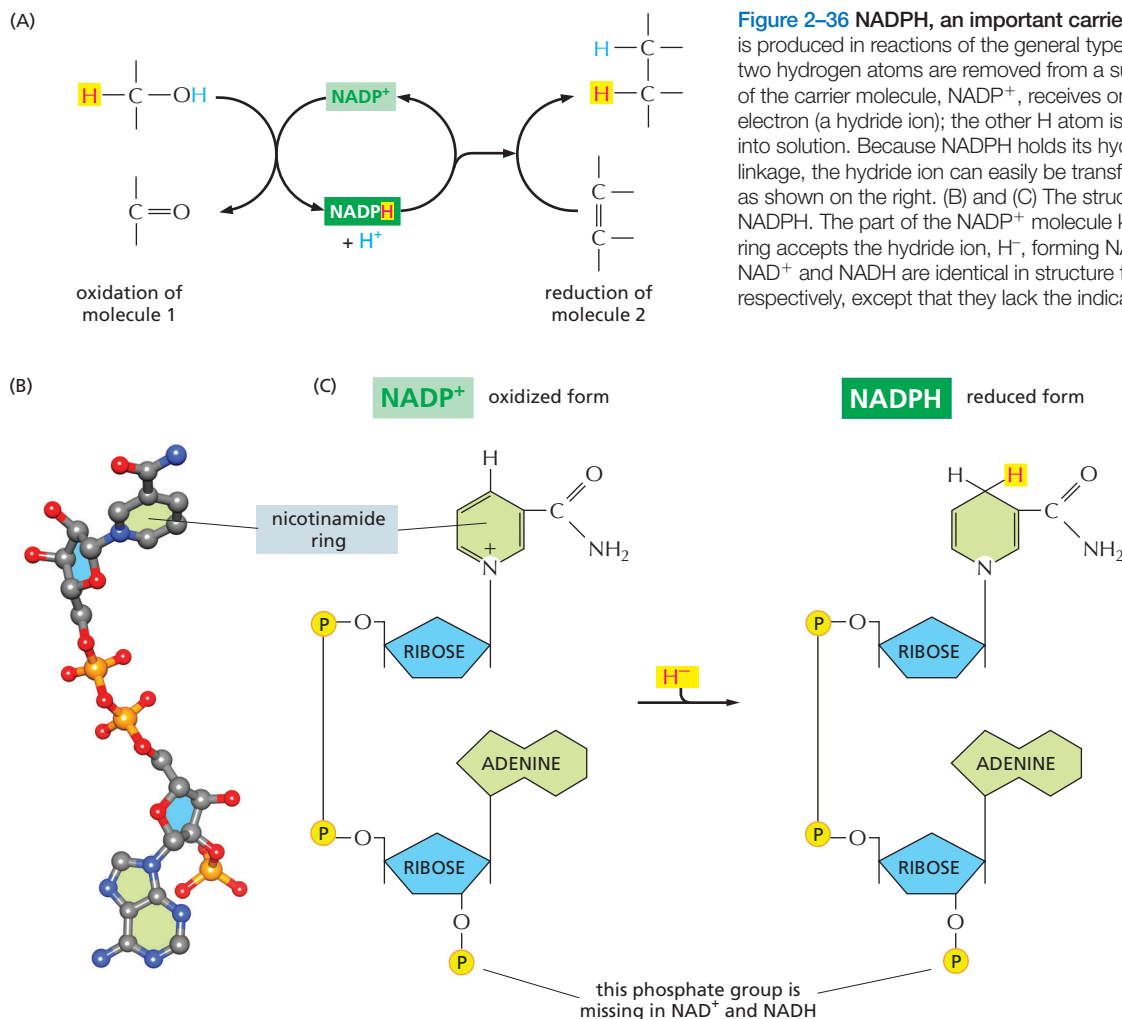


Figure 2–36 NADPH, an important carrier of electrons. (A) NADPH is produced in reactions of the general type shown on the left, in which two hydrogen atoms are removed from a substrate. The oxidized form of the carrier molecule, NADP⁺, receives one hydrogen atom plus an electron (a hydride ion); the other H atom is released as a proton (H⁺) into solution. Because NADPH holds its hydride ion in a high-energy linkage, the hydride ion can easily be transferred to other molecules, as shown on the right. (B) and (C) The structures of NADP⁺ and NADPH. The part of the NADP⁺ molecule known as the nicotinamide ring accepts the hydride ion, H⁻, forming NADPH. The molecules NAD⁺ and NADH are identical in structure to NADP⁺ and NADPH, respectively, except that they lack the indicated phosphate group.

arrangement of electrons without it. In this subsequent reaction, which regenerates NADP⁺, it is the NADPH that is oxidized and the substrate that is reduced. The NADPH is an effective donor of its hydride ion to other molecules for the same reason that ATP readily transfers a phosphate: in both cases the transfer is accompanied by a large negative free-energy change. One example of the use of NADPH in biosynthesis is shown in [Figure 2–37](#).

The extra phosphate group on NADPH has no effect on the electron-transfer properties of NADPH compared with NADH, being far away from the region involved in electron transfer (see [Figure 2–36C](#)). It does, however, give a molecule of NADPH a slightly different shape from that of NADH, making it possible for NADPH and NADH to bind as substrates to completely different sets of enzymes. Thus, the two types of carriers are used to transfer electrons (or hydride ions) between two different sets of molecules.

Why should there be this division of labor? The answer lies in the need to regulate two sets of electron-transfer reactions independently. NADPH operates chiefly with enzymes that catalyze anabolic reactions, supplying the high-energy electrons needed to synthesize energy-rich biological molecules. NADH, by contrast, has a special role as an intermediate in the catabolic system of reactions that generate ATP through the oxidation of food molecules, as we will discuss shortly. The genes of NADH from NAD⁺ and of NADPH from NADP⁺ occur by different pathways and are independently regulated so that the cell can adjust the supply of electrons for these two contrasting purposes. Inside the cell the ratio of NAD⁺ to NADH is kept high, whereas the ratio of NADP⁺ to NADPH is kept low. This

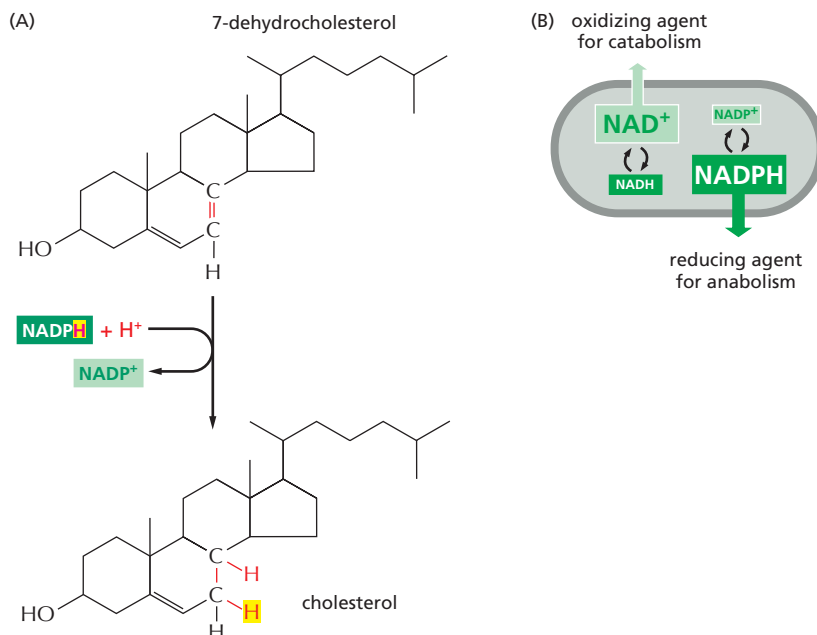


Figure 2-37 NADPH as a reducing agent. (A) The final stage in a biosynthetic route leading to cholesterol. As in many other biosynthetic reactions, the reduction of the $\text{C}=\text{C}$ bond is achieved by the transfer of a hydride ion from the carrier molecule NADPH, plus a proton (H^+) from the solution. (B) Keeping NADPH levels high and NADH levels low alters the redox potential for each of the two redox pairs— $\text{NADPH}:\text{NADP}^+$ and $\text{NADH}:\text{NAD}^+$ (see Panel 14-1, p. 825). It causes NADPH to be a much stronger electron donor (reducing agent) than NADH, and it causes NAD^+ to be a much better electron acceptor (oxidizing agent) than NADP^+ , as indicated.

provides plenty of NAD^+ to act as an oxidizing agent and plenty of NADPH to act as a reducing agent (Figure 2-37B)—as required for their special roles in catabolism and anabolism, respectively.

There Are Many Other Activated Carrier Molecules in Cells

Other activated carriers also pick up and carry a chemical group in an easily transferred, high-energy linkage. For example, coenzyme A carries a readily transferable acetyl group in a thioester linkage and in this activated form is known as **acetyl CoA** (acetyl coenzyme A). Acetyl CoA (Figure 2-38) is used to add two carbon units in the biosynthesis of larger molecules.

In acetyl CoA, as in other carrier molecules, the transferable group makes up only a small part of the molecule. The rest consists of a large organic portion that serves as a convenient “handle,” facilitating the recognition of the carrier molecule by specific enzymes. As with acetyl CoA, this handle portion very often contains a nucleotide derivative (usually adenosine diphosphate), a curious fact that may be a relic from an early stage of evolution. It is currently thought that the main catalysts for early life-forms—before DNA or proteins—were RNA molecules (or

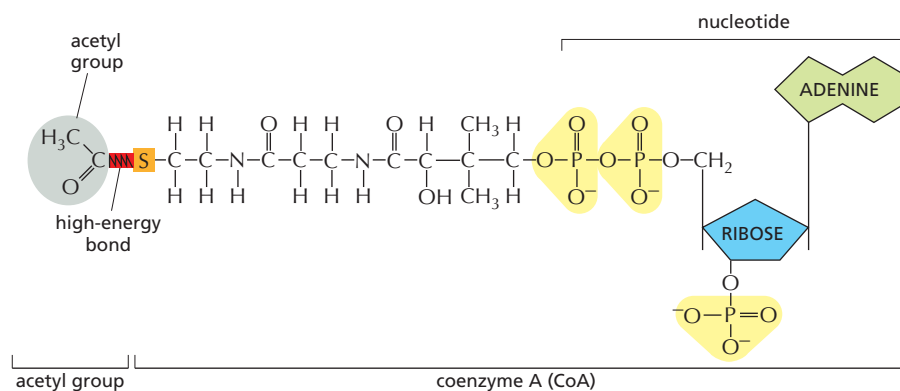


Figure 2-38 The structure of the important activated carrier molecule acetyl CoA. The sulfur atom (orange) forms a thioester bond to acetate. Because this is a high-energy linkage, releasing a large amount of free energy when it is hydrolyzed, the acetate molecule can be readily transferred to other molecules.

TABLE 2-3 Some Activated Carrier Molecules Widely Used in Metabolism

Activated carrier	Group carried in high-energy linkage
ATP	Phosphate
NADH, NADPH, FADH ₂	Electrons and hydrogens
Acetyl CoA	Acetyl group
Carboxylated biotin	Carboxyl group
S-Adenosylmethionine	Methyl group
Uridine diphosphate glucose	Glucose

their close relatives), as described in Chapter 6. It is tempting to speculate that many of the carrier molecules that we find today originated in this earlier RNA world, where their nucleotide portions could have been useful for binding them to RNA enzymes (ribozymes).

Thus, ATP transfers phosphate, NADPH transfers electrons and hydrogen, and acetyl CoA transfers two-carbon acetyl groups. **FADH₂** (reduced flavin adenine dinucleotide) is used like NADH in electron and proton transfers (Figure 2-39). The reactions of other activated carrier molecules involve the transfer of a methyl, carboxyl, or glucose group for biosyntheses (Table 2-3). These activated carriers are generated in reactions that are coupled to ATP hydrolysis, as in the example in Figure 2-40. Therefore, the energy that enables their groups to be used for biosynthesis ultimately comes from the catabolic reactions that generate ATP. Similar processes occur in the synthesis of the very large molecules of the cell—the nucleic acids, proteins, and polysaccharides—that we discuss next.

The Synthesis of Biological Polymers Is Driven by ATP Hydrolysis

As discussed previously, the macromolecules of the cell constitute most of its dry mass (see Figure 2-7). These molecules are made from subunits (or monomers) that are linked together in a *condensation* reaction, in which the constituents of a

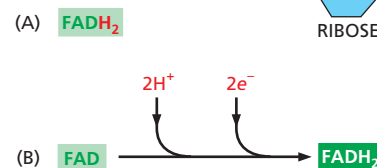
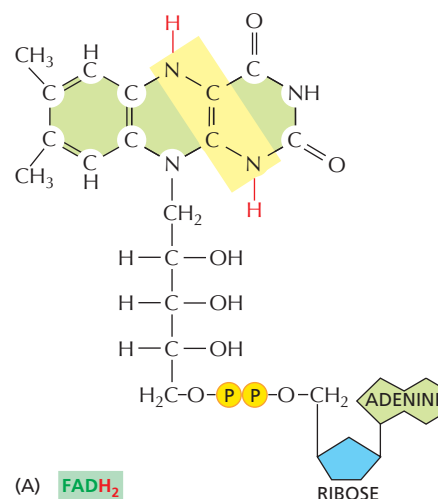


Figure 2-39 FADH₂ is a carrier of hydrogens and high-energy electrons, like NADH and NADPH. (A) Structure of FADH₂, with its hydrogen-carrying atoms highlighted in yellow. (B) The formation of FADH₂ from FAD (flavin adenine dinucleotide).

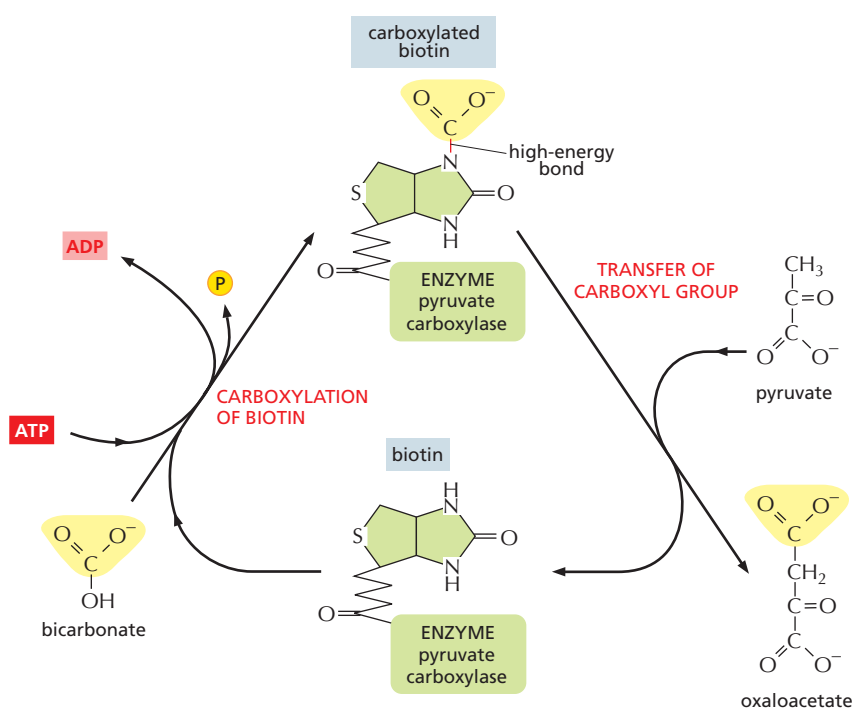


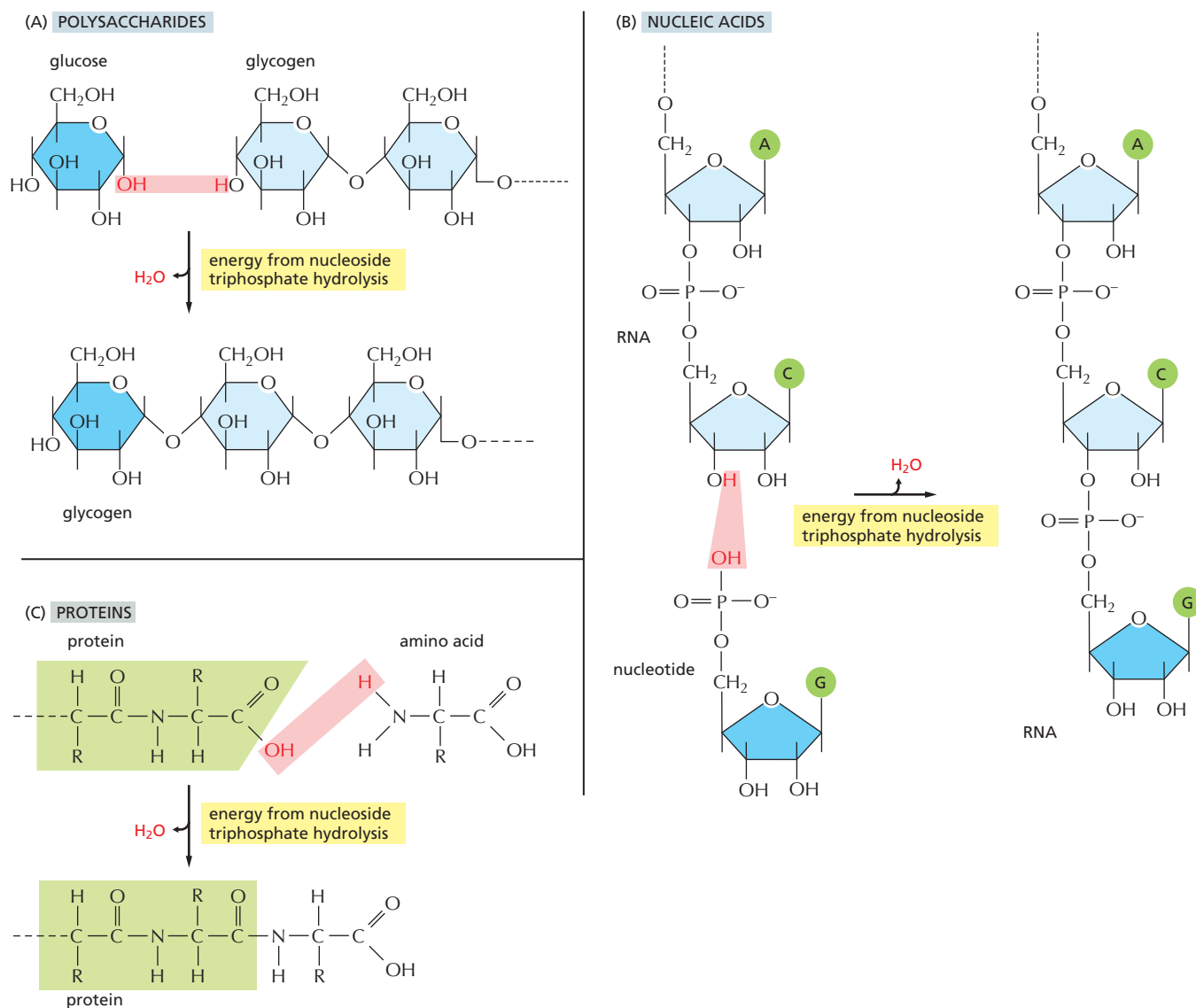
Figure 2-40 A carboxyl group-transfer reaction using an activated carrier molecule. Carboxylated biotin is used by the enzyme *pyruvate carboxylase* to transfer a carboxyl group in the production of oxaloacetate, a molecule needed for the citric acid cycle. The acceptor molecule for this group-transfer reaction is pyruvate. Other enzymes use biotin, a B-complex vitamin, to transfer carboxyl groups to other acceptor molecules. Note that synthesis of carboxylated biotin requires energy that is derived from ATP—a general feature of many activated carriers.

water molecule (OH plus H) are removed from the two reactants. Consequently, the reverse reaction—the breakdown of all three types of polymers—occurs by the enzyme-catalyzed addition of water (*hydrolysis*). This hydrolysis reaction is energetically favorable, whereas the biosynthetic reactions require an energy input (see Figure 2-9).

The nucleic acids (DNA and RNA), proteins, and polysaccharides are all polymers that are produced by the repeated addition of a monomer onto one end of a growing chain. The synthesis reactions for these three types of macromolecules are outlined in **Figure 2-41**. As indicated, the condensation step in each case depends on energy from nucleoside triphosphate hydrolysis. And yet, except for the nucleic acids, there are no phosphate groups left in the final product molecules. How are the reactions that release the energy of ATP hydrolysis coupled to polymer synthesis?

For each type of macromolecule, an enzyme-catalyzed pathway exists, which resembles that discussed previously for the synthesis of the amino acid glutamine (see Figure 2-35). The principle is exactly the same, in that the -OH group that will be removed in the condensation reaction is first activated by becoming involved in a high-energy linkage to a second molecule. However, the actual mechanisms used to link ATP hydrolysis to the synthesis of proteins and polysaccharides are more complex than that used for glutamine synthesis, inasmuch as more than

Figure 2-41 How energy is used to synthesize macromolecules. Synthesis of a portion of (A) a polysaccharide, (B) a nucleic acid, and (C) a protein is shown here. In each case, synthesis involves a condensation reaction in which water is lost; the atoms involved are shaded in pink. Not shown is the consumption of high-energy nucleoside triphosphates that is required to activate each subunit prior to its addition. In contrast, the reverse reaction—the breakdown of all three types of polymers—occurs through the simple addition of water, or hydrolysis (not shown), which is energetically favorable and does not require an energy carrier.



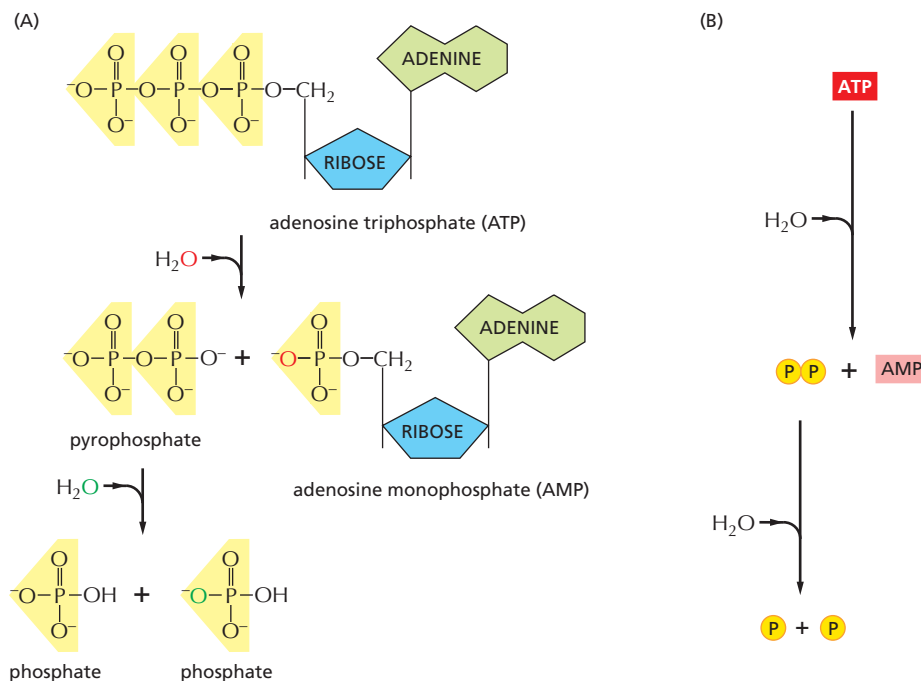


Figure 2-42 An alternative pathway of ATP hydrolysis, in which pyrophosphate is first formed and then hydrolyzed. This route releases about twice as much free energy (approximately -100 kJ/mole) as the reaction shown earlier in Figure 2-33, and it forms AMP instead of ADP. (A) In the two successive hydrolysis reactions, oxygen atoms from the participating water molecules are retained in the products, as indicated, whereas the hydrogen atoms dissociate to form free hydrogen ions (H^+ , not shown). (B) Summary of overall reaction.

one high-energy intermediate is required to generate the final high-energy bond that is broken during the condensation step (discussed in Chapter 6 for protein synthesis).

Each activated carrier has limits in its ability to drive a biosynthetic reaction. The ΔG for the hydrolysis of ATP to ADP and phosphate depends on the concentrations of all of the reactants, but under the usual conditions in a cell it is between -46 and -54 kJ/mole. In principle, this hydrolysis reaction could drive an unfavorable reaction with a ΔG of, perhaps, $+40$ kJ/mole, provided that a suitable reaction path is available. For some biosynthetic reactions, however, even -50 kJ/mole does not provide enough of a driving force. In these cases, the path of ATP hydrolysis can be altered so that it initially produces AMP and *pyrophosphate*, which is itself then hydrolyzed in a subsequent step (Figure 2-42). The whole process makes available a total free-energy change of about -100 kJ/mole. An important type of biosynthetic reaction that is driven in this way is the synthesis of nucleic acids (polynucleotides) from nucleoside triphosphates, as illustrated on the right side of Figure 2-43.

Note that the repetitive condensation reactions that produce macromolecules can be oriented in one of two ways, which differ in the position of the high-energy bond that drives polymerization. In so-called *polymer-end activation*, the reactive bond required for the condensation reaction is carried on the end of the growing polymer, and it must therefore be regenerated each time that a monomer is added. In this case, each monomer brings with it the reactive bond that will be used in adding the *next* monomer in the series. In *direct-monomer activation*, the reactive bond carried by each monomer is instead used immediately for its own addition (Figure 2-44).

We shall see in later chapters that both of these types of polymerization are used. The synthesis of DNA, RNA, and some simple polysaccharides occurs by direct-monomer activation, for example, whereas the synthesis of proteins occurs by a polymer-end activation process.

Summary

Living cells need to create and maintain order within themselves to survive and grow. This is thermodynamically possible only because of a continual input of energy, part of which must be released from the cells to their environment as heat

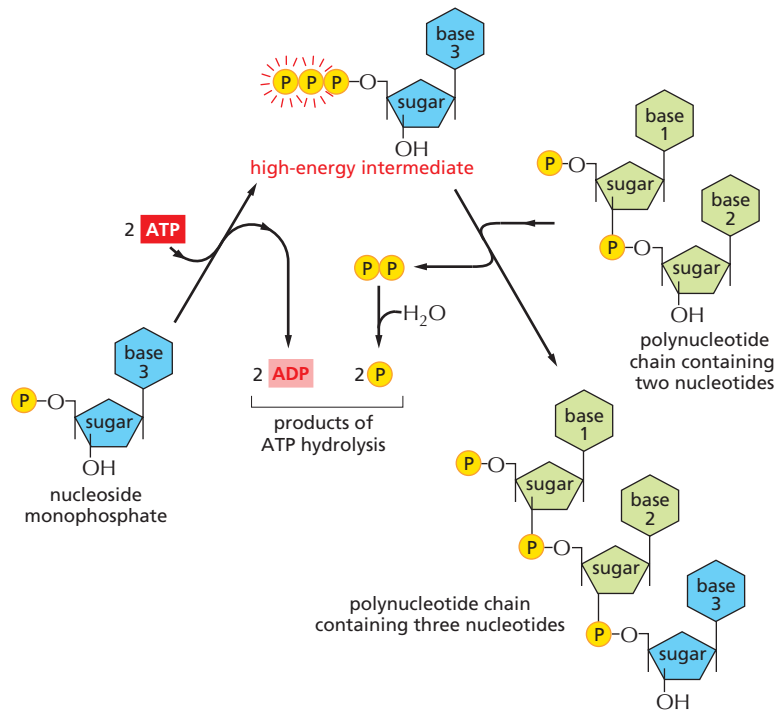


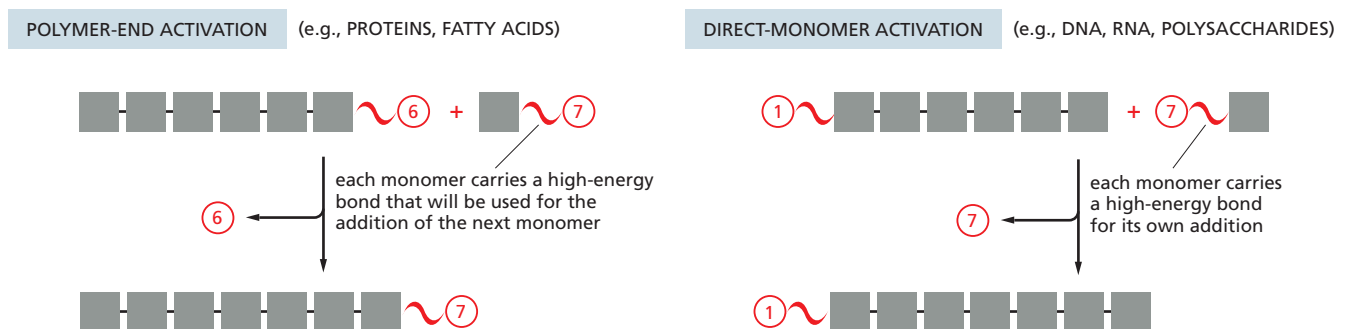
Figure 2–43 Synthesis of a polynucleotide, RNA or DNA, is a multistep process driven by ATP hydrolysis. In the first step, a nucleoside monophosphate (a nucleotide) is activated by the sequential transfer of the terminal phosphate groups from two ATP molecules. The high-energy intermediate formed—a nucleoside triphosphate—exists free in solution until it reacts with the growing end of an RNA or a DNA chain, which leads to release of pyrophosphate. Hydrolysis of the latter to inorganic phosphate is highly favorable and helps to drive the overall reaction in the direction of polynucleotide synthesis. For details, see Chapter 5.

that disorders the surroundings. The only chemical reactions possible are those that increase the total amount of disorder in the universe. The free-energy change for a reaction, ΔG , measures this disorder, and it must be less than zero for a reaction to proceed spontaneously. This ΔG depends both on the intrinsic properties of the reacting molecules and their concentrations, and it can be calculated from these concentrations if either the equilibrium constant (K) for the reaction or its standard free-energy change, ΔG° , is known.

The energy needed for life comes ultimately from the electromagnetic radiation of the Sun, which drives the formation of organic molecules in photosynthetic organisms such as green plants. Animals obtain their energy by eating organic molecules and oxidizing them in a series of enzyme-catalyzed reactions that are coupled to the formation of ATP—a common currency of energy in all cells.

To make possible the continual generation of order in cells, energetically favorable reactions, such as the hydrolysis of ATP, are coupled to energetically unfavorable reactions. In the biosynthesis of macromolecules, ATP is used to form reactive phosphorylated intermediates. Because the energetically unfavorable reaction of biosynthesis now becomes energetically favorable, ATP hydrolysis is said to drive the reaction. Polymeric molecules such as proteins, nucleic acids, and polysaccharides are assembled from small activated precursor molecules by repetitive condensation reactions that are driven in this way. Other reactive

Figure 2–44 Two active-intermediate orientations are utilized for the repetitive condensation reactions that form biological polymers. A polymer-end activation mechanism for polymer synthesis is compared with its alternative, direct-monomer activation. As indicated, these two mechanisms are used to produce different types of biological macromolecules.



molecules, called either activated carriers or coenzymes, transfer other chemical groups in the course of biosynthesis: NADPH transfers hydrogen as a proton plus two electrons (a hydride ion), for example, whereas acetyl CoA transfers an acetyl group.

HOW CELLS OBTAIN ENERGY FROM FOOD

The constant supply of energy that cells need to generate and maintain the biological order that keeps them alive comes from the chemical-bond energy in food molecules.

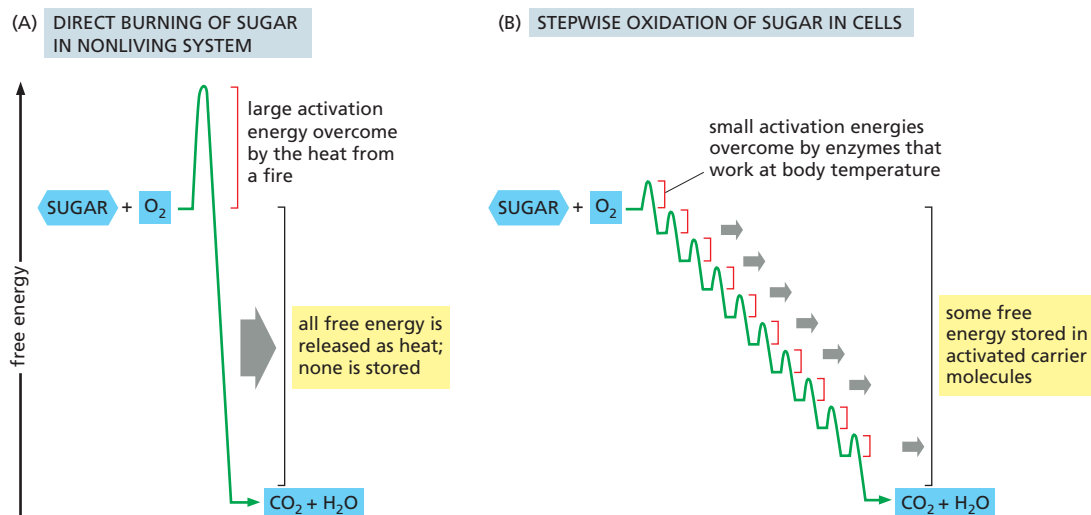
The proteins, lipids, and polysaccharides that make up most of the food we eat must be broken down into smaller molecules before our cells can use them—either as a source of energy or as building blocks for other molecules. Enzymatic digestion breaks down the large polymeric molecules in food into their monomer subunits—proteins into amino acids, polysaccharides into sugars, and fats into fatty acids and glycerol. After digestion, the small organic molecules derived from food enter the cytosol of cells, where their gradual oxidation begins.

Sugars are particularly important fuel molecules, and they are oxidized in small controlled steps to carbon dioxide (CO_2) and water (Figure 2–45). In this section, we trace the major steps in the breakdown, or catabolism, of sugars and show how they produce ATP, NADH, and other activated carrier molecules in animal cells. A very similar pathway also operates in plants, fungi, and many bacteria. As we shall see, the oxidation of fatty acids is equally important for cells. Other molecules, such as proteins, can also serve as energy sources when they are funneled through appropriate enzymatic pathways.

Glycolysis Is a Central ATP-producing Pathway

The major process for oxidizing sugars is the sequence of reactions known as **glycolysis**, which produces ATP without the involvement of molecular oxygen (O_2 gas). It occurs in the cytosol of most cells, including many anaerobic microorganisms. Glycolysis probably evolved early in the history of life, before photosynthetic organisms introduced oxygen into the atmosphere. During glycolysis, a glucose molecule with six carbon atoms is converted into two molecules of *pyruvate*, each of which contains three carbon atoms. For each glucose molecule, two molecules of ATP are hydrolyzed to provide energy to drive the early steps, but four

Figure 2–45 Schematic representation of the controlled stepwise oxidation of sugar in a cell, compared with ordinary burning. (A) If the sugar were oxidized to CO_2 and H_2O in a single step, it would release an amount of energy much larger than could be captured for useful purposes. (B) In the cell, enzymes catalyze oxidation via a series of small steps in which free energy is transferred in conveniently sized packets to carrier molecules—most often ATP and NADH. Heat is also released, enabling the universe to be disordered sufficiently to make the entire pathway energetically favorable (ΔG is negative). Each step is catalyzed by an enzyme that lowers the activation-energy barrier that must be surmounted by the random collision of molecules to allow the reaction to occur at the temperature inside cells (see Figure 2–24). Note that the total free energy released by the complete oxidative breakdown of glucose to CO_2 and H_2O —2880 kJ/mole—is exactly the same in (A) and (B).



molecules of ATP are produced in the later steps. At the end of glycolysis, there is consequently a net gain of two molecules of ATP for each glucose molecule broken down.

Piecing together the complete glycolytic pathway in the 1930s was a major triumph of biochemistry, as the pathway consists of a sequence of 10 separate reactions, each producing a different sugar intermediate and each catalyzed by a different enzyme. These enzymes, like most enzymes, all have names ending in “-ase”—like isomerase and dehydrogenase—which specify the type of reaction they catalyze. The sugar oxidation occurs when electrons are removed by NAD^+ (producing NADH) from some of the carbons derived from the glucose molecule. Some of the energy released by this oxidation drives the direct synthesis of ATP molecules from ADP and phosphate, and some remains with the electrons in the activated electron carrier NADH. The pathway is outlined in [Figure 2-46](#) and shown in detail in [Panel 2-8](#) (pp. 108–109) and [Movie 2.5](#).

Two molecules of NADH are formed per molecule of glucose in the course of glycolysis. As we shall see, in aerobic organisms these NADH molecules donate their electrons to the electron-transport chain described in Chapter 14, which enables the NAD^+ formed by NADH oxidation to be reused for glycolysis.

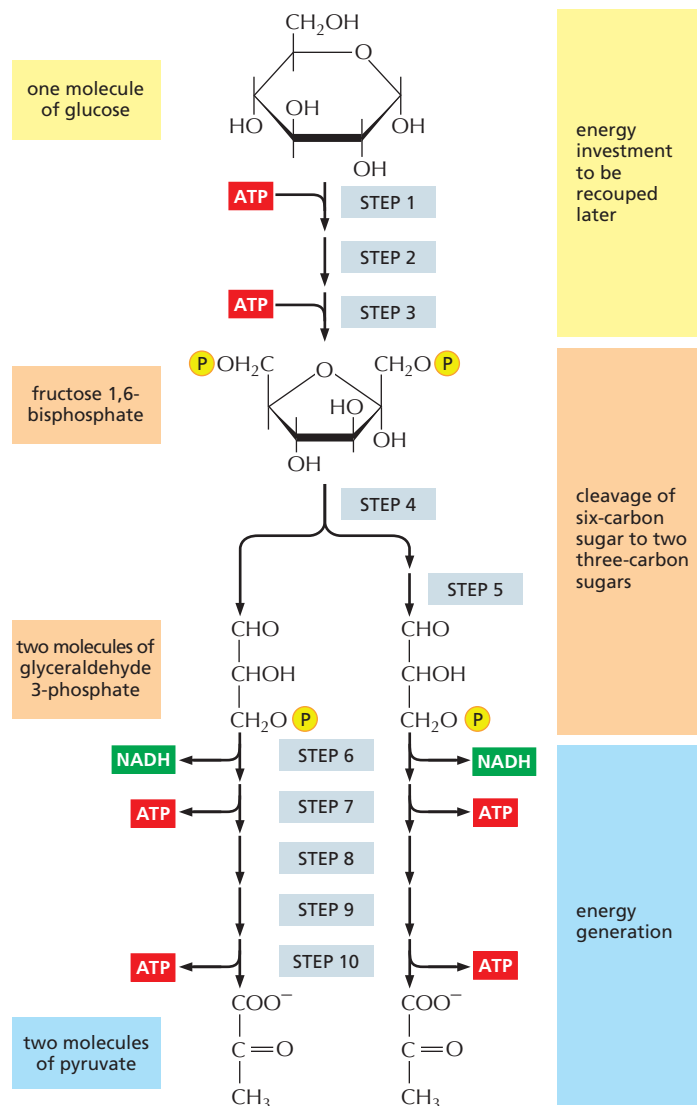


Figure 2-46 An outline of glycolysis. Each of the 10 steps shown is catalyzed by a different enzyme. Note that step 4 cleaves a six-carbon sugar into two three-carbon sugars, so that the number of molecules at every stage after this doubles. Note also that one of the two products of step 4 is modified (isomerized) in step 5 to convert it into a second molecule of glyceraldehyde 3-phosphate, the other product of step 4. As indicated, step 6 begins the energy-generation phase of glycolysis. Because two molecules of ATP are hydrolyzed in the early, energy-investment phase, glycolysis results in the net synthesis of 2 ATP and 2 NADH molecules per molecule of glucose. Glycolysis is also referred to as the Embden–Meyerhof pathway, named for the biochemists who first described it. All the steps of glycolysis are reviewed in [Panel 2-8](#) and [Movie 2.5](#).

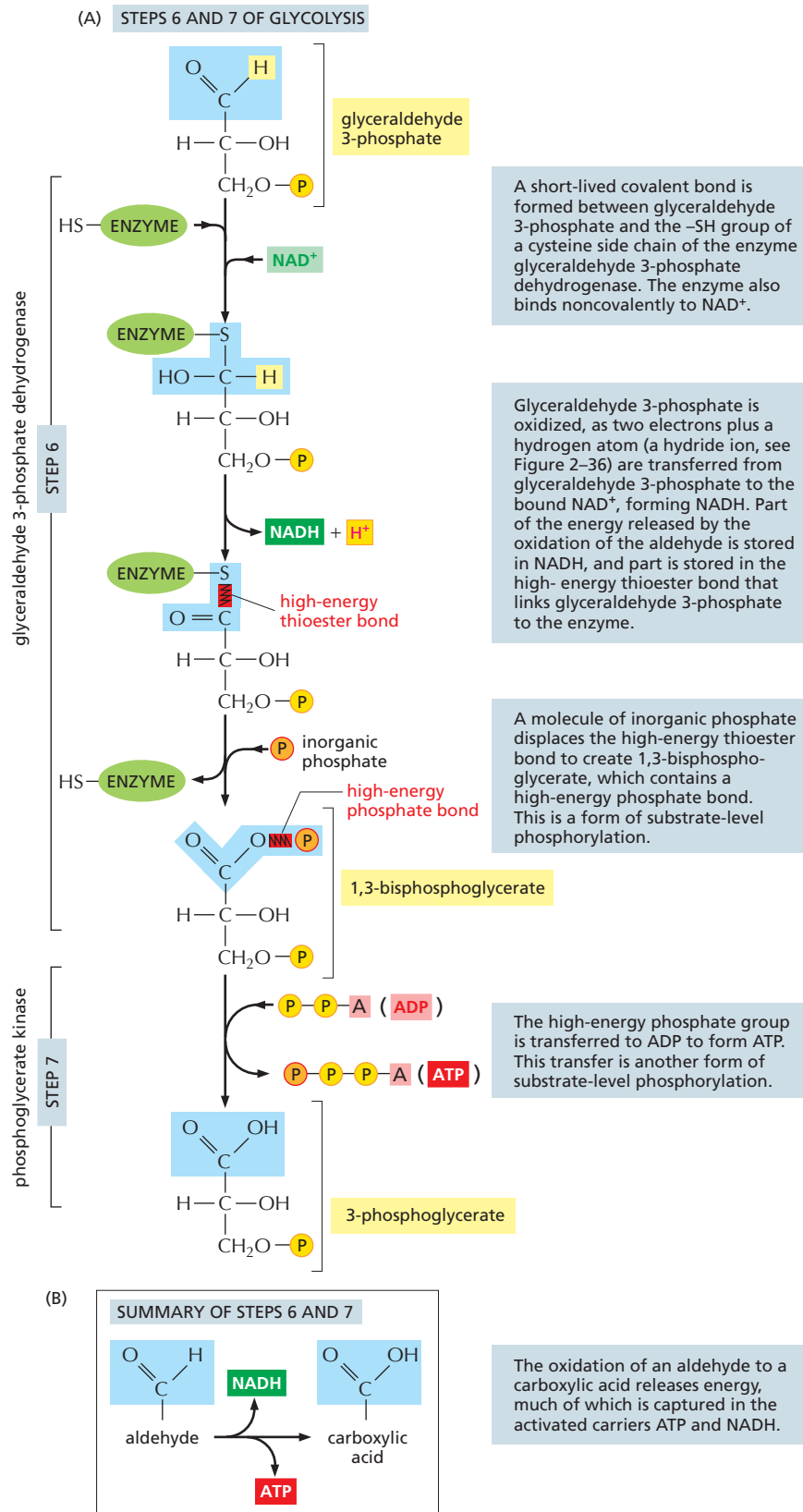


Figure 2-47 How the oxidation of glyceraldehyde 3-phosphate is coupled to the formation of ATP and NADH in steps 6 and 7 of glycolysis. (A) In step 6, the enzyme glyceraldehyde 3-phosphate dehydrogenase couples the energetically favorable oxidation of an aldehyde to the energetically unfavorable formation of a high-energy phosphate bond. At the same time, this oxidation enables energy to be stored in NADH . In step 7, the newly formed high-energy phosphate bond in 1,3-bisphosphoglycerate is transferred to ADP by the enzyme phosphoglycerate kinase, forming a molecule of ATP and leaving a free carboxylic acid group on the oxidized sugar. The part of the molecule that undergoes a change is shaded in blue; the rest of the molecule remains unchanged throughout all these reactions. (B) Summary of the overall chemical change produced by the reactions of steps 6 and 7.

Glycolysis Illustrates How Enzymes Couple Oxidation to Energy Storage

The formation of ATP during glycolysis provides a particularly clear demonstration of how enzymes couple energetically unfavorable reactions with favorable ones, thereby driving the many chemical reactions that make life possible. Two central reactions in glycolysis (steps 6 and 7) convert the three-carbon sugar intermediate glyceraldehyde 3-phosphate (an aldehyde) into 3-phosphoglycerate (a carboxylic acid; see Panel 2–8, pp. 108–109), thus oxidizing an aldehyde group to a carboxylic acid group. This sugar oxidation process releases enough free energy to convert a molecule of ADP to ATP and to transfer two electrons (and a proton) from the aldehyde to NAD^+ to form NADH, while still liberating enough heat to the environment to make the overall reaction energetically favorable (ΔG° for the overall reaction is -12.5 kJ/mole).

Figure 2–47 details this remarkable feat of energy harvesting. The chemical reactions are precisely guided by two enzymes to which the sugar intermediates are tightly bound. The first enzyme (glyceraldehyde 3-phosphate dehydrogenase) forms a covalent bond to the sugar aldehyde group through a reactive $-\text{SH}$ group on the enzyme and then catalyzes the oxidation of this aldehyde group to a carboxylic acid. This creates a highly reactive enzyme–substrate bond that is displaced by an inorganic phosphate ion to produce a high-energy sugar phosphate intermediate. This intermediate (1,3-bisphosphoglycerate) then binds to the second enzyme (phosphoglycerate kinase), which catalyzes an energetically favorable transfer of its high-energy phosphate to ADP, forming ATP and completing the process of oxidizing a sugar aldehyde to a carboxylic acid.

We have shown this particular oxidation process in some detail because it provides a clear example of enzyme-mediated energy storage through coupled reactions (**Figure 2–48**). Steps 6 and 7 are the only reactions in glycolysis that create a high-energy phosphate linkage directly from inorganic phosphate. As such, they account for the net yield of two ATP molecules and two NADH molecules per molecule of glucose (see Panel 2–8, pp. 108–109).

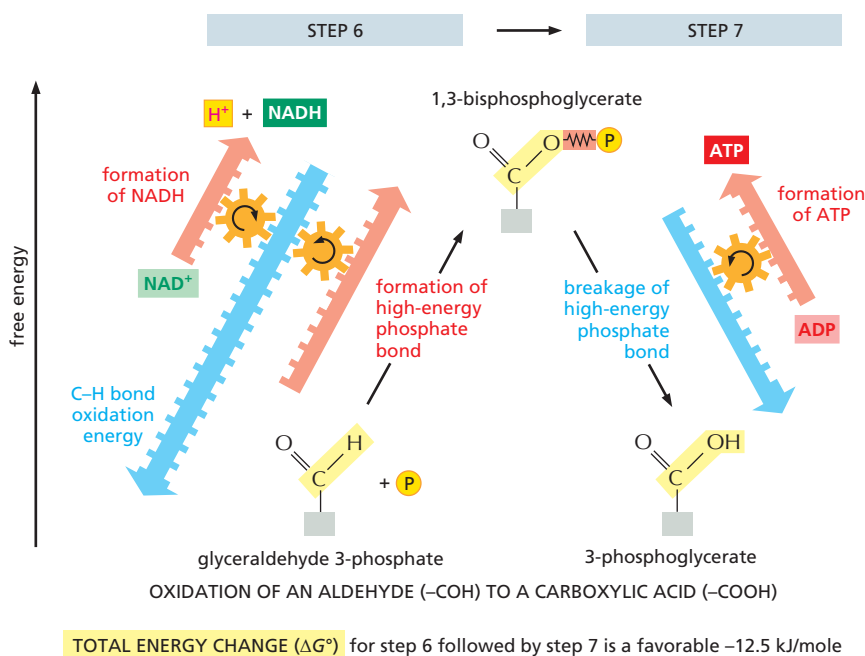
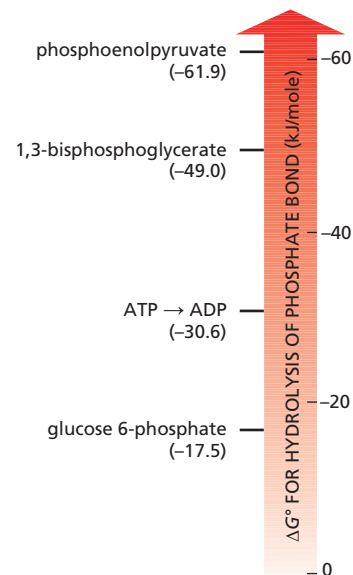


Figure 2–48 How a pair of coupled reactions drives the energetically unfavorable formation of NADH and ATP in steps 6 and 7 of glycolysis. In this diagram, energetically favorable reactions are represented by *blue arrows* and energetically costly reactions by *red arrows*. In step 6, the energy released by the energetically favorable oxidation of a C–H bond in glyceraldehyde 3-phosphate (*blue arrow*) is large enough to drive two energetically costly reactions: the formation of both NADH and a high-energy phosphate bond in 1,3-bisphosphoglycerate (*red arrows*). The subsequent energetically favorable breakage of that high-energy phosphate bond in step 7 then drives the formation of ATP.

Figure 2–49 Phosphate bonds with different hydrolysis energies. Examples of molecules formed during glycolysis that contain different types of phosphate bonds are shown, along with the standard free-energy change for hydrolysis of those bonds in kJ/mole. The transfer of a phosphate group from one molecule to another is energetically favorable if the standard free-energy change (ΔG°) for hydrolysis of the phosphate bond is more negative for the donor molecule than for the acceptor. (The hydrolysis reactions can be thought of as the transfer of the phosphate group to water.) Thus, a phosphate group is readily transferred from 1,3-bisphosphoglycerate to ADP to form ATP. Transfer reactions involving the phosphate groups in these molecules are detailed in Panel 2–8 (pp. 108–109). Note that standard conditions often do *not* pertain to living cells, where the relative concentrations of reactants and products will influence the actual change in free energy (see p. 67).



As we have just seen, ATP can be formed readily from ADP when a reaction intermediate is formed with a phosphate bond of higher energy than that of the terminal phosphate bond in ATP. Phosphate bonds can be ordered in energy by comparing the standard free-energy change (ΔG°) for the breakage of each bond by hydrolysis. **Figure 2–49** compares the high-energy phosphoanhydride bonds in ATP with the energy of other types of phosphate bonds that are generated during glycolysis.

Fermentations Produce ATP in the Absence of Oxygen

For most animal and plant cells, glycolysis is only a prelude to the final stage of the breakdown of food molecules. In these cells, the pyruvate formed by glycolysis is rapidly transported into the mitochondria, where it is converted into CO_2 plus acetyl CoA, whose acetyl group is then completely oxidized to CO_2 and H_2O .

In contrast, for many anaerobic organisms—which do not utilize molecular oxygen and can grow and divide without it—glycolysis is the principal source of the cell's ATP. Certain animal tissues, such as skeletal muscle, can also continue to function when molecular oxygen is limited. In these anaerobic conditions, the pyruvate and the NADH electrons stay in the cytosol. The pyruvate is converted into products excreted from the cell; for example, into ethanol and CO_2 in the yeasts used in brewing and breadmaking or into lactic acid (lactate) in muscle. In this process, the NADH gives up its electrons and is converted back into NAD^+ . This is crucial, because a regeneration of NAD^+ is required to maintain the reactions of glycolysis (**Figure 2–50**). Energy-yielding pathways like these are called **fermentations**.

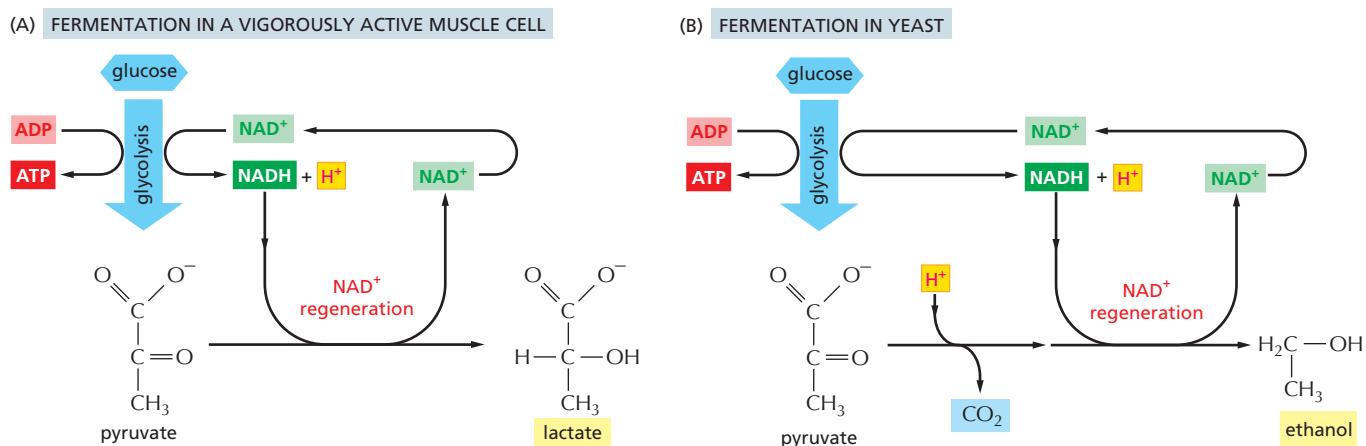


Figure 2–50 Two pathways for the anaerobic breakdown of pyruvate. (A) When there is inadequate oxygen in a muscle cell undergoing vigorous contraction, the pyruvate produced by glycolysis is converted to lactic acid (lactate) as shown. This reaction regenerates the NAD^+ required in step 6 of glycolysis, but the whole pathway yields much less energy overall than complete oxidation does. (B) In some organisms that can grow anaerobically, such as yeasts, pyruvate is converted via acetaldehyde into carbon dioxide and ethanol. Again, this pathway regenerates NAD^+ from NADH , as needed for glycolysis to continue. Both (A) and (B) are examples of *fermentations*.

Organisms Store Food Molecules in Special Reservoirs

All organisms need to maintain a high ATP/ADP ratio to maintain biological order in their cells. Yet animals have only periodic access to food, and plants need to survive overnight without sunlight, when they are unable to produce sugar from photosynthesis. For this reason, both plants and animals convert sugars and fats to special forms for storage.

To compensate for long periods of fasting, animals store fatty acids as **fat droplets** composed of water-insoluble *triacylglycerols* (also called triglycerides). The triacylglycerols in animals are mostly stored in the cytoplasm of specialized fat cells called **adipocytes** (Figure 2-51). For shorter-term storage, sugar is stored as glucose subunits in the large branched polysaccharide **glycogen**, which is present as small granules in the cytoplasm of many cells, with the largest stores in liver and muscle.

The synthesis and degradation of glycogen are rapidly regulated according to need. When cells need more ATP than they can generate from the food molecules taken in from the bloodstream, they break down glycogen in a reaction that produces glucose 1-phosphate, which is rapidly converted to glucose 6-phosphate for glycolysis (Figure 2-52). During fasting, liver cells release glucose derived from breakdown of their glycogen stores into the bloodstream for use by other cells, while muscle cells hoard their supplies for their own use.

Quantitatively, fat is far more important than glycogen as an energy store for animals, because it provides for more efficient storage. The oxidation of a gram of fat releases about twice as much energy as the oxidation of a gram of glycogen does. Moreover, glycogen differs from fat in binding a great deal of water, producing a sixfold difference in the actual mass of glycogen required to store the same amount of energy as fat. An average adult human stores enough glycogen for only about a day of normal activities but enough fat to last for nearly a month.

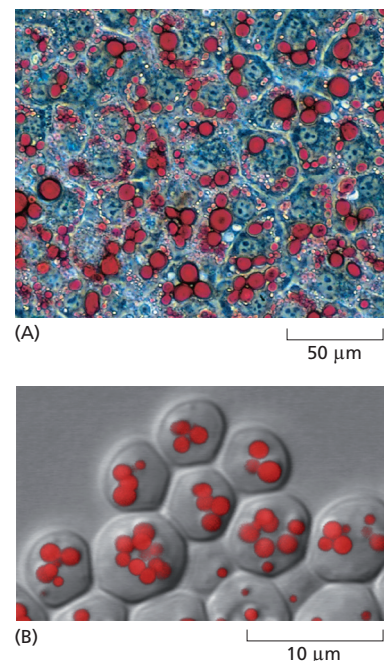


Figure 2-51 Fats stored in the form of lipid droplets in cells. (A) Fat droplets (stained red) in the cytoplasm of developing adipocytes. (B) Lipid droplets (red) in yeast cells, which also use them as a reservoir of energy and as building blocks for membrane lipid biosynthesis. (A, courtesy of Peter Tontonoz and Ronald M. Evans; B, courtesy of Sepp D. Kohlwein.)

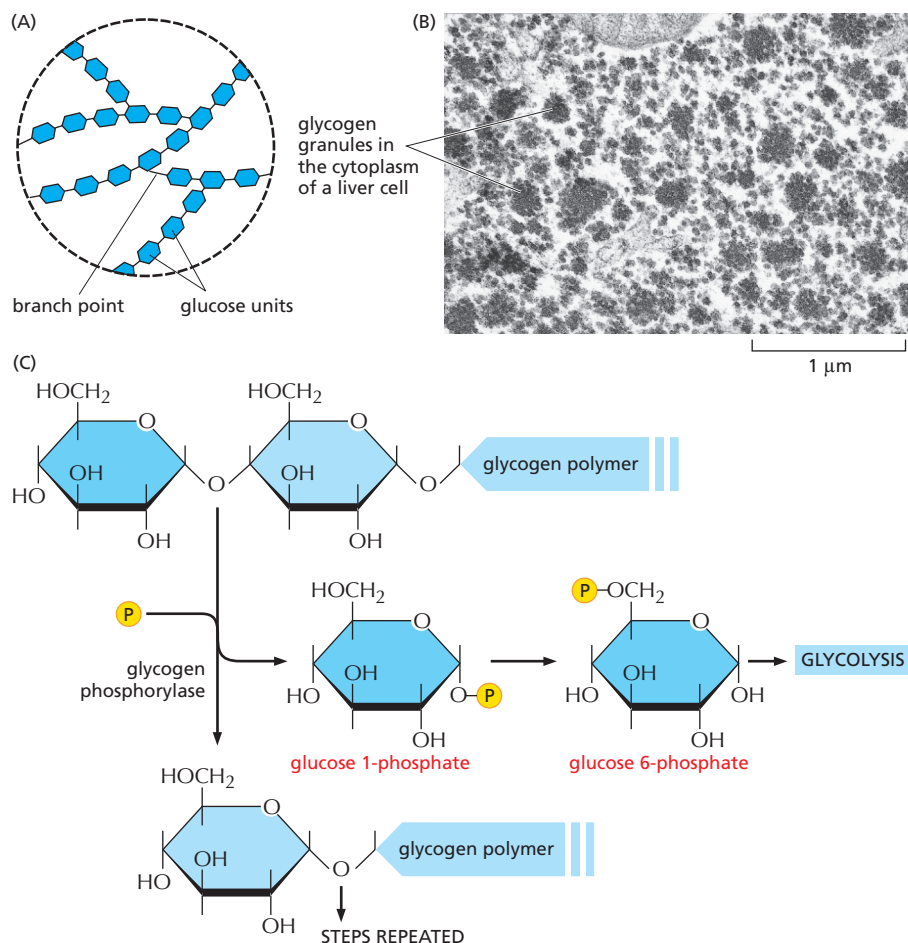


Figure 2-52 How animal cells store glucose in the form of glycogen to provide energy in times of need. (A) The structure of glycogen; starch in plants is a very similar branched polymer of glucose but has many fewer branch points. (B) An electron micrograph showing glycogen granules in the cytoplasm of a liver cell; each granule contains both glycogen and the enzymes required for glycogen synthesis and breakdown. (C) The enzyme glycogen phosphorylase breaks down glycogen when cells need more glucose. (B, courtesy of Robert Fletterick and Daniel S. Friend, by permission of E.L. Bearer.)

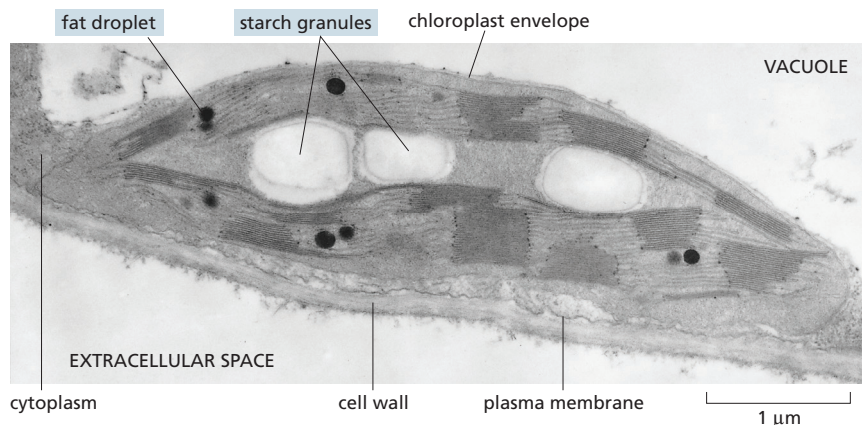


Figure 2–53 Plant cells store both starch and fats in their chloroplasts. An electron micrograph of a single chloroplast in a plant cell shows the starch granules and fat droplets that have been synthesized in the organelle. (Courtesy of K. Plaskitt.)

If our main fuel reservoir had to be carried as glycogen instead of fat, body weight would increase by an average of about 60 pounds.

How do plants store sugars and fats? Plants produce abundant amounts of both ATP and NADPH by the photosynthesis that is carried out in their chloroplasts. But this organelle is isolated from the rest of its plant cell by a membrane that is impermeable to both types of activated carrier molecules. Moreover, the plant contains many cells—such as those in the roots—that lack chloroplasts and therefore cannot produce their own sugars. Thus, sugars are exported from chloroplasts to the mitochondria present in all cells of the plant. Most of the ATP needed for general plant cell metabolism is synthesized in these mitochondria, using exactly the same pathways for the oxidative breakdown of sugars as in nonphotosynthetic organisms; this ATP is then passed to the rest of the cell.

During periods of excess photosynthetic capacity during the day, chloroplasts convert some of the sugars that they make into fats and into **starch**, a polymer of glucose analogous to the glycogen of animals. Both are stored inside the chloroplast until needed for energy-yielding oxidation during periods of darkness (**Figure 2–53**).

The embryos inside plant seeds must live on stored sources of energy for a prolonged period until they germinate and produce leaves that can harvest the energy in sunlight. For this reason, plant seeds often contain especially large amounts of fats and starch—which makes them a major food source for animals, including ourselves (**Figure 2–54**).

Between Meals, Most Animal Cells Derive Their Energy from Fatty Acids Obtained from Fat

After a meal, most of the energy that an animal needs is derived from sugars obtained from food. Excess sugars, if any, are used to replenish depleted glycogen stores or to synthesize fats as a stable longer-term food store. In the absence



Figure 2–54 Some plant seeds that serve as important foods for humans. Corn, nuts, and peas all contain rich stores of starch and fat that provide the young plant embryo in the seed with energy and building blocks for biosynthesis. (Courtesy of the John Innes Foundation.)

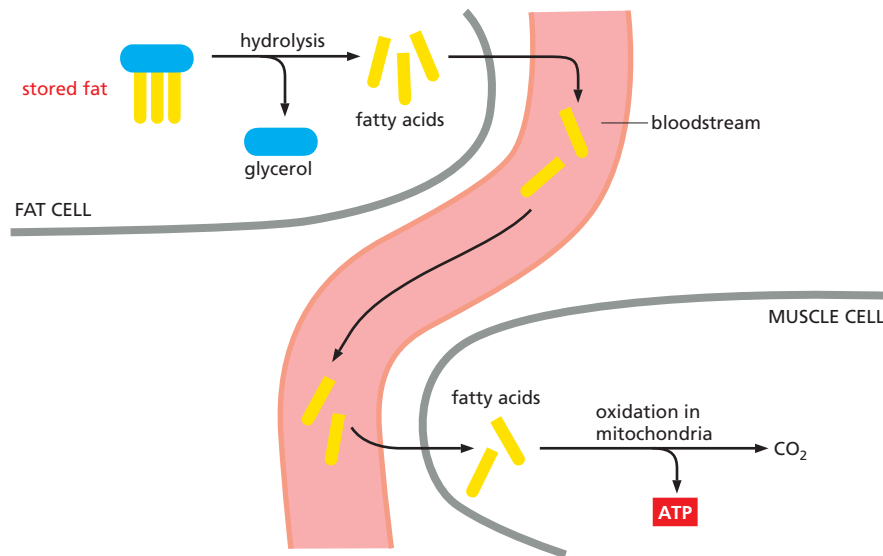


Figure 2–55 How stored fats are mobilized for energy production in animals. Low glucose levels in the blood trigger the hydrolysis of the triacylglycerol molecules in fat droplets to free fatty acids and glycerol. These fatty acids enter the bloodstream, where they bind to the abundant blood protein, serum albumin. Special fatty acid transporters in the plasma membrane of cells that oxidize fatty acids, such as muscle cells, then pass these fatty acids into the cytosol, from which they are moved into mitochondria for energy production.

of another meal, the liver begins to release glucose from its store of glycogen to maintain circulating glucose levels, and the fat stored in adipose tissue is also called into play. By the morning after an overnight fast, the glycogen reserves are exhausted, and fatty acid oxidation generates most of the ATP we need.

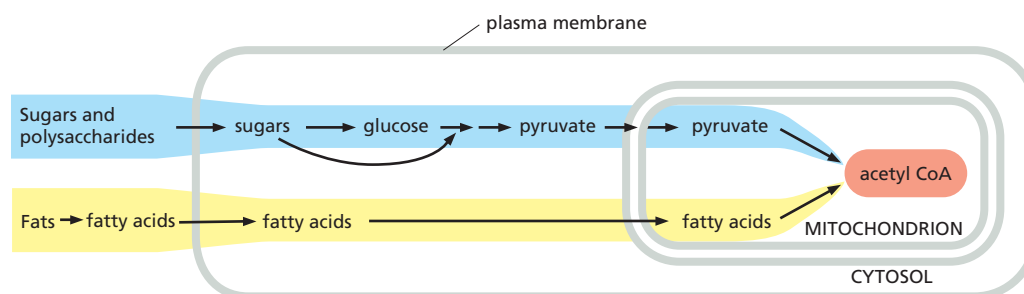
Low glucose levels in the blood trigger the breakdown of fats for energy production. As illustrated in **Figure 2–55**, the triacylglycerols stored in fat droplets in adipocytes are hydrolyzed to produce fatty acids and glycerol, and the fatty acids released are transferred to cells in the body through the bloodstream. Notably, the brain must rely on circulating glucose—or *ketone bodies* when available—because fatty acids are poorly utilized by the brain.

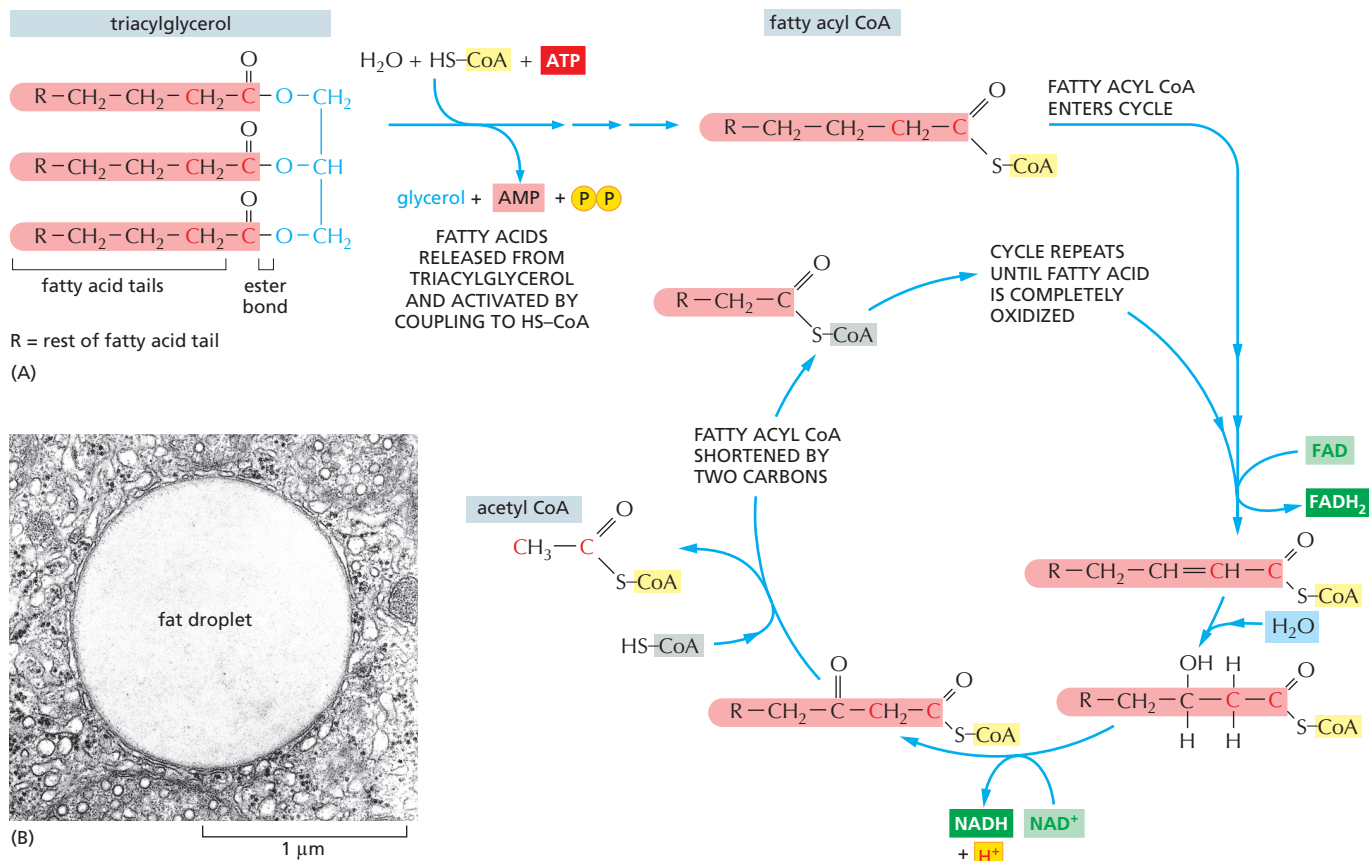
What are ketone bodies? During prolonged periods of fasting, when the circulating supply of glucose is mostly maintained by its synthesis from amino acids derived from the breakdown of proteins in muscle, the liver assumes its role as a central metabolic hub to convert fatty acids to the energy-rich molecules acetoacetate and β -hydroxybutyrate. These ketone bodies are released into the bloodstream to serve as an alternative fuel in heart and brain cells, where they are oxidized through the citric acid cycle to generate ATP. By thereby meeting most of the energy needs of the brain and heart, this process partially spares protein breakdown. Consumption of a diet very low in carbohydrate (a *ketogenic diet*) leads to the production of ketone bodies and can enable weight loss in most individuals. A spontaneous breakdown of acetoacetate to acetone generates the bad breath often associated both with such diets and with prolonged fasting.

Sugars and Fats Are Both Degraded to Acetyl CoA in Mitochondria

The fatty acids imported from the bloodstream are moved into mitochondria, where all of their oxidation takes place (**Figure 2–56**). Each molecule of fatty acid (as the activated molecule *fatty acyl CoA*) is broken down completely by a cycle

Figure 2–56 Pathways for the production of acetyl CoA from sugars and fats. The mitochondrion in eukaryotic cells is where acetyl CoA is produced from both types of food molecules. It is therefore the place where most of the cell's oxidation reactions occur and where most of its ATP is made. Amino acids (not shown) can also enter mitochondria to be converted there into acetyl CoA or another intermediate of the citric acid cycle. The structure and function of mitochondria are discussed in detail in Chapter 14.





of reactions that trims two carbons at a time from its carboxyl end, generating one molecule of acetyl CoA for each turn of the cycle. A molecule of NADH and a molecule of $FADH_2$ are also produced in this process (Figure 2-57).

As was shown in Figure 2-56, in eukaryotes the pyruvate that was produced by glycolysis from sugars in the cytosol is likewise transported into mitochondria, where it is rapidly decarboxylated by a giant complex of three enzymes, called the *pyruvate dehydrogenase complex*. The products are a molecule of CO_2 (a waste product), a molecule of NADH, and acetyl CoA (see Panel 2-9, pp. 110-111).

Sugars and fats are the major energy sources for most nonphotosynthetic organisms, including humans. Most of the useful energy that can be extracted from their oxidation remains stored in the acetyl CoA molecules that are produced from both sources. In the *citric acid cycle* of reactions, the acetyl group ($-COCH_3$) in acetyl CoA is completely oxidized to CO_2 and H_2O . This cycle, to be described next, therefore plays a critical, central role in the energy metabolism of aerobic organisms.

The Citric Acid Cycle Generates NADH by Oxidizing Acetyl Groups to CO_2

In the nineteenth century, biologists noticed that in the absence of air, cells produce lactic acid (for example, in muscle) or ethanol (for example, in yeast), while in its presence they consume O_2 and produce CO_2 and H_2O instead. Efforts to define the pathways of aerobic metabolism eventually focused on the oxidation of pyruvate and led in 1937 to the discovery of the **citric acid cycle**, also known as the *tricarboxylic acid cycle* or the *Krebs cycle*. The citric acid cycle accounts for about two-thirds of the total oxidation of carbon compounds in most cells, and its major end products are CO_2 and high-energy electrons in the form of NADH. The CO_2 is released as a waste product, while the high-energy electrons from NADH are passed to a membrane-bound

Figure 2-57 How the oxidation of fatty acids produces acetyl CoA. (A) Fats are stored in the form of triacylglycerol, the glycerol portion of which is shown in blue. Three fatty acid chains (shaded in red) are linked to this glycerol through ester bonds that can be hydrolyzed by enzymes called lipases to allow the fatty acids to enter the bloodstream (see Figure 2-55). Once the fatty acids enter mitochondria (see Figure 2-56), they are coupled to coenzyme A in a reaction requiring ATP. These activated fatty acids (fatty acyl CoA molecules) are then oxidized in a cycle containing four enzymes, which are not shown. As indicated, each turn of the cycle shortens the fatty acyl CoA molecule by two carbons (red) and generates one molecule each of the energy-rich molecules $FADH_2$, NADH, and acetyl CoA. (B) Fats are insoluble in water and spontaneously form large lipid droplets. This electron micrograph shows a lipid droplet in the cytoplasm of a specialized fat cell, an adipocyte. (B, courtesy of Daniel S. Friend and by permission of E.L. Bearer.)

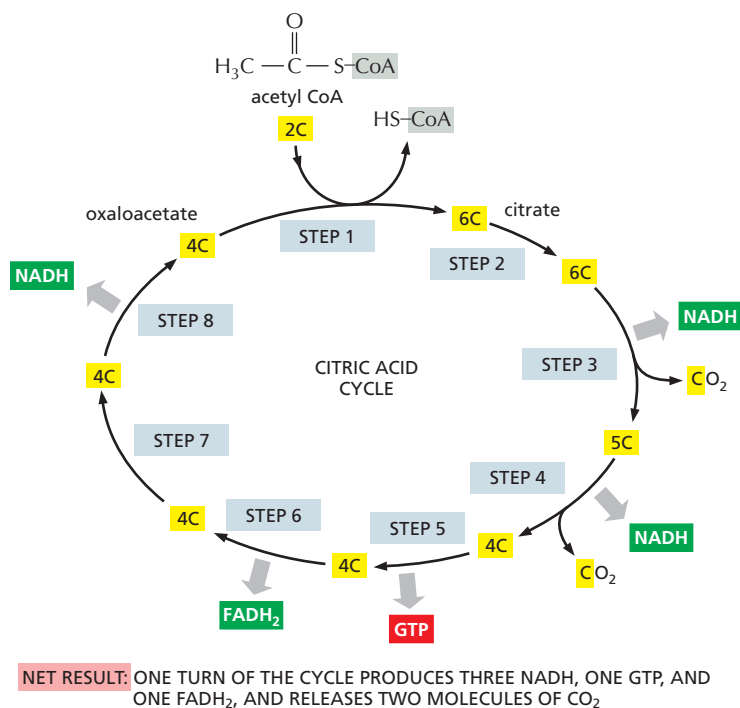


Figure 2-58 Simple overview of the citric acid cycle. The reaction of acetyl CoA with oxaloacetate starts the cycle by producing citrate (citric acid). In each turn of the cycle, two molecules of CO₂ are produced as waste products, plus three molecules of NADH, one molecule of GTP, and one molecule of FADH₂. The number of carbon atoms in each intermediate is shown in a yellow box. For details, see Panel 2-9 (pp. 110–111).

electron-transport chain that will be discussed in Chapter 14, eventually combining with O₂ to produce H₂O.

The citric acid cycle itself does not use gaseous O₂ (it uses oxygen atoms from H₂O). But the cycle does require O₂ in subsequent reactions to keep it going. This is because there is no other efficient way for the NADH to get rid of its electrons and thus regenerate the NAD⁺ that the cycle requires.

The citric acid cycle takes place inside mitochondria in eukaryotic cells. The process begins when the acetyl group is transferred from acetyl CoA to a four-carbon molecule, *oxaloacetate*, to form the six-carbon tricarboxylic acid, *citric acid*, for which the subsequent cycle of reactions is named. This citric acid molecule is then gradually oxidized, allowing the energy of oxidation to be harnessed to produce activated carrier molecules. The chain of eight reactions forms a cycle because at the end the oxaloacetate is regenerated to enter a new turn of the cycle, as shown in outline in [Figure 2-58](#).

We have thus far highlighted only one of the three types of activated carrier molecules that are produced by the citric acid cycle: NADH, the reduced form of the NAD⁺/NADH electron carrier system (see [Figure 2-36](#)). In addition to three molecules of NADH, each turn of the cycle also produces one molecule of FADH₂ from FAD (see [Figure 2-39](#)) and one molecule of GTP, guanosine triphosphate, from GDP. The structure of GTP is illustrated in [Figure 2-59](#). GTP is a close relative

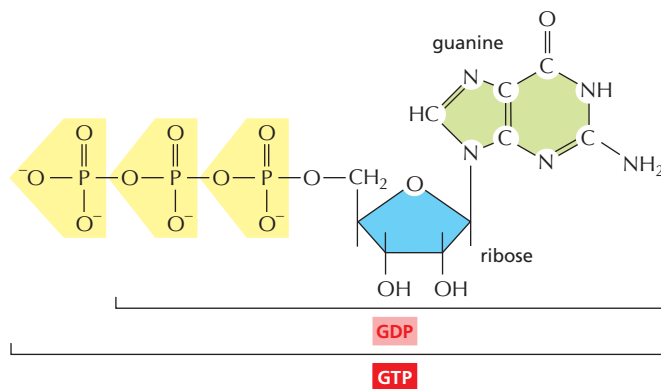


Figure 2-59 The structure of GTP, guanosine triphosphate. GTP and GDP are close relatives of ATP and ADP, respectively.

of ATP, and the transfer of its terminal phosphate group to ADP produces one ATP molecule in each cycle.

Panel 2-9 (pp. 110–111) and **Movie 2.6** present the complete citric acid cycle. Water, rather than molecular oxygen, supplies the extra oxygen atoms required to make CO_2 from the acetyl groups entering the citric acid cycle. As illustrated in the Panel, three molecules of water are taken up in each cycle, and the oxygen atoms of some of them are ultimately used to make CO_2 .

In addition to pyruvate and fatty acids, some amino acids pass from the cytosol into mitochondria, where they are also converted into acetyl CoA or one of the other intermediates of the citric acid cycle. As we discuss next, in mitochondria the large amount of energy stored in the electrons of NADH and FADH_2 is utilized for ATP production through the process of *oxidative phosphorylation*, which is the only step in the oxidative catabolism of foodstuffs that directly requires gaseous oxygen (O_2) from the atmosphere. Thus, in the eukaryotic cell, the mitochondrion is the center toward which all energy-yielding processes lead, whether they begin with sugars, fats, or proteins.

Electron Transport Drives the Synthesis of the Majority of the ATP in Most Cells

Most chemical energy is released in the last stage in the degradation of a food molecule. This process begins when NADH and FADH_2 transfer the electrons that they obtained by oxidizing food-derived organic molecules to an **electron-transport chain** embedded in the inner membrane of the mitochondrion. As the electrons pass along this long chain of specialized electron acceptor and donor molecules, they fall to successively lower energy states, being finally passed to molecules of oxygen gas (O_2) that have diffused into the mitochondrion, reducing the oxygen to produce water. The electrons have now reached a very low energy level, and all the available energy has been extracted from the oxidized food molecule.

The energy released by this chain of electron transfers is used to pump H^+ ions (protons) across the membrane—from the innermost mitochondrial compartment (the matrix) to the intermembrane space (and then to the cytosol). The resulting electrochemical proton gradient across the inner mitochondrial membrane serves as a major source of energy for cells, being tapped like a battery to drive a variety of energy-requiring reactions. The most prominent of these reactions is the generation of ATP by the phosphorylation of ADP, as part of a process known as **oxidative phosphorylation** (see Figure 14–12). This harnessing of chemical energy through membrane-based electron transfers is one of the most remarkable achievements of cell evolution—as such, we shall devote an entire chapter to it (Chapter 14).

Many Biosynthetic Pathways Begin with Glycolysis or the Citric Acid Cycle

Catabolic reactions, such as those of glycolysis and the citric acid cycle, produce both energy for the cell and the building blocks from which many other organic molecules are made. Thus far, we have emphasized energy production rather than the provision of starting materials for biosynthesis. But many of the intermediates formed in glycolysis and the citric acid cycle are siphoned off by such anabolic pathways, in which the intermediates are converted by a series of enzyme-catalyzed reactions into amino acids, nucleotides, lipids, and other small organic molecules that the cell needs. The oxaloacetate and α -ketoglutarate produced during the citric acid cycle, for example (see Panel 2-9, pp. 110–111), are transferred from the mitochondrial matrix back to the cytosol, where they serve as precursors for the production of many essential molecules, such as the amino acids aspartate and glutamate, respectively. An idea of the extent of these anabolic pathways can be gathered from

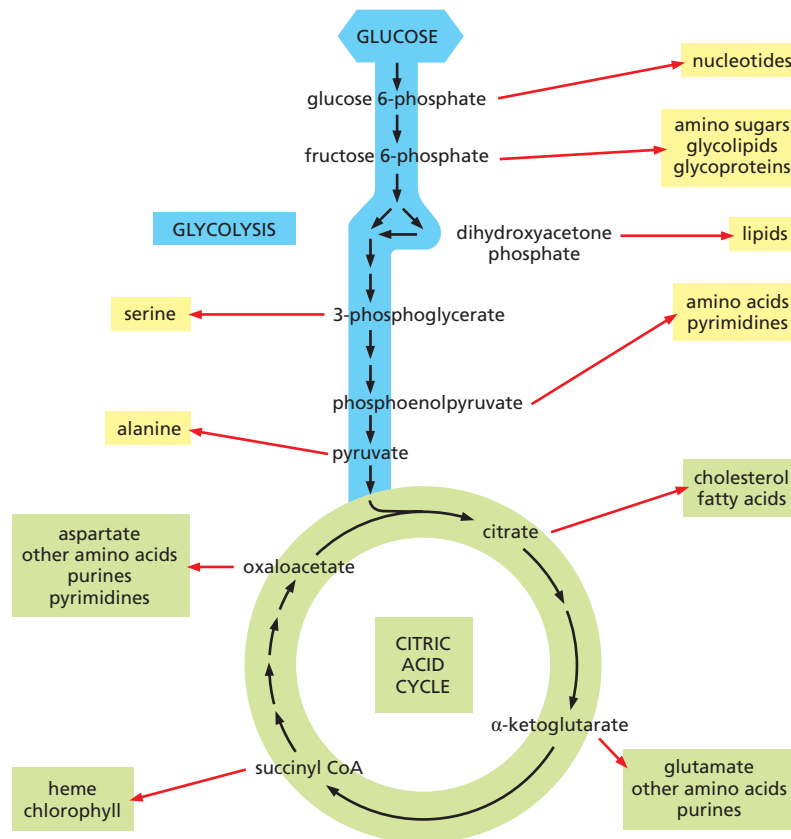


Figure 2-60 Glycolysis and the citric acid cycle provide the precursors needed to synthesize many important biological molecules. The amino acids, nucleotides, lipids, sugars, and other molecules—shown here as products—in turn serve as the precursors for the many macromolecules of the cell. Each *black arrow* in this diagram denotes a single enzyme-catalyzed reaction; the *red arrows* generally represent pathways with many steps that are required to produce the indicated products.

Figure 2-60, which illustrates some of the branches leading from the central catabolic reactions to biosyntheses.

Animals Must Obtain All the Nitrogen and Sulfur They Need from Food

So far we have concentrated mainly on carbohydrate metabolism and have not yet considered the metabolism of nitrogen or sulfur. These two elements are important constituents of biological macromolecules. Nitrogen and sulfur atoms pass from compound to compound and between organisms and their environment in a series of reversible cycles.

Although molecular nitrogen constitutes nearly 80% of Earth's atmosphere, nitrogen is chemically unreactive as a gas. Only a few living species are able to incorporate it into organic molecules, a process called **nitrogen fixation**. Nitrogen fixation occurs both in certain microorganisms and by some geophysical processes, such as lightning discharge. It is essential to the biosphere as a whole, for without it life could not exist on this planet. Only a small fraction of the nitrogenous compounds in today's organisms, however, is due to fresh products of nitrogen fixation from the atmosphere. Most organic nitrogen has been in circulation for some time, passing from one living organism to another. Thus, present-day nitrogen-fixing reactions can be said to perform a "topping-up" function for the total nitrogen supply.

Vertebrates receive virtually all of their nitrogen from their dietary intake of proteins and nucleic acids. In the body, these macromolecules are broken down to amino acids and the components of nucleotides, and the nitrogen they contain is used to produce new proteins and nucleic acids—or other molecules. About

half of the 20 amino acids found in proteins are essential amino acids for vertebrates (Figure 2–61), which means that they cannot be synthesized from other ingredients of the diet. The other amino acids can be so synthesized, using a variety of raw materials that include the intermediates of the citric acid cycle just described. The essential amino acids are made by plants and other organisms, usually by long and energetically expensive pathways that have been lost in the course of vertebrate evolution.

The nucleotides needed to make RNA and DNA are synthesized using specialized biosynthetic pathways. All of the nitrogens in the purine and pyrimidine bases (as well as some of the carbons) are derived from the plentiful amino acids glutamine, aspartic acid, and glycine, whereas the ribose and deoxyribose sugars are derived from glucose. There are no “essential nucleotides” that must be provided in the diet.

As we have seen, the amino acids derived from food that are not used in biosynthesis can be oxidized to generate metabolic energy. Most of their carbon and hydrogen atoms eventually form CO_2 or H_2O , whereas their nitrogen atoms are shuttled through various forms and eventually appear as urea, which is excreted. Each amino acid is processed differently, and a whole constellation of enzymatic reactions exists for their catabolism.

Sulfur is abundant on Earth in its most oxidized form, sulfate (SO_4^{2-}). To be useful for life, sulfate must be reduced to sulfide (S^{2-}), the oxidation state of sulfur required for the synthesis of essential biological molecules, including the amino acids methionine and cysteine, coenzyme A, and the iron–sulfur centers essential for electron transport. The sulfur-reduction process begins in bacteria, fungi, and plants, where a special group of enzymes use ATP and reducing power to create a sulfate assimilation pathway. Humans and other animals cannot reduce sulfate and must therefore acquire the sulfur they need for their metabolism in the food that they eat.

Metabolism Is Highly Organized and Regulated

One can get a sense of the intricacy of a cell as a chemical machine from the relation of glycolysis and the citric acid cycle to the other metabolic pathways sketched out in Figure 2–62. This diagram represents only some of the enzymatic pathways in a human cell. It is obvious that our discussion of cell metabolism has dealt with only a tiny fraction of the broad field of cell chemistry.

All these reactions occur in a cell that is less than 0.1 mm in diameter, and each requires a different enzyme. The same molecule can often be part of many different pathways. Pyruvate, for example, is a substrate for half a dozen or more different enzymes, each of which modifies it chemically in a different way. One enzyme converts pyruvate to acetyl CoA, another to oxaloacetate; a third enzyme changes pyruvate to the amino acid alanine, a fourth to lactate, and so on. All of these different pathways compete for the same pyruvate molecule, and similar competitions for thousands of other small molecules go on at the same time.

The situation is further complicated in a multicellular organism. Different cell types require somewhat different sets of enzymes. And different tissues make distinct contributions to the chemistry of the organism as a whole. All types of cells have their distinctive metabolic traits, and they cooperate extensively in the normal state, as well as in response to stress and starvation. One might think that the whole system would need to be so finely balanced that any minor upset, such as a temporary change in dietary intake, would be disastrous.

In fact, the metabolic balance of a cell is amazingly stable. Whenever the balance is perturbed, the cell reacts so as to restore the initial state. The cell can adapt and continue to function during starvation or disease. Mutations of many kinds can damage or even eliminate particular reaction pathways, and yet—provided that certain minimum requirements are met—the cell survives. It does so because an elaborate network of *control mechanisms* regulates and coordinates the rates of all of its reactions. These controls rest, ultimately, on the remarkable abilities

THE ESSENTIAL AMINO ACIDS

THREONINE
METHIONINE
LYSINE
VALINE
LEUCINE
ISOLEUCINE
HISTIDINE
PHENYLALANINE
TRYPTOPHAN

Figure 2–61 The nine essential amino acids. These cannot be synthesized by human cells and so must be supplied in the diet.



of proteins to change their shape and their chemistry in response to changes in their immediate environment.

The principles that underlie how proteins are built and the chemistry behind their regulation are clearly central to all of biology. And it is proteins that will be our next concern.

Summary

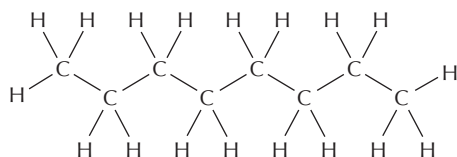
Food molecules are broken down by controlled stepwise oxidation to provide chemical energy in the form of ATP and NADH. Following the breakdown of large food molecules to their simple subunits, three main sets of reactions act in series, the products of each being the starting material for the next: glycolysis (which occurs in the cytosol), the citric acid cycle (in the mitochondrial matrix), and oxidative phosphorylation (on the inner mitochondrial membrane). The intermediate products of glycolysis and the citric acid cycle are used both as sources of metabolic energy and to produce many of the small molecules used as the raw materials for biosynthesis. Cells store sugar molecules as glycogen in animals and starch in plants; both plants and animals also use fats extensively as a food store. These storage materials in turn serve as a major source of food for humans, along with the proteins that comprise the majority of the dry mass of most of the cells in the foods we eat.

Figure 2–62 Glycolysis and the citric acid cycle are at the center of an elaborate set of metabolic pathways in human cells. Some 2000 metabolic reactions are shown schematically with the reactions of glycolysis and the citric acid cycle in red. Many other reactions either lead into these two central pathways—delivering small molecules to be catabolized with production of energy—or they lead outward as in Figure 2–60 to supply carbon compounds for the purpose of biosynthesis. (Adapted from KEGG Database. With permission from Kanehisa Laboratories.)

CARBON SKELETONS

Carbon has a unique role in the cell because of its ability to form strong covalent bonds with other carbon atoms. Thus carbon atoms can join to form:

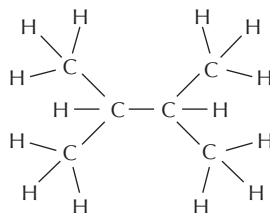
chains



also written as



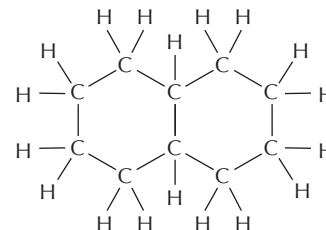
branched trees



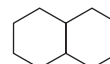
also written as



rings



also written as



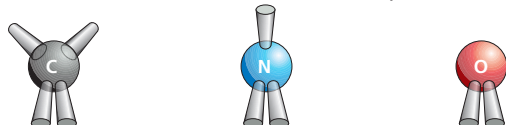
COVALENT BONDS

A covalent bond forms when two atoms come very close together and share one or more of their outer-shell electrons. Each atom forms a fixed number of covalent bonds in a defined spatial arrangement.

SINGLE BONDS: two electrons shared per bond

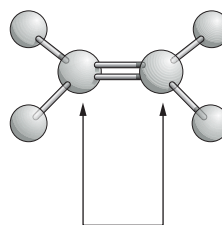


DOUBLE BONDS: four electrons shared per bond



The precise spatial arrangement of covalent bonds influences the three-dimensional structure and chemistry of molecules.

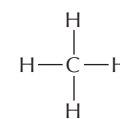
In this review Panel, we see how covalent bonds are used in a variety of biological molecules.



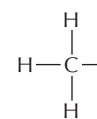
Atoms joined by two or more covalent bonds cannot rotate freely around the bond axis. This restriction has a major influence on the three-dimensional shape of many macromolecules.

C-H COMPOUNDS

Carbon and hydrogen together make stable compounds (or groups) called **hydrocarbons**. These are nonpolar, do not form hydrogen bonds, and are generally insoluble in water.



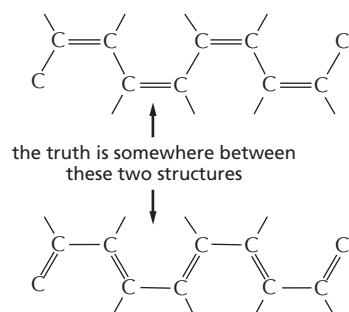
methane



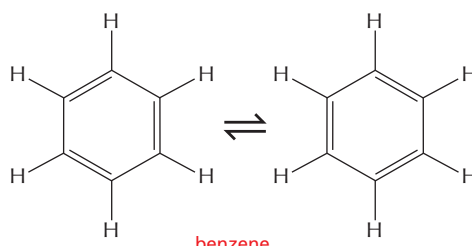
methyl group

ALTERNATING DOUBLE BONDS

A carbon chain can include double bonds. If these are on alternate carbon atoms, the bonding electrons move within the molecule, stabilizing the structure by a phenomenon called **resonance**.

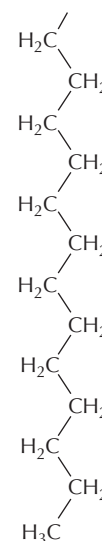


Alternating double bonds in a ring can generate a very stable structure.



benzene

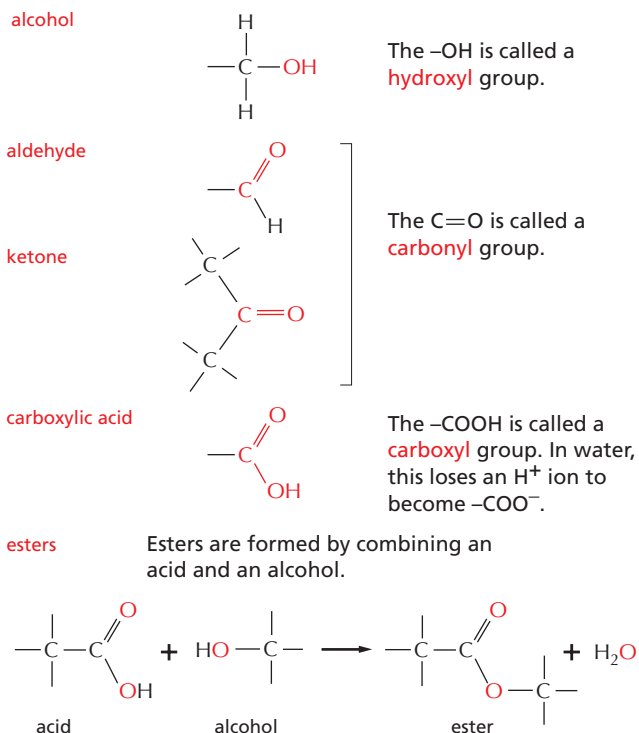
often written as



part of the hydrocarbon "tail" of a fatty acid molecule

C-O COMPOUNDS

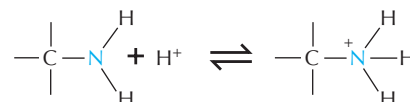
Many biological compounds contain a carbon covalently bonded to an oxygen. For example,



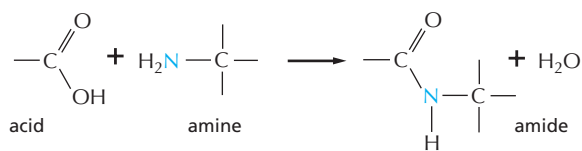
C-N COMPOUNDS

Amines and amides are two important examples of compounds containing a carbon linked to a nitrogen.

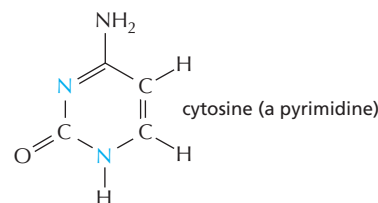
Amines in water combine with an H^+ ion to become positively charged.



Amides are formed by combining an acid and an amine. Unlike amines, amides are uncharged in water. An example is the peptide bond that joins amino acids in a protein.



Nitrogen also occurs in several ring compounds, including important constituents of nucleic acids: purines and pyrimidines.

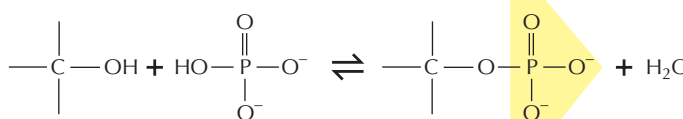
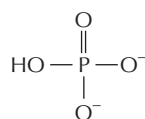


SULFHYDRYL GROUP The $-\text{C}-\text{SH}$ is called a sulfhydryl group. In the amino acid cysteine, the sulfhydryl group may exist in the reduced form $-\text{C}-\text{SH}$ or more rarely in an oxidized, cross-bridging form $-\text{C}-\text{S}-\text{S}-\text{C}-$

PHOSPHATES

Inorganic phosphate is a stable ion formed from phosphoric acid, H_3PO_4 . It is also written as **P**.

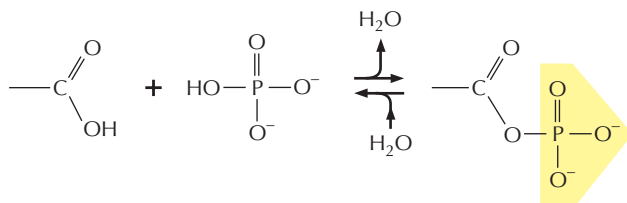
Phosphate esters can form between a phosphate and a free hydroxyl group. **Phosphate groups** are often covalently attached to proteins in this way.



also written as

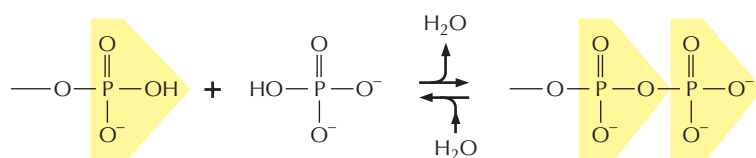
$$\begin{array}{c} | \\ -\text{C}-\text{O}-\text{P} \\ | \\ \text{O}^- \end{array}$$

The combination of a phosphate and a carboxyl group, or two or more phosphate groups, produces an acid anhydride. Because compounds of this type release a large amount of free energy when the bond is broken by hydrolysis in the cell, they are often said to contain a **high-energy** bond.



high-energy acyl phosphate bond (carboxylic-phosphoric acid anhydride) found in some metabolites

also written as

$$\begin{array}{c} \text{O} \\ || \\ -\text{C}-\text{O}-\text{P} \\ | \\ \text{O}^- \end{array}$$


high-energy phosphoanhydride bond found in molecules such as ATP

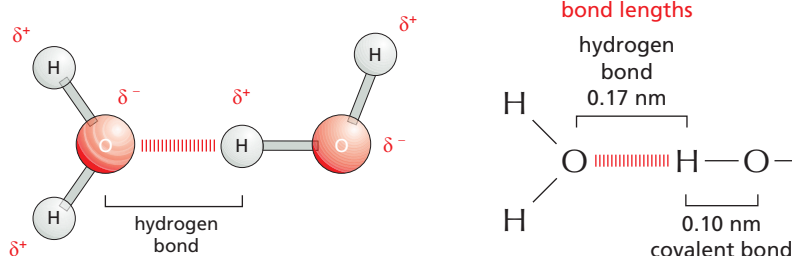
also written as

$$-\text{O}-\text{P}-\text{P}-\text{O}^-$$

HYDROGEN BONDS

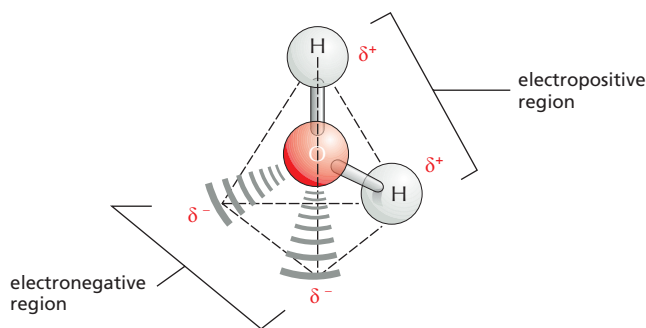
Because they are polarized, two adjacent H_2O molecules can form a noncovalent linkage known as a **hydrogen bond**. Hydrogen bonds have only about 1/20 the strength of a covalent bond.

Hydrogen bonds are strongest when the three atoms lie in a straight line.



WATER

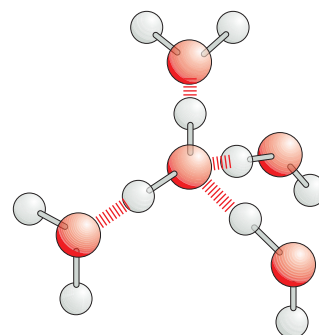
Two atoms connected by a covalent bond may exert different attractions for the electrons of the bond. In such cases, the bond is **polar**, with one end slightly negatively charged (δ^-) and the other slightly positively charged (δ^+).



Although a water molecule has an overall neutral charge (having the same number of electrons and protons), the electrons are asymmetrically distributed, making the molecule polar. The oxygen nucleus draws electrons away from the hydrogen nuclei, leaving the hydrogen nuclei with a small net positive charge. The excess of electron density on the oxygen atom creates weakly negative regions at the other two corners of an imaginary tetrahedron. **On these pages, we review the chemical properties of water and see how water influences the behavior of biological molecules.**

WATER STRUCTURE

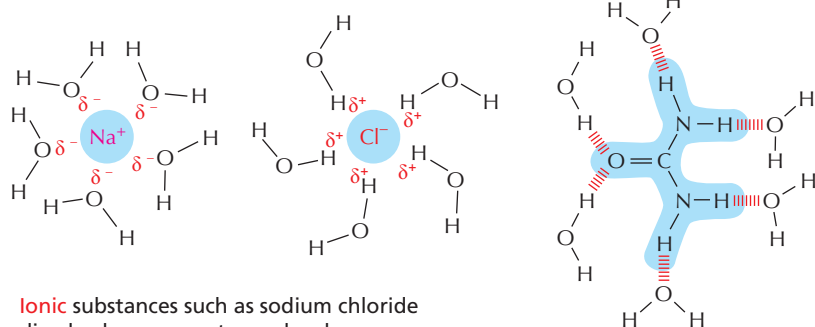
Molecules of water join together transiently in a hydrogen-bonded lattice. Even at 37°C , 15% of the water molecules are joined to four others in a short-lived assembly known as a *flickering cluster*.



The cohesive nature of water is responsible for many of its unusual properties, such as high surface tension, high specific heat capacity, and high heat of vaporization.

HYDROPHILIC MOLECULES

Substances that dissolve readily in water are termed **hydrophilic**. They include ions and polar molecules that attract water molecules through electrical charge effects. Water molecules surround each ion or polar molecule and carry it into solution.

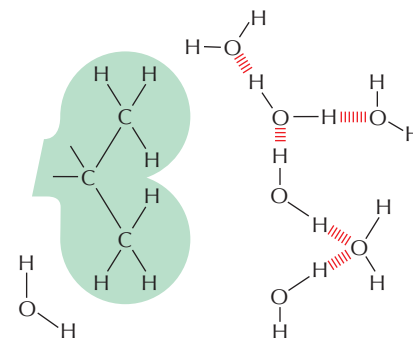


Ionic substances such as sodium chloride dissolve because water molecules are attracted to the positive (Na^+) or negative (Cl^-) charge of each ion.

Polar substances such as urea dissolve because their molecules form hydrogen bonds with the surrounding water molecules.

HYDROPHOBIC MOLECULES

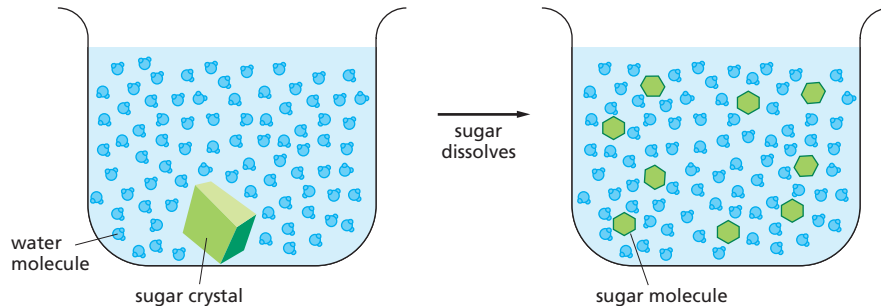
Substances that contain a preponderance of nonpolar bonds are usually insoluble in water and are termed **hydrophobic**. Water molecules are not attracted to such hydrophobic molecules and so have little tendency to surround them and bring them into solution.



Hydrocarbons, which contain many C-H bonds, are especially hydrophobic.

WATER AS A SOLVENT

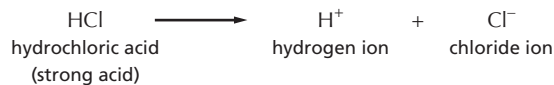
Many substances, such as household sugar (sucrose), **dissolve** in water. That is, their molecules separate from each other, each becoming surrounded by water molecules.



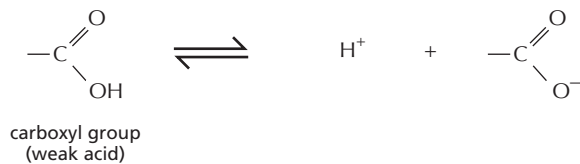
When a substance dissolves in a liquid, the mixture is termed a **solution**. The dissolved substance (in this case sugar) is the **solute**, and the liquid that does the dissolving (in this case water) is the **solvent**. Water is an excellent solvent for hydrophilic substances because of its polar bonds.

ACIDS

Substances that release hydrogen ions (protons) into solution are called **acids**.



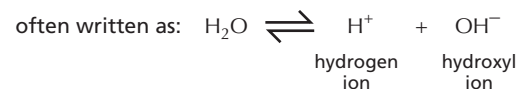
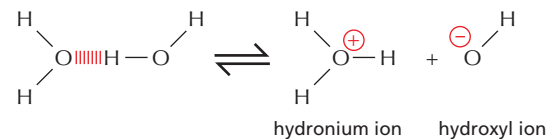
Many of the acids important in the cell are not completely dissociated, and they are therefore **weak acids**; for example, the carboxyl group ($-\text{COOH}$), which dissociates to give a hydrogen ion in solution.



Note that this is a reversible reaction.

HYDROGEN ION EXCHANGE

Positively charged hydrogen ions (H^+) can spontaneously move from one water molecule to another, thereby creating two ionic species.



Because the process is rapidly reversible, hydrogen ions are continually shuttling between water molecules. Pure water contains equal concentrations of hydronium ions and hydroxyl ions (both 10^{-7} M).

pH

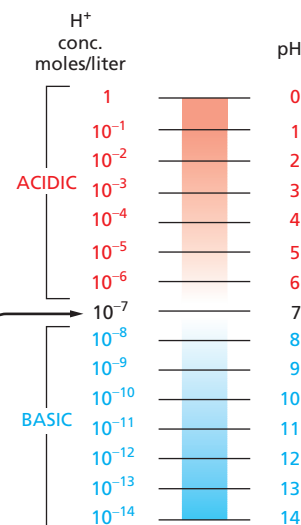
The acidity of a solution is defined by the concentration (conc.) of hydronium ions (H_3O^+) it possesses, generally abbreviated as H^+ . For convenience, we use the pH scale, where

$$\text{pH} = -\log_{10}[\text{H}^+]$$

For pure water

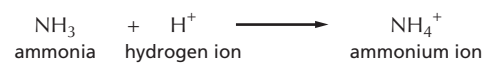
$$[\text{H}^+] = 10^{-7} \text{ moles/liter}$$

$$\text{pH} = 7.0$$

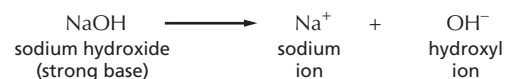


BASES

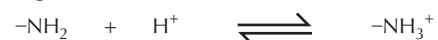
Substances that reduce the number of hydrogen ions in solution are called **bases**. Some bases, such as ammonia, combine directly with hydrogen ions.



Other bases, such as sodium hydroxide, reduce the number of H^+ ions indirectly, by producing OH^- ions that then combine directly with H^+ ions to make H_2O .

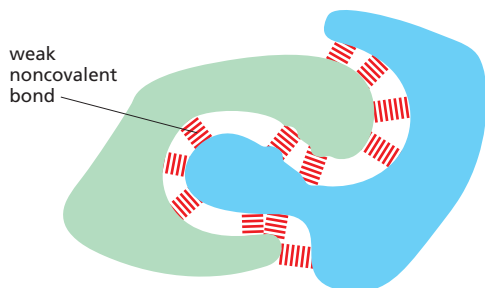


Many bases found in cells are partially associated with H^+ ions and are termed **weak bases**. This is true of compounds that contain an amino group ($-\text{NH}_2$), which has a weak tendency to reversibly accept an H^+ ion from water, thereby increasing the concentration of free OH^- ions.



WEAK NONCOVALENT CHEMICAL BONDS

Organic molecules can interact with other molecules through three types of short-range attractive forces known as *noncovalent bonds*: van der Waals attractions, electrostatic attractions, and hydrogen bonds. The repulsion of hydrophobic groups from water is also important for these interactions and for the folding of biological macromolecules.



Weak noncovalent bonds have less than 1/20 the strength of a strong covalent bond. They are strong enough to provide tight binding only when many of them are formed simultaneously.

HYDROGEN BONDS

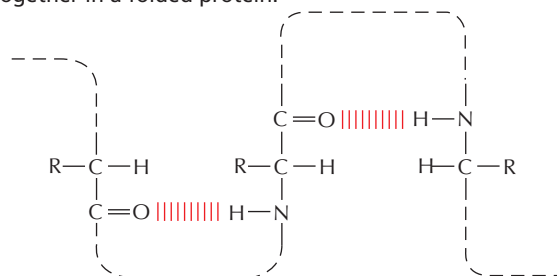
As already described for water (see Panel 2-2, pp. 96-97), **hydrogen bonds** form when a hydrogen atom is "sandwiched" between two electron-attracting atoms (usually oxygen or nitrogen).

Hydrogen bonds are strongest when the three atoms are in a straight line:

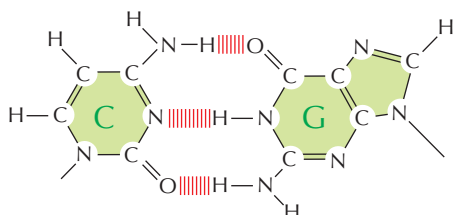


Examples in macromolecules:

Amino acids in a polypeptide chain can be hydrogen-bonded together in a folded protein.

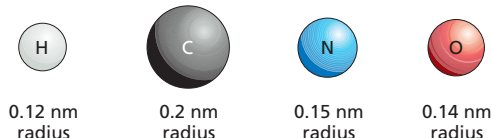


Two bases, G and C, are hydrogen bonded in a DNA double helix.



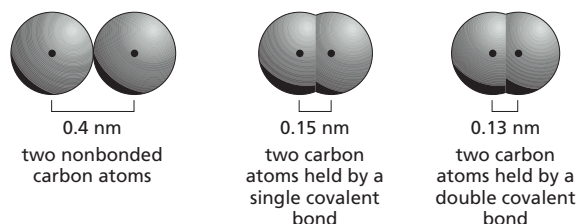
VAN DER WAALS ATTRACTIONS

If two atoms are too close together, they repel each other very strongly. For this reason, an atom can often be treated as a sphere with a fixed radius. The characteristic "size" for each atom is specified by a unique **van der Waals radius**. The contact distance between any two noncovalently bonded atoms is the sum of their van der Waals radii.



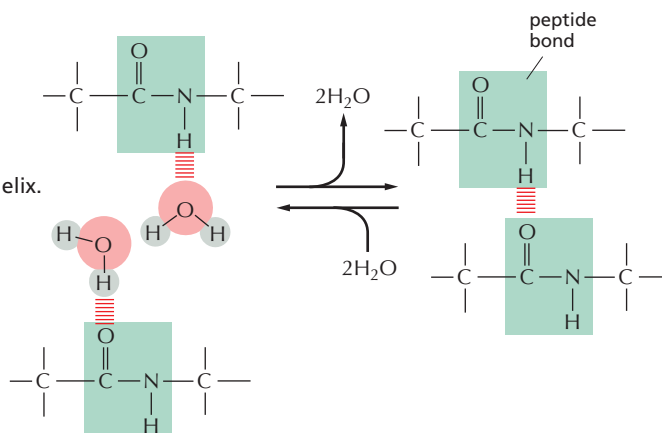
At very short distances, any two atoms show a weak bonding interaction due to their fluctuating electrical charges. The two atoms will be attracted to each other in this way until the distance between their nuclei is approximately equal to the sum of their van der Waals radii. Although they are individually very weak, such **van der Waals attractions** can become important when two macromolecular surfaces fit together very closely, because many atoms are involved.

Note that when two atoms form a covalent bond, the centers of the two atoms (the two atomic nuclei) are much closer together than the sum of the two van der Waals radii. Thus,



HYDROGEN BONDS IN WATER

Any two atoms that can form hydrogen bonds to each other can alternatively form hydrogen bonds to water molecules. Because of this competition with water molecules, the hydrogen bonds formed in water between two peptide bonds, for example, are relatively weak.



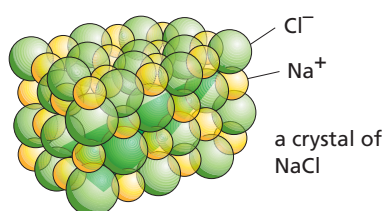
ELECTROSTATIC ATTRACTIONS

Electrostatic attractions occur both between fully charged groups (ionic bond) and between partially charged groups on polar molecules.



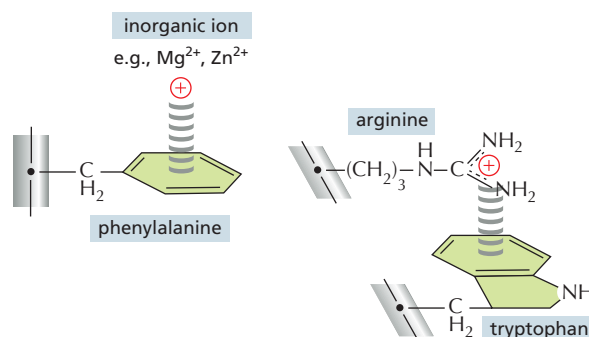
The force of attraction between the two partial charges, δ^+ and δ^- , falls off rapidly as the distance between the charges increases.

In the absence of water, ionic bonds are very strong. They are responsible for the strength of such minerals as marble and agate and for crystal formation in common table salt, NaCl.

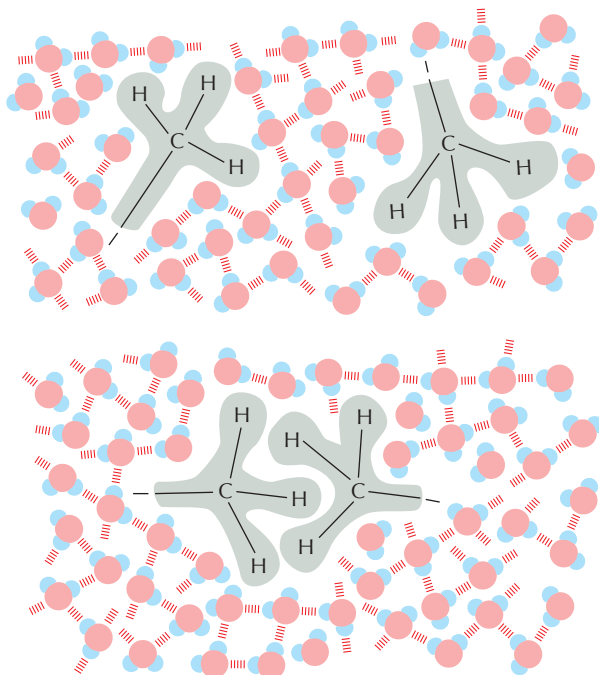


CATION- π INTERACTIONS

The structure of a typical protein reveals that it contains several cation- π (π) interactions, these electrostatic attractions being about half as abundant as the electrostatic attractions between positively and negatively charged amino-acid side chains. In this energetically favorable interaction, a cation is paired with the aromatic (π) electrons of either a tryptophan, tyrosine, or phenylalanine side chain. As shown, the cation can either be an inorganic ion or a positively charged lysine or arginine side chain. A tryptophan-arginine pair is the most common, as illustrated here.



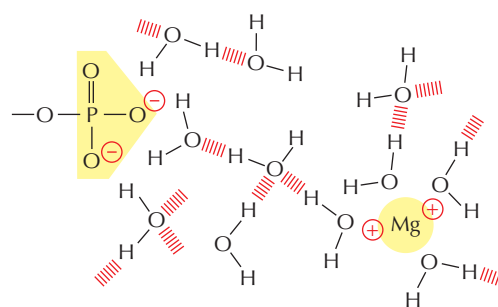
HYDROPHOBIC FORCES



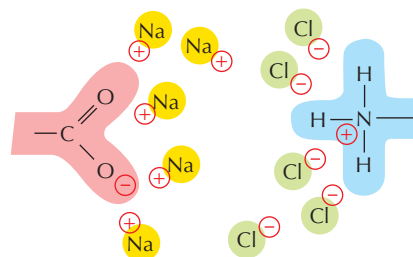
Water forces hydrophobic groups together in order to minimize their disruptive effects on the water network formed by the hydrogen bonds between water molecules. Hydrophobic groups held together in this way are sometimes said to be held together by "hydrophobic bonds," even though the attraction is actually caused by a repulsion from water.

ELECTROSTATIC ATTRACTIONS IN WATER

Charged groups are shielded by their interactions with water molecules. Electrostatic attractions are therefore quite weak in water.



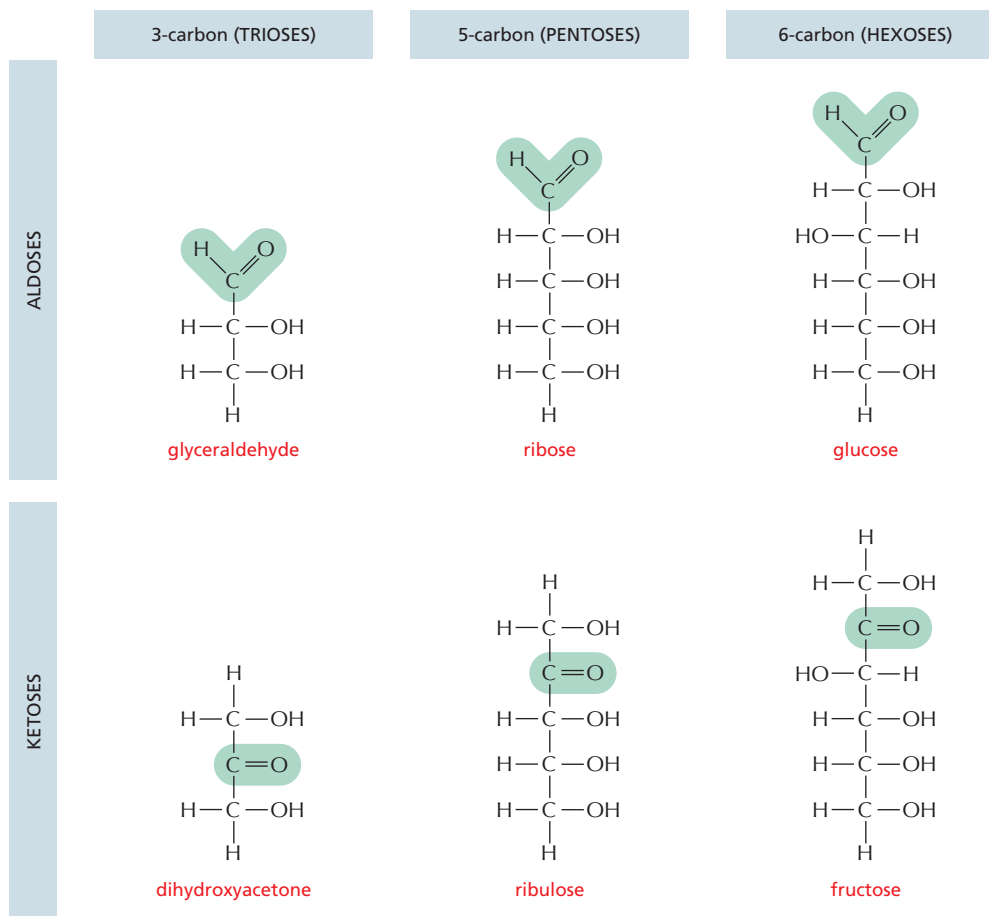
Inorganic ions in solution can also cluster around charged groups and further weaken these electrostatic attractions.



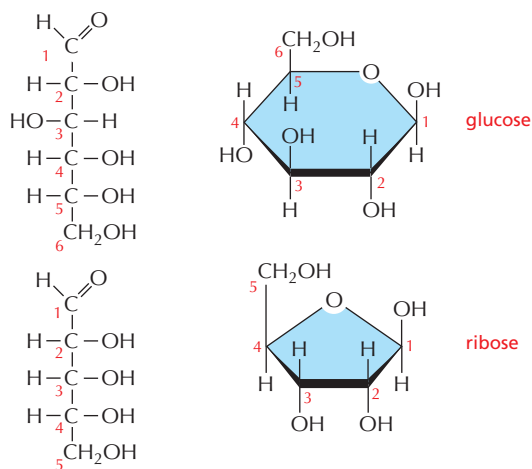
Despite being weakened by water and inorganic ions, electrostatic attractions are very important in biological systems.

MONOSACCHARIDES

Monosaccharides usually have the general formula $(\text{CH}_2\text{O})_n$, where n can be 3, 4, 5, 6, 7, or 8, and have two or more hydroxyl groups. They contain either an aldehyde group ($-\text{C}(=\text{O})\text{H}$) and are called aldoses or a ketone group ($>\text{C}=\text{O}$) and are called ketoses.

**RING FORMATION**

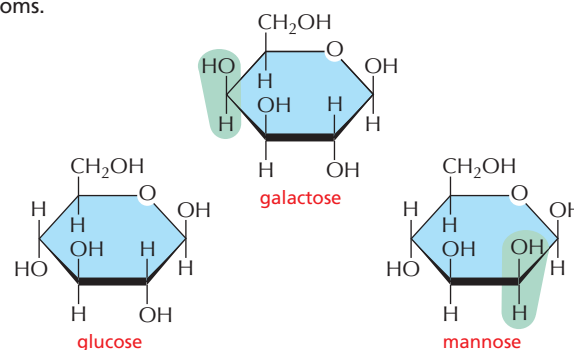
In aqueous solution, the aldehyde or ketone group of a sugar molecule tends to react with a hydroxyl group of the same molecule, thereby closing the molecule into a ring.



Note that each carbon atom has a number.

ISOMERS

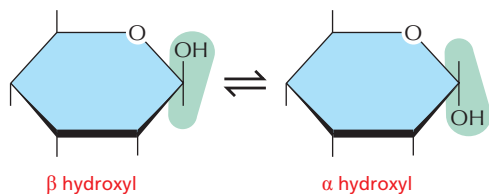
Many monosaccharides differ only in the spatial arrangement of atoms; that is, they are **isomers**. For example, glucose, galactose, and mannose have the same formula ($\text{C}_6\text{H}_{12}\text{O}_6$) but differ in the arrangement of groups around one or two carbon atoms.



These small differences make only minor changes in the chemical properties of the sugars. But the differences are recognized by enzymes and other proteins and therefore can have major biological effects.

α AND β LINKS

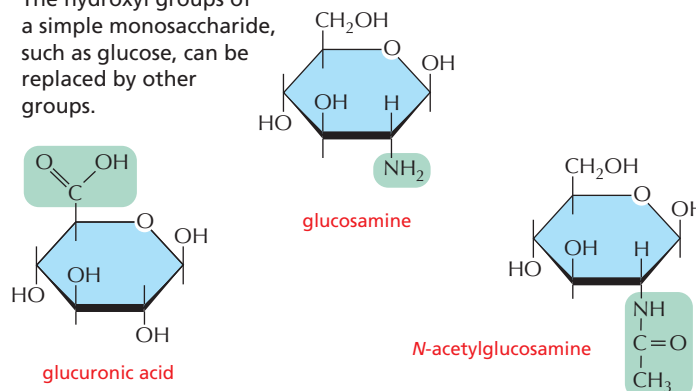
The hydroxyl group on the carbon that carries the aldehyde or ketone can rapidly change from one position to the other. These two positions are called α and β.



As soon as one sugar is linked to another, the α or β form is frozen.

SUGAR DERIVATIVES

The hydroxyl groups of a simple monosaccharide, such as glucose, can be replaced by other groups.



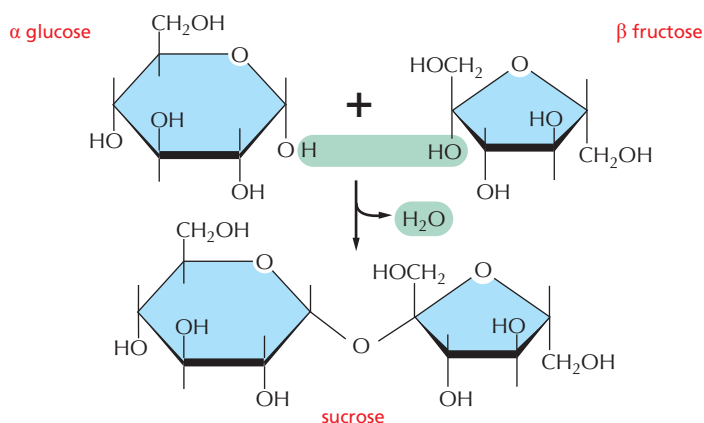
DISACCHARIDES

The carbon that carries the aldehyde or the ketone can react with any hydroxyl group on a second sugar molecule to form a **disaccharide**.

Three common disaccharides are

- maltose (glucose + glucose)
- lactose (galactose + glucose)
- sucrose (glucose + fructose)

The reaction forming sucrose is shown here.

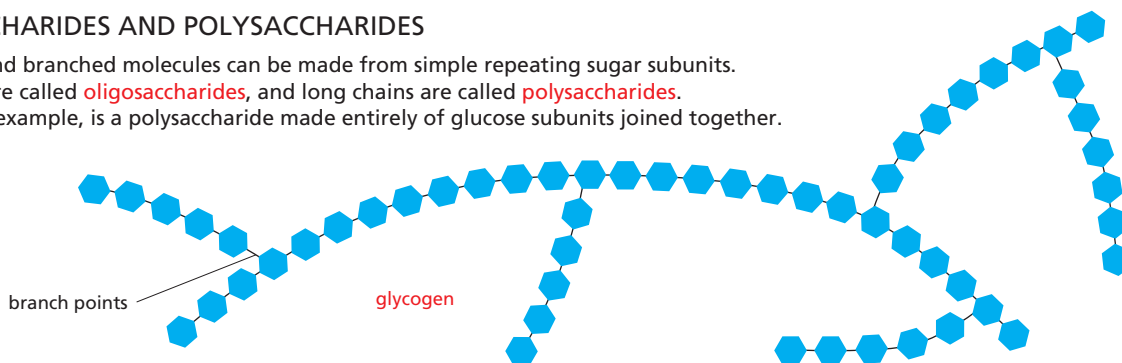


OLIGOSACCHARIDES AND POLYSACCHARIDES

Large linear and branched molecules can be made from simple repeating sugar subunits.

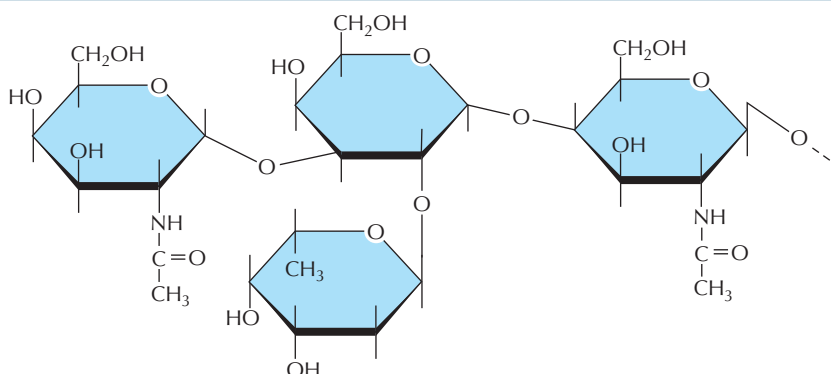
Short chains are called **oligosaccharides**, and long chains are called **polysaccharides**.

Glycogen, for example, is a polysaccharide made entirely of glucose subunits joined together.



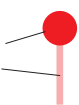
COMPLEX OLIGOSACCHARIDES

In many cases, a sugar sequence is nonrepetitive. Many different molecules are possible. Such complex oligosaccharides are usually linked to proteins or to lipids, as is this oligosaccharide, which is part of a cell-surface molecule that defines a particular blood group.

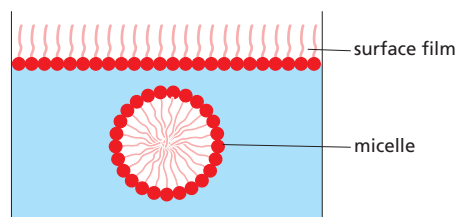


LIPID AGGREGATES

Fatty acids have a hydrophilic head and a hydrophobic tail.



In water, they can form either a surface film or small, spherical micelles.

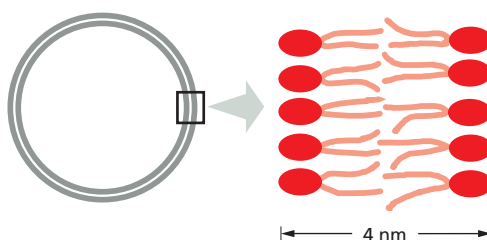


Their derivatives can form larger aggregates held together by hydrophobic forces:

Triacylglycerols form large, spherical fat droplets in the cell cytoplasm.

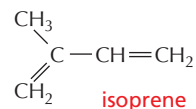


Phospholipids and **glycolipids** form self-sealing **lipid bilayers**, which are the basis for all cell membranes.



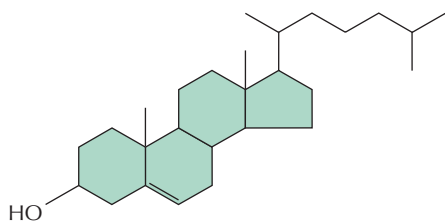
OTHER LIPIDS

Lipids are defined as water-insoluble molecules that are soluble in organic solvents. Two other common types of lipids are steroids and polyisoprenoids. Both are made from isoprene units.

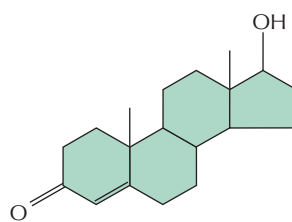


STERIODS

Steroids have a common multiple-ring structure.



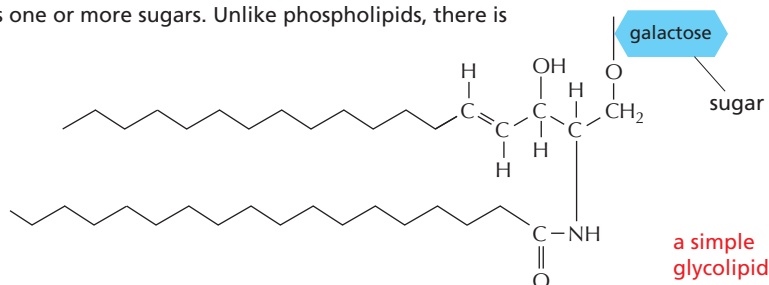
cholesterol—found in many cell membranes



testosterone—male sex hormone

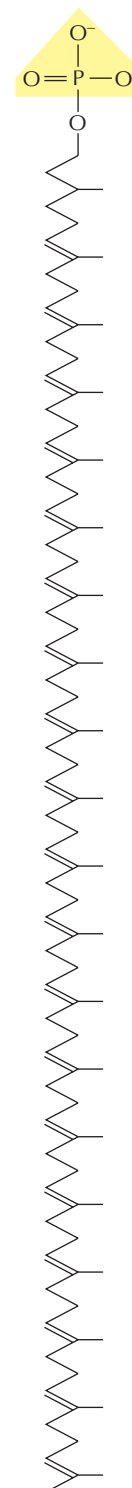
GLYCOLIPIDS

Like phospholipids, these compounds are composed of a hydrophobic region, containing two long hydrocarbon tails, and a polar region, which contains one or more sugars. Unlike phospholipids, there is no phosphate.

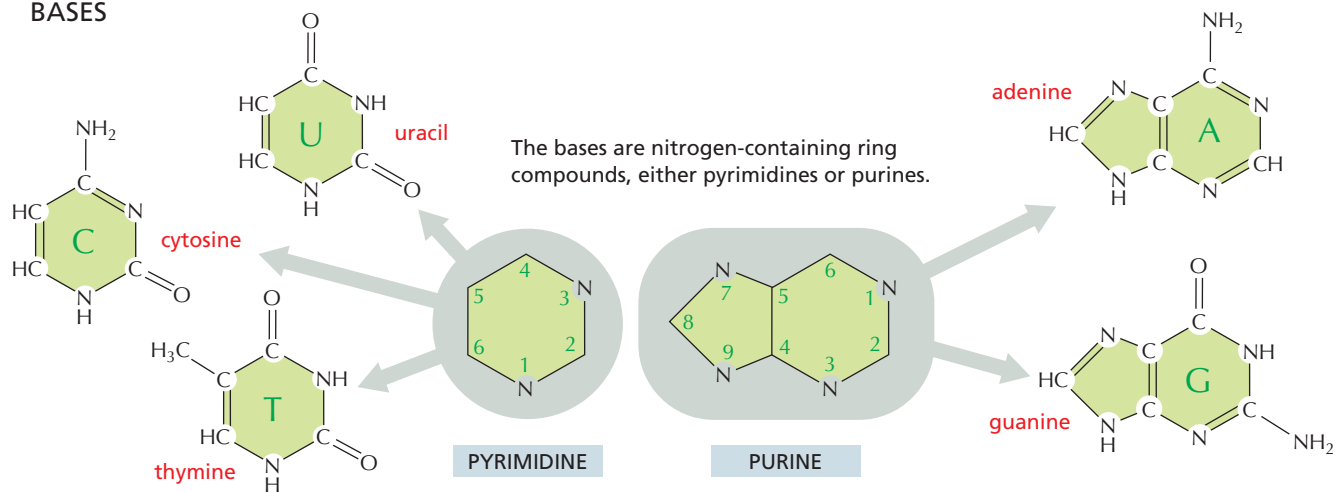


POLYISOPRENOIDS

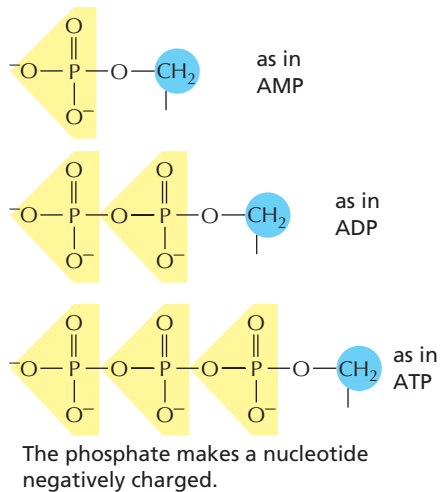
Long-chain polymers of isoprene



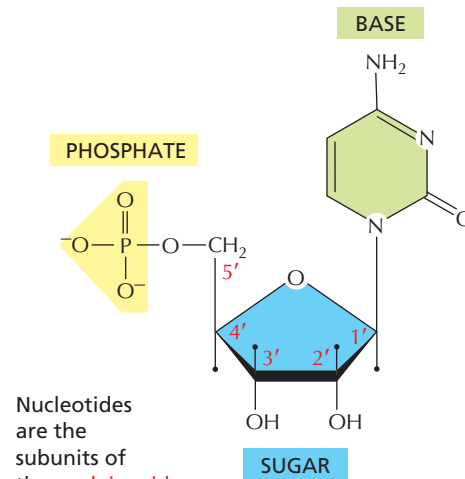
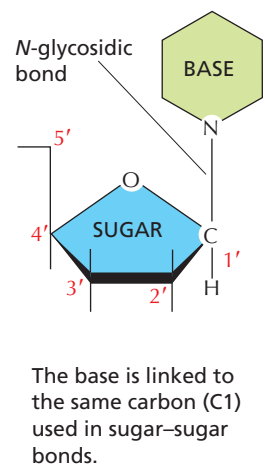
dolichol phosphate—used to carry activated sugars in the membrane-associated synthesis of glycoproteins and some polysaccharides

BASES**PHOSPHATES**

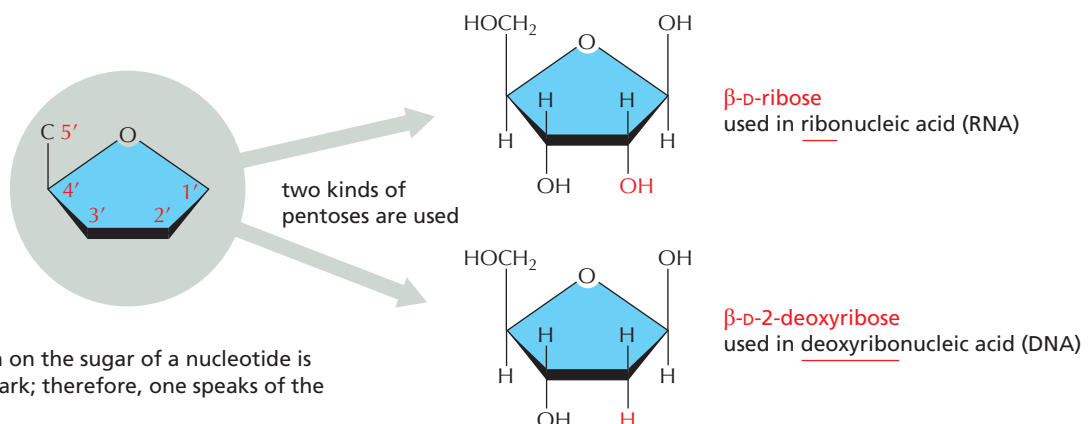
The phosphates are normally joined to the C5 hydroxyl of the ribose or deoxyribose sugar (designated 5'). Mono-, di-, and triphosphates are common.

**NUCLEOTIDES**

A nucleotide consists of a nitrogen-containing base, a five-carbon sugar, and a phosphate group.

**BASE-SUGAR LINKAGE****SUGARS**

PENTOSE
a five-carbon sugar



Each numbered carbon on the sugar of a nucleotide is followed by a prime mark; therefore, one speaks of the "5-prime carbon," etc.

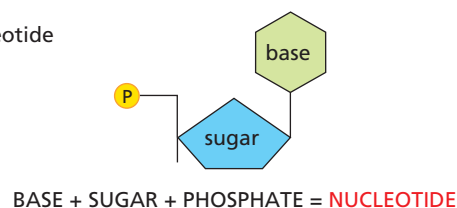
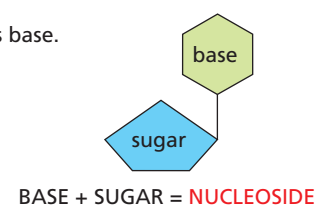
NOMENCLATURE

A nucleoside or nucleotide is named according to its nitrogenous base.

BASE	NUCLEOSIDE	ABBR.
adenine	adenosine	A
guanine	guanosine	G
cytosine	cytidine	C
uracil	uridine	U
thymine	thymidine	T

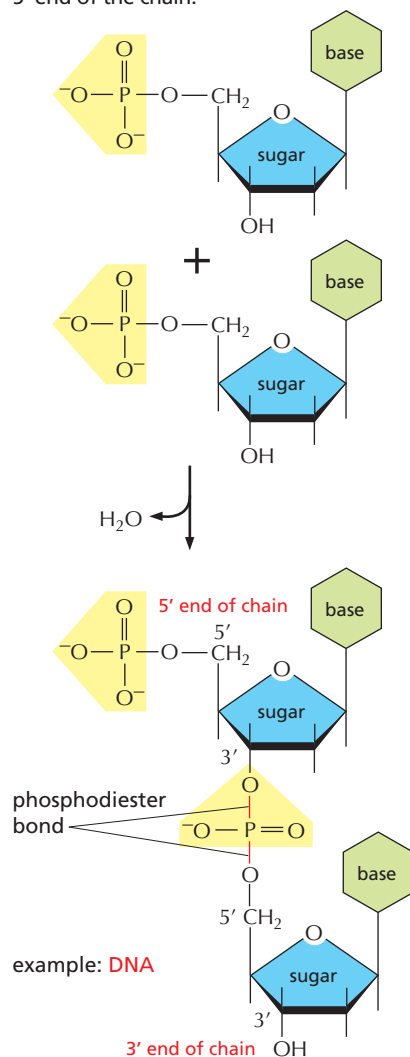
Single-letter abbreviations are used variously as shorthand for (1) the base alone, (2) the nucleoside, or (3) the whole nucleotide—the context will usually make clear which of the three entities is meant. When the context is not sufficient, we will add the terms “base,” “nucleoside,” “nucleotide,” or—as in the examples below—use the full 3-letter nucleotide code.

AMP = adenosine monophosphate
dAMP = deoxyadenosine monophosphate
UDP = uridine diphosphate
ATP = adenosine triphosphate



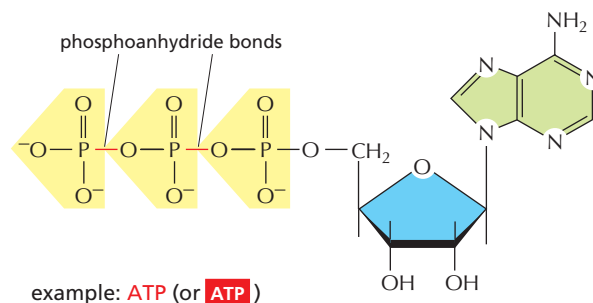
NUCLEIC ACIDS

To form nucleic acid polymers, nucleotides are joined together by **phosphodiester bonds** between the 5' and 3' carbon atoms of adjacent sugar rings. The linear sequence of nucleotides in a nucleic acid chain is abbreviated using a one-letter code, such as AGCTT, starting with the 5' end of the chain.

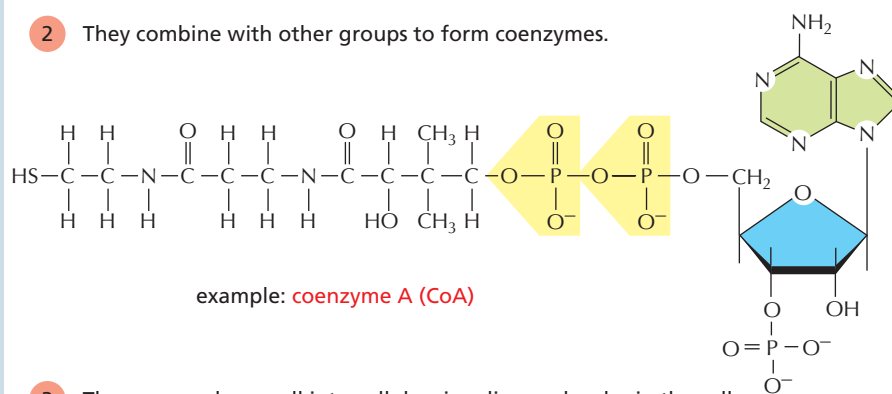


NUCLEOTIDES AND THEIR DERIVATIVES HAVE MANY OTHER FUNCTIONS

- 1 As nucleoside di- and triphosphates, they carry chemical energy in their easily hydrolyzed phosphoanhydride bonds.

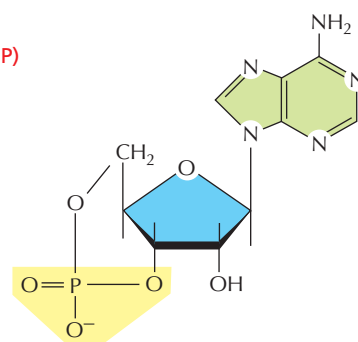


- 2 They combine with other groups to form coenzymes.



- 3 They are used as small intracellular signaling molecules in the cell.

example: **cyclic AMP (cAMP)**

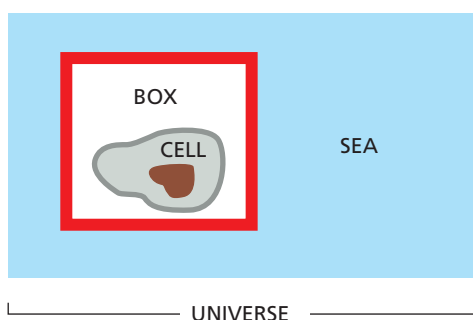


THE IMPORTANCE OF FREE ENERGY FOR CELLS

Life is possible because of the complex network of interacting chemical reactions occurring in every cell. In viewing the metabolic pathways that comprise this network, one might suspect that the cell has had the ability to evolve an enzyme to carry out any reaction that it needs. But this is not so. Although enzymes are powerful catalysts, they can promote only those reactions that are thermodynamically possible; other reactions proceed in cells only because they are *coupled* to very favorable reactions that drive them.

The question of whether a reaction can occur spontaneously, or instead needs to be coupled to another reaction, is central to cell biology. The answer is obtained by reference to a quantity called the *free energy*. In this Panel, we shall explain some of the fundamental ideas—derived from a special branch of chemistry and physics called *thermodynamics*—that are required for understanding what free energy is and why it is so important to cells.

ENERGY RELEASED BY CHANGES IN CHEMICAL BONDING IS CONVERTED INTO HEAT



An *enclosed system* is defined as a collection of molecules that does not exchange matter with the rest of the universe (e.g., the “cell in a box” shown above). Any such system will contain molecules with a total energy E . This energy will be distributed in a variety of ways: some as the translational energy of the molecules, some as their vibrational and rotational energies, but most as the bonding energies between the individual atoms that make up the molecules. Suppose that a reaction occurs in the system. The **first law of thermodynamics** places a constraint on what types of reactions are possible: it states that “**in any process, the total energy of the universe remains constant.**” For example, suppose that reaction $A \rightarrow B$ occurs somewhere in the box and releases a great deal of chemical-bond energy. This energy will initially increase the intensity of molecular motions (translational, vibrational, and rotational) in the system, which is equivalent to raising its temperature. However, these increased motions will soon be transferred out of the system by a series

of molecular collisions that heat up first the walls of the box and then the outside world (represented by the sea in our example). In the end, the system returns to its initial temperature, by which time all the chemical-bond energy released in the box has been converted into heat energy and transferred out of the box to the surroundings. According to the first law, the change in the energy in the box (ΔE_{box} , which we shall denote as ΔE) must be equal and opposite to the amount of heat energy transferred, which we shall designate as h ; that is, $\Delta E = -h$. Thus, the energy in the box (E) decreases when heat leaves the system.

E also can change during a reaction as a result of work being done on the outside world. For example, suppose that there is a small increase in the volume (ΔV) of the box during a reaction. Because the walls of the box must push against the constant pressure (P) in the surroundings in order to expand, this does work on the outside world and requires energy. The energy used is $P(\Delta V)$, which according to the first law must decrease the energy in the box (E) by the same amount. In most reactions, chemical-bond energy is converted into both work and heat. **Enthalpy** (H) is a composite function that includes both of these ($H = E + PV$). To be rigorous, it is the change in enthalpy (ΔH) in an enclosed system, and not the change in energy, that is equal to the heat transferred to the outside world during a reaction. Reactions in which H decreases release heat to the surroundings and are said to be “exothermic,” while reactions in which H increases absorb heat from the surroundings and are said to be “endothermic.” Thus, $-h = \Delta H$. However, the volume change is negligible in most biological reactions, so to a good approximation

$$-h = \Delta H \approx \Delta E$$

THE SECOND LAW OF THERMODYNAMICS

Consider a container in which 1000 coins are all lying heads-up. If the container is shaken vigorously, subjecting the coins to the types of random motions that all molecules experience due to their frequent collisions with other molecules, one will end up with about half the coins oriented heads-down. The reason for this reorientation is that there is only a single way in which the original orderly state of the coins can be reinstated (every coin must lie heads-up), whereas there are many different ways (about 10^{298}) to achieve a disorderly state in which there is an equal mixture of heads and tails; in fact, there are more ways

to achieve a 50–50 state than to achieve any other state. Each state has a probability of occurrence that is proportional to the number of ways it can be realized. The **second law of thermodynamics** states that “**systems will change spontaneously from states of lower probability to states of higher probability.**” Because states of lower probability are more ordered than states of higher probability, the second law can be restated: “**the universe constantly changes so as to become more disordered.**”

THE ENTROPY, S

The second law (but not the first law) allows one to predict the *direction* of a particular reaction. But to make it useful for this purpose, one needs a convenient measure of the probability or, equivalently, the degree of disorder of a state. The entropy (S) is such a measure. It is a logarithmic function of the probability. Thus the *change in entropy* (ΔS) that occurs when the reaction $A \rightarrow B$ converts 1 mole of A into 1 mole of B is

$$\Delta S = R \ln p_B/p_A$$

where p_A and p_B are the probabilities of the two states A and B, R is the gas constant ($8.31 \text{ J K}^{-1} \text{ mole}^{-1}$), and ΔS is measured in entropy units (eu).

For 1000 coins, the relative probability of all heads (state A) versus half heads and half tails (state B) is equal to the ratio of the number of different ways that the two results can be obtained. One can calculate that $p_A = 1$ and $p_B = 1000!(500! \times 500!) = 10^{298}$. Therefore, the entropy change for the reorientation of the coins

when their container is vigorously shaken and an equal mixture of heads and tails is obtained is $R \ln (10^{298})$, or about 1370 eu per mole of such containers (6×10^{23} containers). Because ΔS defined above is positive for the transition from state A to state B ($p_B/p_A > 1$), **reactions with an increase in S (i.e., for which $\Delta S > 0$) are favored and will occur spontaneously.**

Heat energy causes the random commotion of molecules. Because the transfer of heat from an enclosed system to its surroundings increases the number of different arrangements that the molecules in the outside world can have, it increases their entropy. It can be shown that the release of a fixed quantity of heat energy has a greater disordering effect at low temperature than at high temperature, and that the value of ΔS for the surroundings, as described above (ΔS_{sea}), is precisely equal to h , the amount of heat transferred to the surroundings from the system, divided by the absolute temperature (T):

$$\Delta S_{\text{sea}} = h/T$$

THE GIBBS FREE ENERGY, G

When dealing with an enclosed biological system, one would like to have a simple way of predicting whether a given reaction will or will not occur spontaneously in the system. We have seen that the crucial question is whether the entropy change for the universe is positive or negative when that reaction occurs. In our idealized system, the cell in a box, there are two separate components to the entropy change of the universe—the entropy change for the system enclosed in the box and the entropy change for the surrounding “sea”—and both must be added together before any prediction can be made. For example, it is possible for a reaction to absorb heat and thereby decrease the entropy of the sea ($\Delta S_{\text{sea}} < 0$) and at the same time cause such a large degree of disordering inside the box ($\Delta S_{\text{box}} > 0$) that the total $\Delta S_{\text{universe}} = \Delta S_{\text{sea}} + \Delta S_{\text{box}}$ is greater than zero. In this case, the reaction will occur spontaneously, even though the sea gives up heat to the box during the reaction. An example of such a reaction is the dissolving of sodium chloride in a beaker containing water (the “box”), which is a spontaneous process even though the temperature of the water drops as the salt goes into solution.

Chemists have found it useful to define a number of new “composite functions” that describe *combinations* of physical properties of a system. The properties that can be combined include the temperature (T), pressure (P), volume (V), energy (E), and entropy (S). The enthalpy (H) is one such composite function. But **by far the most useful composite function for biologists is the Gibbs free energy, G .** It serves as an accounting device that allows one to deduce the entropy change of the universe resulting from a chemical reaction in the box, while avoiding any separate consideration of the entropy change in the sea. The definition of G is

$$G = H - TS$$

where, for a box of volume V , H is the enthalpy described earlier ($E + PV$), T is the absolute temperature, and S is the entropy. Each of these quantities applies to the inside of the box only. The change in free energy during a reaction in the box (the G of the products minus the G of the starting materials) is denoted as ΔG and, as we shall now demonstrate, it is a direct measure of the amount of disorder that is created in the universe when the reaction occurs.

At constant temperature the change in free energy (ΔG) during a reaction equals $\Delta H - T\Delta S$. Remembering that $\Delta H = -h$, the heat absorbed from the sea, we have

$$\begin{aligned} -\Delta G &= -\Delta H + T\Delta S \\ -\Delta G &= h + T\Delta S, \text{ so } -\Delta G/T = h/T + \Delta S \end{aligned}$$

But h/T is equal to the entropy change of the sea (ΔS_{sea}), and the ΔS in the above equation is ΔS_{box} . Therefore

$$-\Delta G/T = \Delta S_{\text{sea}} + \Delta S_{\text{box}} = \Delta S_{\text{universe}}$$

We conclude that **the free-energy change is a direct measure of the entropy change of the universe.** A reaction will proceed in the direction that causes the change in the free energy (ΔG) to be less than zero, because in this case there will be a positive entropy change in the universe when the reaction occurs.

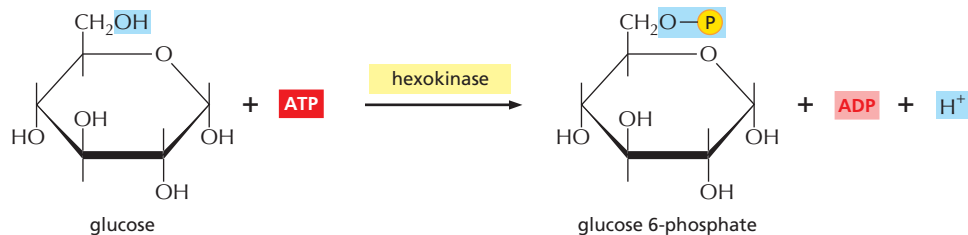
For a complex set of coupled chemical reactions involving many different molecules, the total free-energy change can be computed simply by adding up the free energies of all the different molecular species after the reaction and comparing this value with the sum of free energies before the reaction. For common substances, the required free-energy values can be found from published tables. In this way, one can predict the direction of a reaction and thereby readily check the feasibility of any proposed mechanism. (Thus, for example, from the magnitude of the electrochemical proton gradient across the inner mitochondrial membrane and the ΔG for ATP hydrolysis inside the mitochondrion, one can be certain that the ATP synthase enzyme requires that more than one proton pass through it for each molecule of ATP that it synthesizes.)

The value of ΔG for a reaction is a direct measure of how far the reaction is from equilibrium. The large negative value for ATP hydrolysis in a cell merely reflects the fact that cells keep the ATP hydrolysis reaction as much as 10 orders of magnitude away from equilibrium. If a reaction reaches equilibrium, $\Delta G = 0$, that reaction then proceeds at precisely equal rates in the forward and backward directions. For ATP hydrolysis, equilibrium is reached when the vast majority of the ATP has been hydrolyzed, as occurs in a dead cell.

For each step, the part of the molecule that undergoes a change is shadowed in **blue**, and the name of the enzyme that catalyzes the reaction is in a **yellow box**. Reactions represented by double arrows ($\rightleftharpoons$) are readily reversible, whereas those represented by single arrows ($\rightarrow$) are effectively irreversible. To watch a video of the reactions of glycolysis, see Movie 2.5.

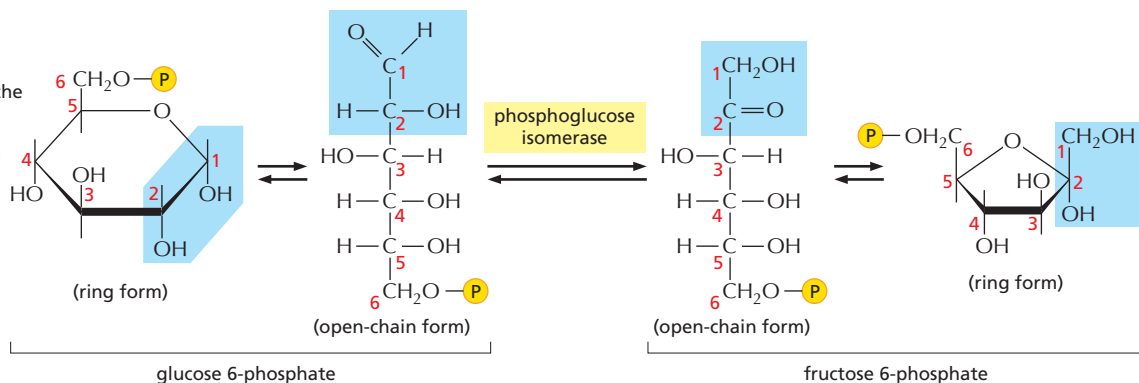
STEP 1

Glucose is phosphorylated by ATP to form a sugar phosphate. The negative charge of the phosphate prevents passage of the sugar phosphate through the plasma membrane, trapping glucose inside the cell.



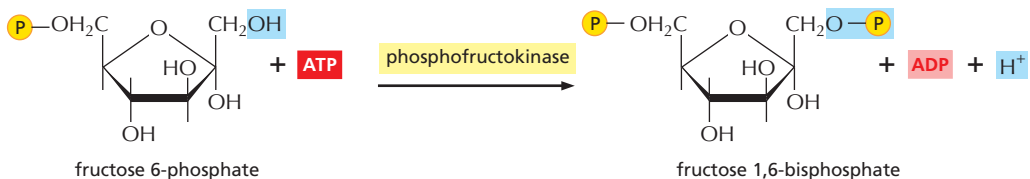
STEP 2

A readily reversible rearrangement of the chemical structure (isomerization) moves the carbonyl oxygen from carbon 1 to carbon 2, forming a ketose from an aldose sugar. (See Panel 2-4, pp. 100-101.)



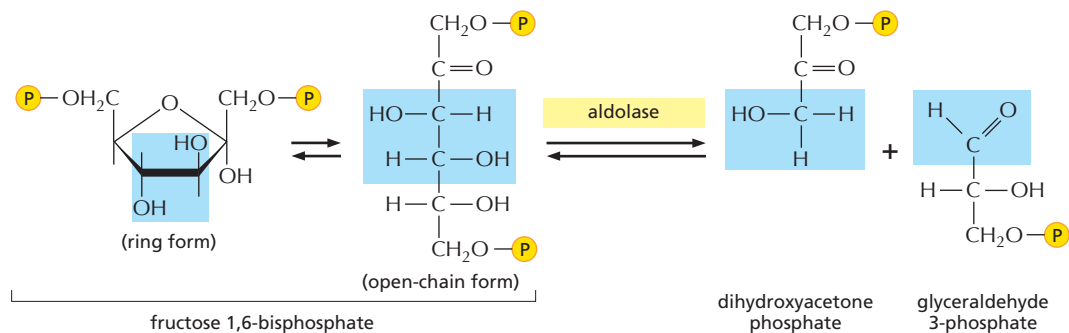
STEP 3

The new hydroxyl group on carbon 1 is phosphorylated by ATP, in preparation for the formation of two three-carbon sugar phosphates. The entry of sugars into glycolysis is controlled at this step, through regulation of the enzyme *phosphofructokinase*.



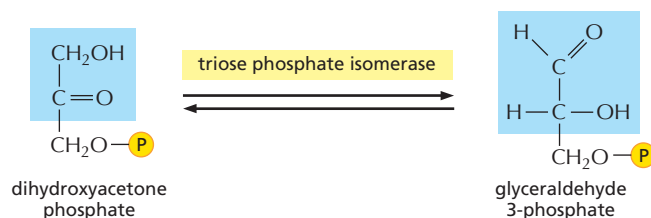
STEP 4

The six-carbon sugar is cleaved to produce two three-carbon molecules. Only the glyceraldehyde 3-phosphate can proceed immediately through glycolysis.



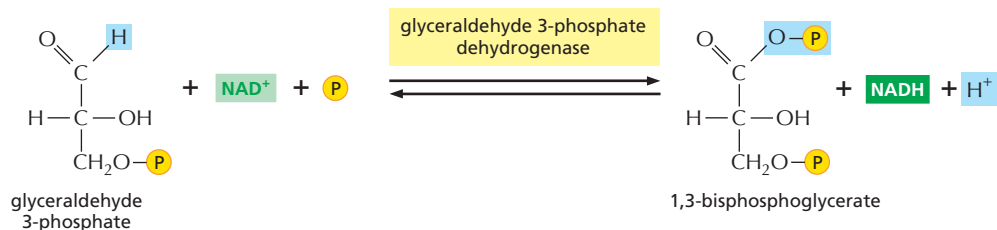
STEP 5

The other product of step 4, dihydroxyacetone phosphate, is isomerized to form a second molecule of glyceraldehyde 3-phosphate.

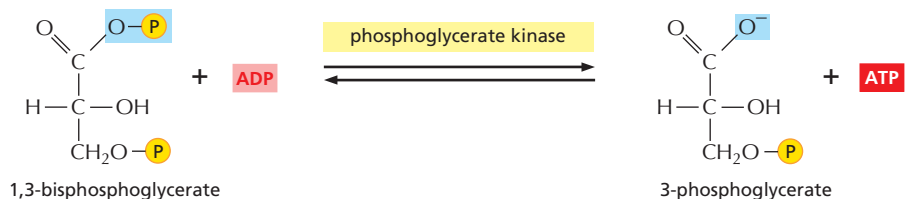


STEP 6

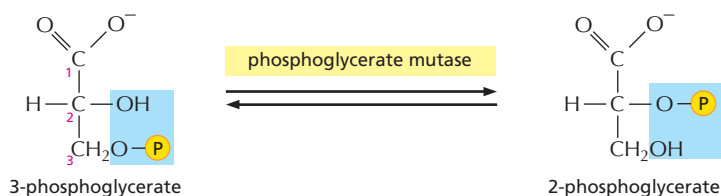
The two molecules of glyceraldehyde 3-phosphate produced in steps 4 and 5 are oxidized. The energy-generation phase of glycolysis begins, as NADH and a new high-energy anhydride linkage to phosphate are formed (see Figure 2-47).

**STEP 7**

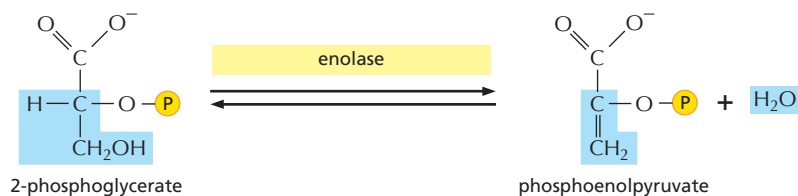
The transfer to ADP of the high-energy phosphate group that was generated in step 6 forms ATP.

**STEP 8**

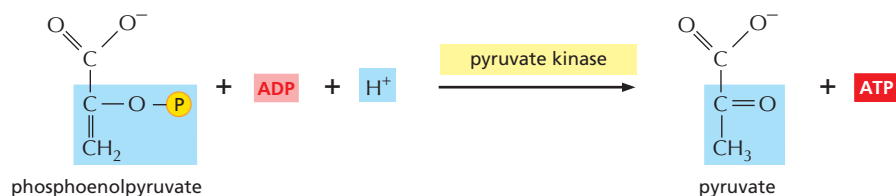
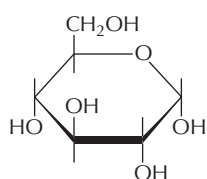
The remaining phosphate ester linkage in 3-phosphoglycerate, which has a relatively low free energy of hydrolysis, is moved from carbon 3 to carbon 2 to form 2-phosphoglycerate.

**STEP 9**

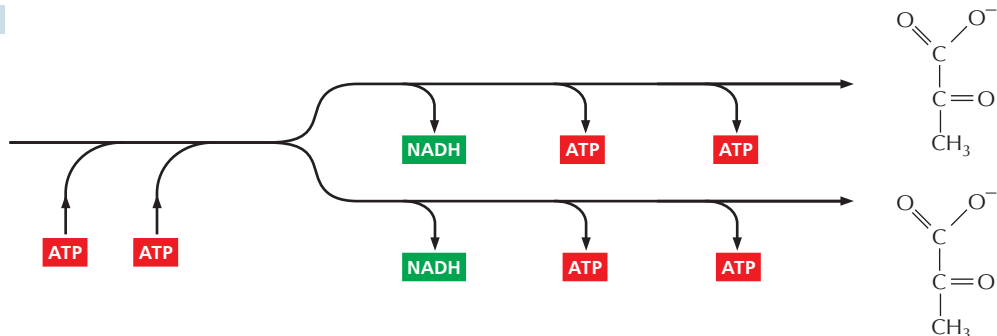
The removal of water from 2-phosphoglycerate creates a high-energy enol phosphate linkage.

**STEP 10**

The transfer to ADP of the high-energy phosphate group that was generated in step 9 forms ATP, completing glycolysis.

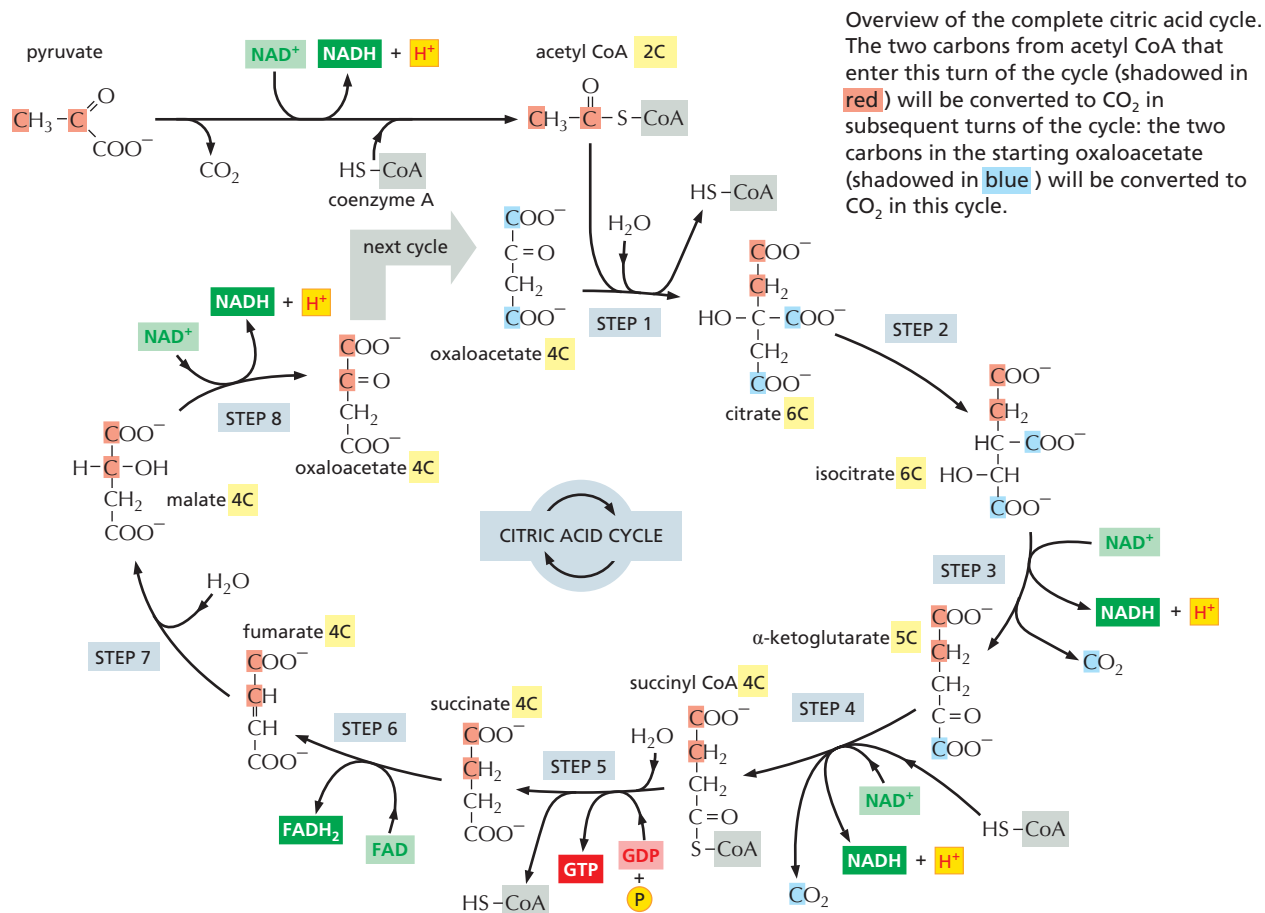
**NET RESULT OF GLYCOLYSIS**

one molecule of glucose



In addition to the pyruvate, the net products of glycolysis are two molecules of ATP and two molecules of NADH.

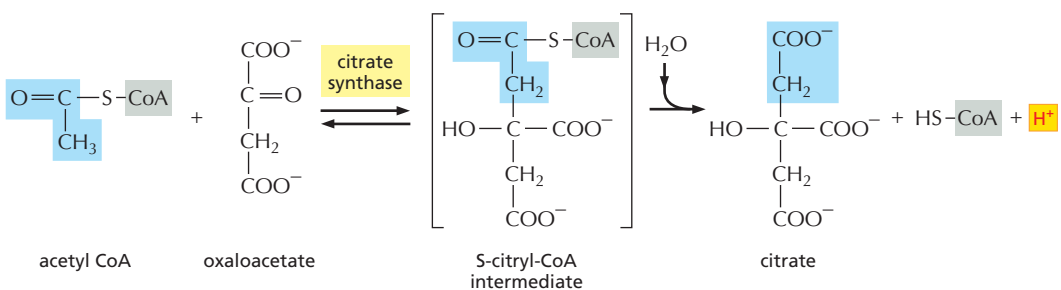
two molecules of pyruvate



Details of these eight steps are shown below. In this part of the Panel, for each step, the part of the molecule that undergoes a change is shadowed in blue, and the name of the enzyme that catalyzes the reaction is in a yellow box. To watch a video of the reactions of the citric acid cycle, see Movie 2.6.

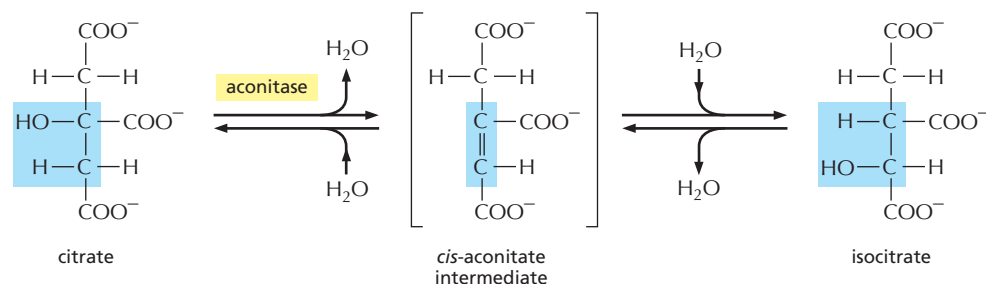
STEP 1

After the enzyme removes a proton from the CH₃ group on acetyl CoA, the negatively charged CH₂⁻ forms a bond to a carbonyl carbon of oxaloacetate. The subsequent loss by hydrolysis of the coenzyme A (HS-CoA) drives the reaction strongly forward.



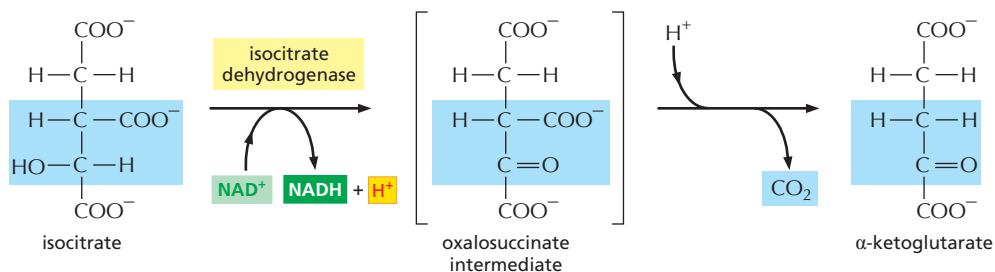
STEP 2

An isomerization reaction, in which water is first removed and then added back, moves the hydroxyl group from one carbon atom to its neighbor.



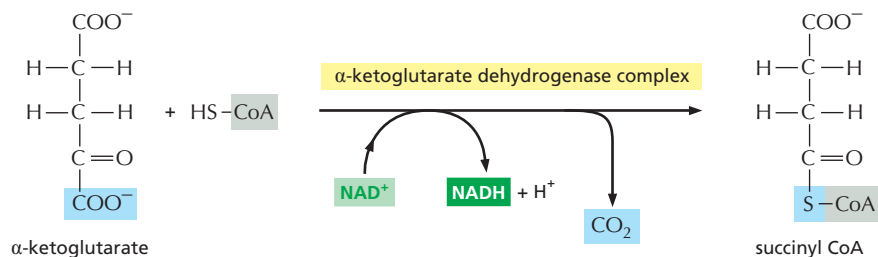
STEP 3

In the first of four oxidation steps in the cycle, the carbon carrying the hydroxyl group is converted to a carbonyl group. The immediate product is unstable, losing CO_2 while still bound to the enzyme.



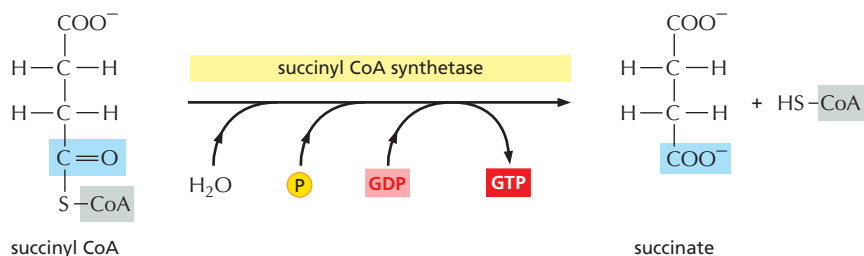
STEP 4

The α -ketoglutarate dehydrogenase complex closely resembles the large enzyme complex that converts pyruvate to acetyl CoA, the pyruvate dehydrogenase complex. It likewise catalyzes an oxidation that produces NADH , CO_2 , and a high-energy thioester bond to coenzyme A (CoA).



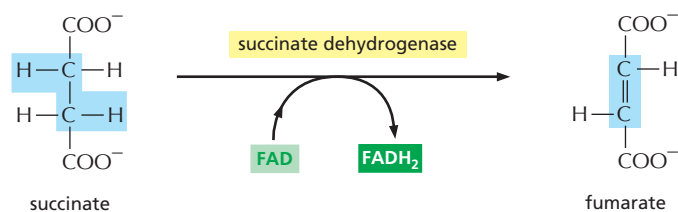
STEP 5

An inorganic phosphate displaces the CoA, forming a high-energy phosphate linkage to succinate. This phosphate is then passed to GDP to form GTP. (In bacteria and plants, ATP is formed instead.)



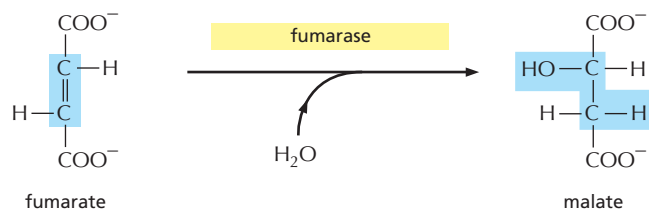
STEP 6

In the third oxidation step of the cycle, FAD accepts two hydrogen atoms from succinate.



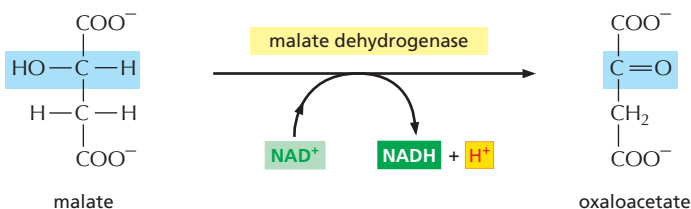
STEP 7

The addition of water to fumarate places a hydroxyl group next to a carbonyl carbon.



STEP 8

In the last of four oxidation steps in the cycle, the carbon carrying the hydroxyl group is converted to a carbonyl group, regenerating the oxaloacetate needed for step 1.



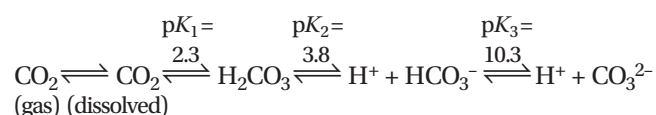
PROBLEMS

Which statements are true? Explain why or why not.

- 2-1** A 10^{-8} M solution of HCl has a pH of 8.
- 2-2** Most of the interactions between macromolecules could be mediated just as well by covalent bonds as by noncovalent bonds.
- 2-3** Animals and plants use oxidation to extract energy from food molecules.
- 2-4** If an oxidation occurs in a reaction, it must be accompanied by a reduction.
- 2-5** Linking the energetically unfavorable reaction $A \rightarrow B$ to a second, favorable reaction $B \rightarrow C$ will shift the equilibrium constant for the first reaction.
- 2-6** The criterion for whether a reaction proceeds spontaneously is ΔG , not ΔG° , because ΔG takes into account the concentrations of the substrates and products.
- 2-7** The oxygen consumed during the oxidation of glucose in animal cells is returned as CO_2 to the atmosphere.

Discuss the following problems.

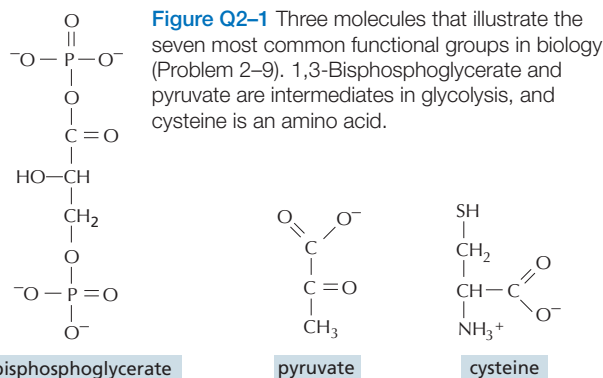
2-8 During an all-out sprint, muscles produce a high concentration of lactic acid, which lowers the pH of the blood and of the cytosol. The lower pH inside the cell reduces the rate of ATP production and contributes to the fatigue that sprinters experience well before their fuel reserves are exhausted. The main blood buffer against pH changes is the bicarbonate/ CO_2 system.



To improve their performance, would you advise sprinters to hold their breath or to breathe rapidly just before the race? Why?

- A.** Breathe rapidly to decrease the pH of the blood.
- B.** Breathe rapidly to increase the pH of the blood.
- C.** Hold their breath to decrease the pH of the blood.
- D.** Hold their breath to increase the pH of the blood.

2-9 The three molecules in **Figure Q2-1** contain the seven most common functional groups in biology. Most molecules in the cell are built from these functional groups. Indicate and name the functional groups in these molecules.



2-10 The molecular weight of ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) is 46 and its density is 0.789 g/cm^3 .

- A.** What is the molarity of ethanol in beer that is 5% ethanol by volume? [Alcohol content of beer varies from about 4% (lite beer) to 8% (stout beer).]
- B.** The legal limit for a driver's blood alcohol content varies, but 80 mg of ethanol per 100 mL of blood (usually referred to as a blood alcohol level of 0.08) is typical. What is the molarity of ethanol in a person at this legal limit?
- C.** How many 12-oz (355-mL) bottles of 5% beer could a 70-kg person drink and remain under the legal limit? A 70-kg person contains about 40 liters of water. Ignore the metabolism of ethanol and assume that the water content of the person remains constant and that ethanol distributes evenly in that volume.

2-11 A histidine side chain is known to play an important role in the catalytic mechanism of an enzyme; however, it is not clear whether histidine is required in its protonated (charged) or unprotonated (uncharged) state. To answer this question, you measure enzyme activity over a range of pH, with the results shown in **Figure Q2-2**. Which form of histidine is required for enzyme activity?

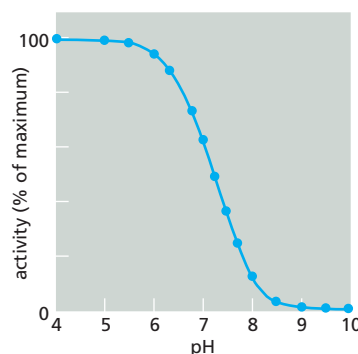


Figure Q2-2 Enzyme activity as a function of pH (Problem 2-11).

2-12 The organic chemistry of living cells is said to be special for two reasons: it occurs in an aqueous environment, and it accomplishes some very complex reactions. But do you suppose it is really all that much different from the organic chemistry carried out in the top laboratories in the world? Why or why not?

2-13 Polymerization of tubulin subunits into microtubules occurs with an increase in the orderliness of the subunits. Yet tubulin polymerization occurs with an increase in entropy (decrease in order). How can that be?

2-14 “Diffusion” sounds slow—and over everyday distances it is—but on the scale of a cell it is very fast. The average instantaneous velocity of a particle in solution—that is, the velocity between the very frequent collisions—is

$$v = (kT/m)^{1/2}$$

where $k = 1.38 \times 10^{-16}$ g cm²/K sec², T = temperature in K (37°C is 310 K), and m = mass in g/molecule.

Calculate the instantaneous velocity of a water molecule (molecular mass = 18 daltons), a glucose molecule (molecular mass = 180 daltons), and a myoglobin molecule (molecular mass = 15,000 daltons) at 37°C.

2-15 Each phosphoanhydride bond between the phosphate groups in ATP is a high-energy linkage with a ΔG° value of -30.5 kJ/mole. Hydrolysis of this bond in cells normally liberates usable energy in the range of 45 to 55 kJ/mole. Why do you think a range of values for released energy is given for ΔG , rather than a precise number, as for ΔG° ?

- A. ΔG cannot be accurately measured and can only be estimated as a range of values.
- B. Differences in the concentrations of ATP, ADP, and phosphate can significantly change ΔG .
- C. The specific enzyme catalyzing hydrolysis determines how much energy is released.
- D. The temperature fluctuations occurring inside the cell can significantly change ΔG .

2-16 A 70-kg adult human (154 lb) could meet his or her entire energy needs for one day by eating 3 moles of glucose (540 g). (We do not recommend this.) Each molecule of glucose generates 30 molecules of ATP when it is oxidized to CO₂. The concentration of ATP is maintained in cells at about 2 mM, and a 70-kg adult has about 25 liters of intracellular fluid. Given that the ATP concentration remains constant in cells, calculate how many times per day, on average, each ATP molecule in the body is hydrolyzed and resynthesized.

2-17 What is the “packet of energy” that NADH and NADPH carry?

- A. A hydride ion (two electrons and one proton)
- B. A hydrogen atom (one electron and one proton)
- C. A hydrogen ion (one proton and no electrons)
- D. A hydronium ion (a protonated water molecule)

2-18 Cancer cells can increase the rate of glycolysis up to 200-fold relative to normal differentiated cells. This effect, known as the Warburg effect after its discoverer, is exploited in an imaging technique that is commonly used to detect tumors. An individual is dosed with a molecule labeled with a radioactive isotope of fluorine (¹⁸F). The labeled molecule is preferentially taken up by cancer cells and is detected by positron emission tomography (PET) scanning. Which one of the following molecules would you label with ¹⁸F to detect tumors via the Warburg effect?

- A. Acetyl CoA
- B. Glucose
- C. Lactate
- D. Pyruvate

2-19 Does a Snickers candy bar (65 g, 1360 kJ) provide enough energy to climb from Zermatt (elevation 1660 m) to the top of the Matterhorn (4478 m; [Figure Q2-3](#)), or might you need to stop at Hörnli Hut (3260 m) to eat another one? Imagine that you and your gear have a mass of 75 kg and that all of your work is done against gravity (that is, you are just climbing straight up). Remember from your introductory physics course that

$$\text{work (J)} = \text{mass (kg)} \times g (\text{m/sec}^2) \times \text{height gained (m)}$$

where g is acceleration due to gravity (9.8 m/sec²). One joule is 1 kg m²/sec². What assumptions made here will greatly underestimate how much candy you need?

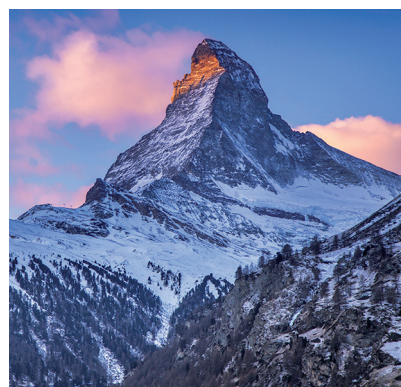


Figure Q2-3 The Matterhorn (Problem 2-19). (Earth Trotter Photos/Shutterstock.)

2-20 Assuming that there are 5×10^{13} cells in the human body and that ATP is turning over at a rate of 10^9 ATP molecules per minute in each cell, how many watts is the human body consuming? (A watt is a joule per second.) Assume that hydrolysis of ATP yields 50 kJ/mole.

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